Multiverse analysis

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The authors made the following contributions. First Author: Conceptualization, Writing - Original Draft Preparation, Writing - Review & Editing; Ernst-August Doelle: Writing - Review & Editing, Supervision.

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Abstract

One or two sentences providing a **basic introduction** to the field, comprehensible to a scientist in any discipline.

Keywords: keywords

Word count: X

Multiverse analysis

Results

Study characteristics

Table 1 describes each study in the primary analytic sample, and Table 2 shows how each study's measure of socioeconomic status was harmonised.

Model diagnostics

The model diagnostics for the main analysis, and for all models presented in this document are in Table 3

Fixed Effect Model

In this document we show the results using another analytic technique. Namely treating study ID as a fixed effect. This is what we wrote in the protocol originally.

The association between physical activity volume and sleep. We estimated the associations between physical activity and sleep (RQ1) using fixed-effects models. We estimated the association between physical activity volume and sleep by age, and the results are presented in Table 4 and Figure 1.

The association between physical activity intensity and sleep. We estimated how physical activity intensity is associated with sleep across different age groups. We present the results controlling for SES, sex, BMI z-score, and the fixed effects of study ID, in Table 4 and Figure 2.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by age.

Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 3.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 4.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 5.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 6.

Log Transforming PA Volume

Based on preliminary modelling on a subset of the data, we chose to leave physical activity volume in its natural form to enhance interpretability. The log scale meant that the variable became unintepretable towards the edge of its range. Below, we present the results had we not made this decision. These results are all moderated by age, as this was our primary analysis. The associated figures are back-transformed to the scale range.

The association between physical activity volume and sleep. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by age, and the results are presented in Table 6 and Figure 7.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 8.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 9.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 10.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 11.

All Eligible Participants Regardless of Valid Wear Days

The primary analysis required at least 4 days of valid wear time. Here we refit the primary models to every eligible participant, regardless of how many valid days they contributed. Both analyses share the same imputations and standardisation constants. Results, controlling for SES, sex, and BMI z-score, are presented in Tables 8 and 9 and Figures 12 to 17.

Participants With at Least Seven Valid Days

Some recommend at least seven nights to estimate sleep regularity. Here we refit the primary models to the 14,961 participants with at least seven days of valid wear time. Results, controlling for SES, sex, and BMI z-score, are presented in Tables 10 and 11 and Figures 18 to 23.

Moderation

In the following section we present models for a range of moderators which were pre-defined in our protocol.

BMI z-score

The association between physical activity volume and sleep by BMI z-score. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by BMI z-score, and the results are presented in Table 12 and Figure 24

The association between physical activity intensity and sleep by BMI z-score. We estimated how physical activity intensity is associated with sleep across BMI z-score. We present the results in Table 12 and Figure 25.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 13 and Figure 26.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 13 and Figure 27.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 13 and Figure 28.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by BMI z-score. Results are presented in Table 13 and Figure 29.

SES

The association between physical activity volume and sleep by SES. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by SES, and the results are presented in Table 14 and Figure 30

The association between physical activity intensity and sleep by SES. We estimated how physical activity intensity is associated with sleep across SES. We present the results in Table 14 and Figure 31.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 15 and Figure 32.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 15 and Figure 33.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 15 and Figure 34.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by SES. Results are presented in Table 15 and Figure 35.

sex

The association between physical activity volume and sleep by sex. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by sex, and the results are presented in Table 16 and Figure 36

The association between physical activity intensity and sleep by sex. We estimated how physical activity intensity is associated with sleep across sex. We present the results in Table 16 and Figure 37.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 17 and Figure 38.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 17 and Figure 39.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 17 and Figure 40.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by sex. Results are presented in Table 17 and Figure 41.

weekday

The association between physical activity volume and sleep by weekday. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by weekday, and the results are presented in Table 18 and Figure 42

The association between physical activity intensity and sleep by weekday. We estimated how physical activity intensity is associated with sleep across weekday. We present the results in Table 18 and Figure 43.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 19 and Figure 44.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 19 and Figure 45.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 19 and Figure 46.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by weekday. Results are presented in Table 19 and Figure 47.

season

The association between physical activity volume and sleep by season. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by season, and the results are presented in Table 20 and Figure 48

The association between physical activity intensity and sleep by season. We estimated how physical activity intensity is associated with sleep across season. We present the results in Table 20 and Figure 49.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 21 and Figure 50.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 21 and Figure 51.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 21 and Figure 52.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by season. Results are presented in Table 21 and Figure 53.

region

The association between physical activity volume and sleep by region. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by region, and the results are presented in Table 22 and Figure 54

The association between physical activity intensity and sleep by region. We estimated how physical activity intensity is associated with sleep across region. We present the results in Table 22 and Figure 55.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 23 and Figure 56.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 23 and Figure 57.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 23 and Figure 58.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by region. Results are presented in Table 23 and Figure 59.

daylight hours

The association between physical activity volume and sleep by daylight hours. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by daylight hours, and the results are presented in Table 24 and Figure 60

The association between physical activity intensity and sleep by daylight hours. We estimated how physical activity intensity is associated with sleep across daylight hours. We present the results in Table 24 and Figure 61.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 25 and Figure 62.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 25 and Figure 63.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 25 and Figure 64.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by daylight hours. Results are presented in Table 25 and Figure 65.

wear location

The association between physical activity volume and sleep by wear location. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by wear location, and the results are presented in Table 26 and Figure 66

The association between physical activity intensity and sleep by wear location. We estimated how physical activity intensity is associated with sleep across wear location. We present the results in Table 26 and Figure 67.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 68.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 69.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 70.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by wear location. Results are presented in Table 27 and Figure 71.

most active hour

The association between physical activity volume and sleep by most active hour. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by most active hour, and the results are presented in Table 28 and Figure 72

The association between physical activity intensity and sleep by most active hour. We estimated how physical activity intensity is associated with sleep across most active hour. We present the results in Table 28 and Figure 73.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 74.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 75.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 76.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by most active hour. Results are presented in Table 29 and Figure 77.

ethnicity

The association between physical activity volume and sleep by ethnicity. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by ethnicity, and the results are presented in Table 30 and Figure 78

The association between physical activity intensity and sleep by ethnicity. We estimated how physical activity intensity is associated with sleep across ethnicity. We present the results in Table 30 and Figure 79.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 80.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 81.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 82.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by ethnicity. Results are presented in Table 31 and Figure 83.

Table 1

Characteristics of the included studies (primary analytic sample)

Study ID	Study	Reference	Country	Device	Wear location	N	Age, mean (range
100			12 countries	GT3X	Hip	6,352	10.4 (9-11)
101			Switzerland	GENEActiv	Wrist	2,927	61.9 (45-86)
102			Denmark	GT3X	Hip	691	10.0 (8-11)
103			New Zealand	Actigraph	Wrist	650	19.3 (8-58)
104			Brazil	GT3X+	Wrist	737	16.4 (14-22)
105			Czech Republic	GT3X+, GT9X Link, wGT3X-BT, wGT3X+	Wrist	1,170	14.0 (8-19)
106			Australia	GENEActiv	Wrist	297	41.9 (18-65)
107			Spain	GT3X+	Wrist	2,423	72.6 (64-94)
108			Australia	GENEActiv	Wrist	1,035	8.8 (7-11)
110			Finland	Actisleep	Wrist	801	62.6 (58-64)
111			United Kingdom	Axivity AX3	Wrist	692	44.7 (20-70)
112			Spain	GENEActiv	Wrist	144	9.5 (8-12)
113			Spain	Actisleep	Wrist	353	8.4 (8-9)
114			United States	WGT3X-BT	Wrist	154	3.5 (2-4)
115			Ecuador	wGT3X-BT	Hip	1,019	8.8 (7-13)
117			Spain	GENEActiv	Wrist	175	8.8 (4-11)
118			Australia	Actigraph	Wrist	302	16.6 (15-18)
212			Spain	WGT3X	Hip	272	4.7 (2-6)
221			United Kingdom	GENEActiv	Wrist	4,070	69.4 (59-82)
222			-	-	-	0	-

Table 2

Harmonisation of socioeconomic status

Study ID	Original category
100	1
100	2
100	4
100	3
101	High school
101	Apprenticeship
101	Mandatory education
101	University education
102	Vocational basic training/vocational internship and main course
102	Short higher education
102	Long higher education/ Research degree
102	Regular secondary education and vocational secondary education
102	Medium higher education
102	Primary school
102	Unanswered
104	Incomplete elementary school
104	Complete Bachelor degree
104	Incomplete bachelor degree
104	Complete high school
104	Complete elementary school
104	Unknown
104	Has not studied
105	vocational education
105	high school education
105	master's degree or higher
105	bachelor's degree or equivalent
105	10th grade or less
105	short tertiary education
105	26-100 books
105	11-25 books
105	>200 books
105	101-200 books
105	0-10 books
106	High school or less

Table 2 continued

Study ID	Original category
106	Technical or further education institutions (e.g., TAFE)
106	University degree
107	Secondary education 2nd stage (i.e., High shool graduate, complete 12th grade or equivalent
107	Incomplete primary education (i.e., less than 6th grade or equivalent)
107	Complete primary education (i.e., complete 6th grade or equivalent)
107	Medium university studies (i.e., university diplomature or equivalent)
107	Higher university studies (i.e., Bachelor degree, Master degree, Doctoral degree or equivalen
107	Secondary education 1st stage (i.e., complete 10th grade or equivalent)
107	Illiterate (i.e., not able to read or write)
108	Enough to fill two bookcases (101-200 Books)
108	Enough to fill one shelf (11-25 Books)
108	Enough to fill 3 or more bookcases (more than 200 Books)
108	Enough to fill one bookcases (26-100 books)
108	None or very few (0-10 Books)
110	lower-level non-manual occupation
110	higher-level non-manual occupation
110	manual occupation
110	service occupation
111	10
113	primary school
113	university degree
113	bachelor's degree
113	high school
113	master's degree or higher
113	vocational training
114	poor
114	O level
114	A/S level
114	ba/bsc
114	higher degree
114	No academic quals
115	Medium
115	High
115	Low
117	secondary education

Table 2 continued

Study ID	Original category
117	higher education
117	primary education
118	2
118	7
118	5
118	3
118	10
118	8
118	9
118	6
118	4
118	1
221	O level
221	A/S level
221	ba/bsc
221	higher degree
221	No academic quals
222	q5-wealthiest
222	q4
222	q3
222	q2
222	q1-poorest

Table 3

Model diagnostics

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Models moderated by age				
Scale sleep duration by PA volume	-0.01	2.23	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.93	3.65	100.00%	0.00%
Scale sleep onset by PA volume	0.38	10.10	100.00%	0.00%
Scale sleep regularity by PA volume	-0.69	1.64	100.00%	0.00%
Scale sleep duration by PA intensity	-0.01	2.22	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.93	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.39	10.10	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.70	1.63	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	6.06	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	6.10	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	6.04	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.71	6.22	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.27	1.58	100.00%	0.00%
Models moderated by age fixedef				
Scale sleep duration by PA volume	-0.01	2.23	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.93	3.65	100.00%	0.00%
Scale sleep onset by PA volume	0.38	10.10	100.00%	0.00%
Scale sleep regularity by PA volume	-0.69	1.64	100.00%	0.00%
Scale sleep duration by PA intensity	-0.01	2.22	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.93	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.39	10.10	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.70	1.62	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	6.06	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	6.10	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	6.03	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.71	6.22	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.27	1.58	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Models moderated by age log				
Scale sleep duration by log PA volume	-0.01	2.23	100.00%	0.00%
Scale sleep efficiency by log PA volume	-0.93	3.65	100.00%	0.00%
Scale sleep onset by log PA volume	0.38	10.12	100.00%	0.00%
Scale sleep regularity by log PA volume	-0.70	1.66	100.00%	0.00%
Scale sleep duration by PA intensity	-0.01	2.22	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.93	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.39	10.10	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.70	1.63	100.00%	0.00%
Log PA volume by sleep duration(lagged)	-0.47	2.26	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.57	100.00%	0.00%
Log PA volume by sleep efficiency(lagged)	-0.47	2.28	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.56	100.00%	0.00%
Log PA volume by sleep onset(lagged)	-0.47	2.28	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.57	100.00%	0.00%
Log PA volume by sleep regularity(lagged)	-0.37	2.05	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.27	1.58	100.00%	0.00%
Models moderated by age allwear				
Scale sleep duration by PA volume	-0.01	2.23	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.94	3.72	100.00%	0.00%
Scale sleep onset by PA volume	0.38	10.17	100.00%	0.00%
Scale sleep regularity by PA volume	-0.69	1.64	100.00%	0.00%
Scale sleep duration by PA intensity	-0.01	2.22	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.94	3.73	100.00%	0.00%
Scale sleep onset by PA intensity	0.39	10.17	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.70	1.62	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	6.06	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	6.10	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	6.04	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.71	6.22	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.27	1.58	100.00%	0.00%
Models moderated by age 7days				

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep duration by PA volume	0.03	2.53	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.82	2.94	100.00%	0.00%
Scale sleep onset by PA volume	0.33	9.76	100.00%	0.00%
Scale sleep regularity by PA volume	-0.70	1.61	100.00%	0.00%
Scale sleep duration by PA intensity	0.03	2.52	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.82	2.93	100.00%	0.00%
Scale sleep onset by PA intensity	0.34	9.75	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.70	1.59	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.56	5.83	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.21	1.58	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.55	5.90	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.21	1.57	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.56	5.84	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.21	1.57	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.59	6.03	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.58	100.00%	0.00%
Models moderated by bmi				
Scale sleep duration by PA volume	-0.01	2.23	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.93	3.64	100.00%	0.00%
Scale sleep onset by PA volume	0.39	10.07	100.00%	0.00%
Scale sleep regularity by PA volume	-0.70	1.64	100.00%	0.00%
Scale sleep duration by PA intensity	-0.01	2.22	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.93	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.39	10.08	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.70	1.61	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	6.08	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	6.10	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	6.04	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.71	6.17	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.57	100.00%	0.00%
Models moderated by ses				
Scale sleep duration by PA volume	-0.01	2.23	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep efficiency by PA volume	-0.93	3.64	100.00%	0.00%
Scale sleep onset by PA volume	0.39	10.07	100.00%	0.00%
Scale sleep regularity by PA volume	-0.69	1.64	100.00%	0.00%
Scale sleep duration by PA intensity	-0.01	2.22	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.93	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.39	10.08	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.70	1.61	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	6.08	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	6.10	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	6.05	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.71	6.16	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.57	100.00%	0.00%
Models moderated by weekday				
Scale sleep duration by PA volume	-0.03	2.23	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.92	3.64	100.00%	0.00%
Scale sleep onset by PA volume	0.34	10.41	100.00%	0.00%
Scale sleep regularity by PA volume	-0.73	1.87	100.00%	0.00%
Scale sleep duration by PA intensity	-0.03	2.23	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.92	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.35	10.31	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.74	1.90	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.71	6.15	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.28	1.63	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.71	6.18	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.28	1.62	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.72	6.15	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.29	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.74	6.21	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.29	1.61	100.00%	0.00%
Models moderated by season				
Scale sleep duration by PA volume	-0.01	2.23	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.93	3.64	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep onset by PA volume	0.39	10.06	100.00%	0.00%
Scale sleep regularity by PA volume	-0.69	1.63	100.00%	0.00%
Scale sleep duration by PA intensity	-0.02	2.22	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.93	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.39	10.07	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.70	1.61	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	6.08	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	6.10	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	6.04	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.23	1.56	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.71	6.16	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.57	100.00%	0.00%
Models moderated by region				
Scale sleep duration by PA volume	-0.01	2.23	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.93	3.67	100.00%	0.00%
Scale sleep onset by PA volume	0.38	10.07	100.00%	0.00%
Scale sleep regularity by PA volume	-0.69	1.64	100.00%	0.00%
Scale sleep duration by PA intensity	-0.01	2.21	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.93	3.66	100.00%	0.00%
Scale sleep onset by PA intensity	0.39	10.10	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.70	1.61	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	6.05	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	6.11	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	6.01	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.71	6.20	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.58	100.00%	0.00%
Models moderated by daylight				
Scale sleep duration by PA volume	-0.01	2.23	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.93	3.64	100.00%	0.00%
Scale sleep onset by PA volume	0.39	10.06	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep regularity by PA volume	-0.69	1.64	100.00%	0.00%
Scale sleep duration by PA intensity	-0.02	2.23	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.93	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.39	10.08	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.70	1.61	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	6.08	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	6.10	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	6.05	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.71	6.16	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.56	100.00%	0.00%
Models moderated by wear location				
Scale sleep duration by PA volume	-0.01	2.23	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.93	3.65	100.00%	0.00%
Scale sleep onset by PA volume	0.38	10.08	100.00%	0.00%
Scale sleep regularity by PA volume	-0.70	1.65	100.00%	0.00%
Scale sleep duration by PA intensity	-0.01	2.22	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.93	3.64	100.00%	0.00%
Scale sleep onset by PA intensity	0.39	10.09	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.70	1.63	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	6.11	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	6.10	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	6.04	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.71	6.20	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.28	1.58	100.00%	0.00%
Models moderated by PA mostactivehr				
Scale sleep duration by PA volume	-0.01	2.23	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.93	3.64	100.00%	0.00%
Scale sleep onset by PA volume	0.39	10.04	100.00%	0.00%
Scale sleep regularity by PA volume	-0.69	1.64	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep duration by PA intensity	-0.01	2.22	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.93	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.39	10.05	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.70	1.61	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	6.08	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	6.11	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	6.05	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.23	1.56	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.70	6.18	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.56	100.00%	0.00%
Models moderated by sex				
Scale sleep duration by PA volume	-0.01	2.23	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.93	3.64	100.00%	0.00%
Scale sleep onset by PA volume	0.39	10.07	100.00%	0.00%
Scale sleep regularity by PA volume	-0.69	1.64	100.00%	0.00%
Scale sleep duration by PA intensity	-0.01	2.22	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.93	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.39	10.08	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.70	1.61	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	6.07	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	6.10	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	6.02	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.70	6.16	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.56	100.00%	0.00%
Models moderated by ethnicity				
Scale sleep duration by PA volume	-0.01	2.24	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.93	3.64	100.00%	0.00%
Scale sleep onset by PA volume	0.39	10.07	100.00%	0.00%
Scale sleep regularity by PA volume	-0.69	1.63	100.00%	0.00%
Scale sleep duration by PA intensity	-0.01	2.22	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep efficiency by PA intensity	-0.93	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.39	10.08	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.70	1.60	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	6.08	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.57	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	6.10	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.56	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	6.04	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.23	1.56	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.71	6.17	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.57	100.00%	0.00%

Table 4

Physical activity predicting sleep controlling for SES, sex, BMI z-score, and the fixed effects of study IDs.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.66 [0.62, 0.71]	< .001	0.58 [0.53, 0.63]	< .001
Physical activity	0.08 [0.06, 0.09]	< .001	0.05 [0.04, 0.06]	< .001
Age	-0.02 [-0.03, -0.01]	.005	-0.02 [-0.03, -0.01]	.004
Physical activity^2	-0.01 [-0.02, 0.00]	.001	0.00 [-0.01, 0.00]	.212
Physical activity × Age	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, -0.01]	< .001
Age × Physical activity^2	0.00 [0.00, 0.00]	.390	0.00 [-0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	-0.53 [-0.58, -0.48]	< .001	-0.61 [-0.65, -0.56]	< .001
Physical activity	0.03 [0.02, 0.04]	< .001	0.02 [0.01, 0.03]	< .001
Age	-0.01 [-0.02, 0.01]	.317	-0.01 [-0.03, 0.00]	.061
Physical activity^2	-0.01 [-0.02, -0.01]	< .001	-0.02 [-0.02, -0.01]	< .001
Physical activity × Age	-0.02 [-0.02, -0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
Age × Physical activity^2	0.00 [0.00, 0.00]	.016	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	-1.08 [-1.12, -1.05]	< .001	-1.02 [-1.05, -0.98]	< .001
Physical activity	-0.09 [-0.10, -0.09]	< .001	-0.05 [-0.06, -0.05]	< .001
Age	-0.01 [-0.02, 0.00]	.068	-0.01 [-0.02, 0.01]	.357
Physical activity^2	0.01 [0.01, 0.02]	< .001	0.00 [-0.01, 0.00]	.171
Physical activity × Age	-0.02 [-0.02, -0.02]	< .001	0.00 [0.00, 0.00]	.069
Age × Physical activity^2	0.01 [0.00, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	-0.09 [-0.18, 0.00]	.049	-0.43 [-0.52, -0.34]	< .001
Physical activity	0.24 [0.22, 0.25]	< .001	0.12 [0.11, 0.13]	< .001

Table 4 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	0.01 [-0.02, 0.04]	.696	-0.05 [-0.08, -0.02]	$< .001$
Physical activity^2	-0.08 [-0.10, -0.07]	$< .001$	-0.07 [-0.08, -0.06]	$< .001$
Physical activity $\times$ Age	0.00 [-0.01, 0.00]	.193	-0.04 [-0.05, -0.04]	$< .001$
Age $\times$ Physical activity^2	-0.02 [-0.02, -0.01]	$< .001$	0.01 [0.01, 0.01]	$< .001$

Note. Adjusted for SES, sex, BMI z-score, and the fixed effects of study IDs. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 39 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 39 and age interactions are the change per 10 years.

Table 5

Sleep predicting physical activity controlling for SES, sex, BMI z-score, and the fixed effects of study IDs

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-1.15 [-1.20, -1.11]	< .001	-0.41 [-0.44, -0.37]	< .001
Sleep duration	0.01 [0.00, 0.01]	.003	0.03 [0.02, 0.03]	< .001
Age	-0.26 [-0.27, -0.24]	< .001	-0.28 [-0.29, -0.26]	< .001
Sleep duration^2	-0.02 [-0.03, -0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
Sleep duration × Age	-0.01 [-0.01, -0.01]	< .001	0.00 [0.00, 0.00]	.004
Age × Sleep duration^2	0.00 [0.00, 0.00]	.006	0.00 [0.00, 0.00]	.009
Sleep efficiency (z)				
(Intercept)	-1.13 [-1.18, -1.09]	< .001	-0.38 [-0.41, -0.34]	< .001
Sleep efficiency	0.01 [0.00, 0.01]	.005	0.01 [0.00, 0.01]	.036
Age	-0.25 [-0.27, -0.24]	< .001	-0.28 [-0.29, -0.26]	< .001
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.00]	.040	0.00 [0.00, 0.00]	.132
Age × Sleep efficiency^2	0.00 [0.00, 0.00]	.008	0.00 [0.00, 0.00]	.422
Sleep onset (z)				
(Intercept)	-1.26 [-1.30, -1.21]	< .001	-0.46 [-0.50, -0.42]	< .001
Sleep onset	-0.06 [-0.07, -0.06]	< .001	-0.02 [-0.03, -0.01]	< .001
Age	-0.26 [-0.27, -0.25]	< .001	-0.28 [-0.29, -0.27]	< .001
Sleep onset^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.00]	.138
Sleep onset × Age	0.02 [0.02, 0.02]	< .001	0.01 [0.01, 0.02]	< .001
Age × Sleep onset^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.972
Sleep regularity (z)				
(Intercept)	-1.09 [-1.13, -1.05]	< .001	-0.36 [-0.39, -0.32]	< .001
Sleep regularity	0.09 [0.09, 0.10]	< .001	0.05 [0.05, 0.06]	< .001

Table 5 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.25 [-0.26, -0.23]	$< .001$	-0.27 [-0.28, -0.26]	$< .001$
Sleep regularity^2	0.00 [0.00, 0.01]	.125	0.00 [0.00, 0.00]	.820
Sleep regularity $\times$ Age	-0.02 [-0.02, -0.02]	$< .001$	-0.02 [-0.02, -0.01]	$< .001$
Age $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.220	0.00 [0.00, 0.00]	.401

Note. Adjusted for SES, sex, BMI z-score, and the fixed effects of study IDs. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 39 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 39 and age interactions are the change per 10 years.

Table 6

Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.04 [-0.24, 0.32]	.779	-0.01 [-0.17, 0.14]	.862
Log pa volume	-0.17 [-0.30, -0.03]	.017	0.05 [0.04, 0.06]	< .001
Age	0.03 [-0.04, 0.11]	.396	-0.02 [-0.04, -0.01]	< .001
Log pa volume^2	0.04 [0.02, 0.06]	< .001	0.00 [-0.01, 0.00]	.202
Log pa volume × Age	-0.02 [-0.06, 0.02]	.371	-0.01 [-0.01, -0.01]	< .001
Age × Log pa volume^2	0.00 [-0.01, 0.01]	.744	0.00 [-0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	-1.05 [-1.31, -0.79]	< .001	0.15 [-0.03, 0.32]	.095
Log pa volume	0.64 [0.52, 0.76]	< .001	0.02 [0.01, 0.03]	< .001
Age	0.26 [0.19, 0.33]	< .001	-0.01 [-0.03, 0.00]	.136
Log pa volume^2	-0.08 [-0.10, -0.07]	< .001	-0.02 [-0.02, -0.01]	< .001
Log pa volume × Age	-0.12 [-0.16, -0.08]	< .001	-0.01 [-0.02, -0.01]	< .001
Age × Log pa volume^2	0.01 [0.01, 0.02]	< .001	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	-0.16 [-0.47, 0.16]	.334	0.08 [-0.18, 0.34]	.546
Log pa volume	0.30 [0.20, 0.39]	< .001	-0.05 [-0.06, -0.05]	< .001
Age	0.22 [0.17, 0.28]	< .001	0.00 [-0.01, 0.01]	.457
Log pa volume^2	-0.06 [-0.08, -0.05]	< .001	0.00 [-0.01, 0.00]	.174
Log pa volume × Age	-0.10 [-0.13, -0.07]	< .001	0.00 [0.00, 0.00]	.066
Age × Log pa volume^2	0.01 [0.01, 0.02]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	-3.88 [-4.23, -3.53]	< .001	0.08 [-0.08, 0.23]	.328
Log pa volume	1.92 [1.71, 2.13]	< .001	0.12 [0.11, 0.13]	< .001

Table 6 continued

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.33 [-0.44, -0.21]	$< .001$	-0.05 [-0.08, -0.02]	$< .001$
Log pa volume^2	-0.22 [-0.25, -0.19]	$< .001$	-0.07 [-0.08, -0.06]	$< .001$
Log pa volume $\times$ Age	0.20 [0.13, 0.27]	$< .001$	-0.04 [-0.05, -0.04]	$< .001$
Age $\times$ Log pa volume^2	-0.03 [-0.04, -0.02]	$< .001$	0.01 [0.01, 0.01]	$< .001$

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised and (ln) natural-log transformed; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 39 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 39 and age interactions are the change per 10 years.

Table 7

Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	3.61 [3.46, 3.75]	< .001	-0.07 [-0.20, 0.06]	.270
Sleep duration	0.00 [0.00, 0.00]	.629	0.03 [0.02, 0.03]	< .001
Age	-0.13 [-0.14, -0.12]	< .001	-0.28 [-0.29, -0.27]	< .001
Sleep duration^2	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.02, -0.01]	< .001
Sleep duration × Age	-0.01 [-0.01, -0.01]	< .001	0.00 [0.00, 0.00]	.004
Age × Sleep duration^2	0.00 [0.00, 0.00]	.495	0.00 [0.00, 0.00]	.009
Sleep efficiency (z)				
(Intercept)	3.60 [3.46, 3.74]	< .001	-0.08 [-0.20, 0.05]	.246
Sleep efficiency	0.01 [0.01, 0.01]	< .001	0.01 [0.00, 0.01]	.035
Age	-0.13 [-0.14, -0.12]	< .001	-0.28 [-0.29, -0.26]	< .001
Sleep efficiency^2	0.00 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.00]	.665	0.00 [0.00, 0.00]	.128
Age × Sleep efficiency^2	0.00 [0.00, 0.00]	.002	0.00 [0.00, 0.00]	.423
Sleep onset (z)				
(Intercept)	3.60 [3.45, 3.75]	< .001	-0.09 [-0.22, 0.04]	.190
Sleep onset	-0.02 [-0.03, -0.02]	< .001	-0.02 [-0.03, -0.01]	< .001
Age	-0.13 [-0.14, -0.12]	< .001	-0.28 [-0.29, -0.27]	< .001
Sleep onset^2	-0.01 [-0.01, -0.01]	< .001	0.00 [-0.01, 0.00]	.135
Sleep onset × Age	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.02]	< .001
Age × Sleep onset^2	0.00 [0.00, 0.00]	.008	0.00 [0.00, 0.00]	.963
Sleep regularity (z)				
(Intercept)	3.59 [3.45, 3.73]	< .001	-0.09 [-0.21, 0.03]	.140
Sleep regularity	0.06 [0.06, 0.06]	< .001	0.05 [0.05, 0.06]	< .001

Table 7 continued

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.13 [-0.13, -0.12]	$< .001$	-0.27 [-0.28, -0.26]	$< .001$
Sleep regularity^2	0.00 [-0.01, 0.00]	$< .001$	0.00 [0.00, 0.00]	.828
Sleep regularity $\times$ Age	-0.01 [-0.01, -0.01]	$< .001$	-0.02 [-0.02, -0.01]	$< .001$
Age $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.002	0.00 [0.00, 0.00]	.396

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised and (ln) natural-log transformed; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 39 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 39 and age interactions are the change per 10 years.

Table 8

Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-0.02 [-0.18, 0.15]	.846	-0.02 [-0.17, 0.13]	.817
Physical activity	0.08 [0.06, 0.09]	< .001	0.05 [0.04, 0.06]	< .001
Age	-0.02 [-0.04, -0.01]	< .001	-0.02 [-0.04, -0.01]	< .001
Physical activity^2	-0.01 [-0.02, 0.00]	.002	0.00 [-0.01, 0.00]	.164
Physical activity × Age	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, -0.01]	< .001
Age × Physical activity^2	0.00 [0.00, 0.00]	.483	0.00 [-0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	0.14 [-0.01, 0.30]	.071	0.14 [-0.03, 0.31]	.100
Physical activity	0.03 [0.02, 0.04]	< .001	0.02 [0.01, 0.03]	< .001
Age	0.00 [-0.02, 0.01]	.685	-0.01 [-0.02, 0.00]	.173
Physical activity^2	-0.01 [-0.02, -0.01]	< .001	-0.02 [-0.02, -0.01]	< .001
Physical activity × Age	-0.02 [-0.02, -0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
Age × Physical activity^2	0.00 [0.00, 0.00]	.015	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	0.06 [-0.21, 0.33]	.658	0.08 [-0.18, 0.34]	.549
Physical activity	-0.09 [-0.10, -0.09]	< .001	-0.05 [-0.06, -0.04]	< .001
Age	-0.01 [-0.02, 0.00]	.181	0.00 [-0.01, 0.01]	.623
Physical activity^2	0.01 [0.01, 0.02]	< .001	0.00 [-0.01, 0.00]	.140
Physical activity × Age	-0.02 [-0.02, -0.02]	< .001	0.00 [0.00, 0.00]	.075
Age × Physical activity^2	0.01 [0.00, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.15 [0.03, 0.27]	.016	0.08 [-0.07, 0.23]	.313
Physical activity	0.24 [0.22, 0.25]	< .001	0.12 [0.11, 0.13]	< .001

Table 8 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	0.01 [-0.02, 0.04]	.615	-0.05 [-0.08, -0.02]	$< .001$
Physical activity^2	-0.08 [-0.10, -0.07]	$< .001$	-0.07 [-0.08, -0.06]	$< .001$
Physical activity $\times$ Age	0.00 [-0.01, 0.00]	.232	-0.04 [-0.05, -0.04]	$< .001$
Age $\times$ Physical activity^2	-0.02 [-0.02, -0.01]	$< .001$	0.01 [0.01, 0.01]	$< .001$

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 39 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 39 and age interactions are the change per 10 years.

Table 9

Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.18 [-0.17, 0.52]	.319	-0.07 [-0.20, 0.05]	.265
Sleep duration	0.01 [0.00, 0.01]	.003	0.03 [0.02, 0.03]	< .001
Age	-0.26 [-0.27, -0.24]	< .001	-0.28 [-0.29, -0.27]	< .001
Sleep duration^2	-0.02 [-0.03, -0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
Sleep duration × Age	-0.01 [-0.01, -0.01]	< .001	0.00 [0.00, 0.00]	.004
Age × Sleep duration^2	0.00 [0.00, 0.00]	.006	0.00 [0.00, 0.00]	.011
Sleep efficiency (z)				
(Intercept)	0.17 [-0.18, 0.51]	.343	-0.08 [-0.20, 0.05]	.242
Sleep efficiency	0.01 [0.00, 0.01]	.005	0.01 [0.00, 0.01]	.036
Age	-0.25 [-0.27, -0.24]	< .001	-0.28 [-0.29, -0.26]	< .001
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.00]	.040	0.00 [0.00, 0.00]	.131
Age × Sleep efficiency^2	0.00 [0.00, 0.00]	.007	0.00 [0.00, 0.00]	.431
Sleep onset (z)				
(Intercept)	0.16 [-0.19, 0.52]	.366	-0.09 [-0.22, 0.04]	.186
Sleep onset	-0.06 [-0.07, -0.06]	< .001	-0.02 [-0.03, -0.01]	< .001
Age	-0.26 [-0.27, -0.25]	< .001	-0.28 [-0.29, -0.27]	< .001
Sleep onset^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.00]	.141
Sleep onset × Age	0.02 [0.02, 0.02]	< .001	0.01 [0.01, 0.02]	< .001
Age × Sleep onset^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.961
Sleep regularity (z)				
(Intercept)	0.15 [-0.19, 0.48]	.390	-0.09 [-0.21, 0.03]	.137
Sleep regularity	0.09 [0.09, 0.10]	< .001	0.05 [0.05, 0.06]	< .001

Table 9 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.25 [-0.26, -0.23]	$< .001$	-0.27 [-0.28, -0.26]	$< .001$
Sleep regularity^2	0.00 [0.00, 0.01]	.134	0.00 [0.00, 0.00]	.800
Sleep regularity $\times$ Age	-0.02 [-0.02, -0.02]	$< .001$	-0.02 [-0.02, -0.01]	$< .001$
Age $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.222	0.00 [0.00, 0.00]	.363

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 39 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 39 and age interactions are the change per 10 years.

Table 10

Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-0.05 [-0.31, 0.22]	.734	-0.05 [-0.30, 0.20]	.697
Physical activity	0.08 [0.07, 0.10]	< .001	0.05 [0.04, 0.07]	< .001
Age	0.03 [0.01, 0.05]	< .001	0.03 [0.01, 0.05]	< .001
Physical activity^2	-0.01 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.00]	.209
Physical activity × Age	0.00 [-0.01, 0.00]	.062	-0.01 [-0.01, 0.00]	.005
Age × Physical activity^2	0.00 [0.00, 0.00]	.364	0.00 [-0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	0.08 [-0.21, 0.38]	.575	0.07 [-0.24, 0.38]	.644
Physical activity	0.04 [0.03, 0.05]	< .001	0.03 [0.02, 0.05]	< .001
Age	-0.04 [-0.06, -0.02]	< .001	-0.05 [-0.07, -0.03]	< .001
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	-0.02 [-0.03, -0.01]	< .001
Physical activity × Age	-0.03 [-0.03, -0.02]	< .001	-0.02 [-0.02, -0.01]	< .001
Age × Physical activity^2	0.00 [0.00, 0.00]	.020	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	0.02 [-0.36, 0.40]	.915	0.03 [-0.34, 0.39]	.891
Physical activity	-0.08 [-0.09, -0.08]	< .001	-0.05 [-0.05, -0.04]	< .001
Age	-0.07 [-0.08, -0.05]	< .001	-0.06 [-0.07, -0.05]	< .001
Physical activity^2	0.01 [0.01, 0.01]	< .001	0.00 [-0.01, 0.00]	.097
Physical activity × Age	-0.02 [-0.03, -0.02]	< .001	0.00 [-0.01, 0.00]	.003
Age × Physical activity^2	0.01 [0.00, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.12 [0.00, 0.25]	.058	0.06 [-0.14, 0.25]	.571
Physical activity	0.22 [0.20, 0.24]	< .001	0.13 [0.12, 0.15]	< .001

Table 10 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	0.01 [-0.03, 0.05]	.549	-0.05 [-0.09, -0.01]	.018
Physical activity^2	-0.07 [-0.09, -0.06]	$<$ .001	-0.09 [-0.10, -0.08]	$<$ .001
Physical activity $\times$ Age	-0.01 [-0.01, 0.00]	.003	-0.05 [-0.05, -0.04]	$<$ .001
Age $\times$ Physical activity^2	-0.01 [-0.02, -0.01]	$<$ .001	0.02 [0.01, 0.02]	$<$ .001

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 39 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 39 and age interactions are the change per 10 years.

Table 11

Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.06 [-0.33, 0.46]	.752	-0.08 [-0.23, 0.06]	.255
Sleep duration	0.02 [0.01, 0.02]	< .001	0.04 [0.03, 0.04]	< .001
Age	-0.21 [-0.23, -0.20]	< .001	-0.24 [-0.25, -0.22]	< .001
Sleep duration^2	-0.03 [-0.03, -0.02]	< .001	-0.02 [-0.02, -0.01]	< .001
Sleep duration × Age	-0.01 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.00]	< .001
Age × Sleep duration^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	0.06 [-0.33, 0.44]	.771	-0.09 [-0.23, 0.06]	.243
Sleep efficiency	0.01 [0.00, 0.02]	< .001	0.01 [0.00, 0.01]	.030
Age	-0.21 [-0.22, -0.19]	< .001	-0.23 [-0.25, -0.22]	< .001
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.00]	.168	0.00 [0.00, 0.00]	.346
Age × Sleep efficiency^2	0.00 [0.00, 0.00]	.015	0.00 [0.00, 0.00]	.026
Sleep onset (z)				
(Intercept)	0.06 [-0.35, 0.47]	.781	-0.10 [-0.25, 0.05]	.189
Sleep onset	-0.06 [-0.07, -0.05]	< .001	-0.02 [-0.03, -0.01]	< .001
Age	-0.22 [-0.23, -0.20]	< .001	-0.24 [-0.25, -0.22]	< .001
Sleep onset^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.00]	.219
Sleep onset × Age	0.02 [0.01, 0.02]	< .001	0.01 [0.01, 0.01]	< .001
Age × Sleep onset^2	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.334
Sleep regularity (z)				
(Intercept)	0.04 [-0.34, 0.41]	.841	-0.10 [-0.23, 0.03]	.139
Sleep regularity	0.08 [0.07, 0.09]	< .001	0.05 [0.05, 0.06]	< .001

Table 11 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.20 [-0.22, -0.19]	$< .001$	-0.23 [-0.25, -0.22]	$< .001$
Sleep regularity^2	0.00 [0.00, 0.01]	.387	0.00 [0.00, 0.00]	.662
Sleep regularity $\times$ Age	-0.02 [-0.02, -0.01]	$< .001$	-0.02 [-0.02, -0.02]	$< .001$
Age $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.544	0.00 [0.00, 0.00]	.378

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 39 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 39 and age interactions are the change per 10 years.

Table 12

Physical activity predicting sleep controlling for SES, age, and sex.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.00 [-0.17, 0.16]	.953	0.00 [-0.16, 0.15]	.986
Physical activity	0.08 [0.07, 0.09]	< .001	0.06 [0.05, 0.07]	< .001
BMI z-score	-0.01 [-0.02, 0.00]	.007	-0.02 [-0.03, -0.01]	< .001
Physical activity^2	-0.01 [-0.01, 0.00]	< .001	0.00 [0.00, 0.01]	.067
Physical activity × BMI z-score	0.01 [0.00, 0.01]	.112	0.01 [0.00, 0.02]	< .001
BMI z-score × Physical activity^2	0.00 [0.00, 0.00]	.875	0.01 [0.00, 0.01]	.007
Sleep efficiency (z)				
(Intercept)	0.17 [0.01, 0.33]	.033	0.16 [0.00, 0.33]	.049
Physical activity	0.05 [0.04, 0.06]	< .001	0.01 [0.00, 0.01]	.059
BMI z-score	-0.02 [-0.03, -0.01]	< .001	-0.03 [-0.04, -0.02]	< .001
Physical activity^2	-0.01 [-0.01, -0.01]	< .001	0.00 [0.00, 0.00]	.759
Physical activity × BMI z-score	0.02 [0.01, 0.03]	< .001	0.01 [0.00, 0.01]	.002
BMI z-score × Physical activity^2	0.00 [-0.01, 0.00]	.027	0.00 [0.00, 0.00]	.899
Sleep onset (z)				
(Intercept)	0.09 [-0.18, 0.36]	.519	0.08 [-0.18, 0.34]	.548
Physical activity	-0.08 [-0.09, -0.07]	< .001	-0.07 [-0.08, -0.07]	< .001
BMI z-score	0.00 [-0.01, 0.01]	.876	0.00 [-0.01, 0.01]	.800
Physical activity^2	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.479
Physical activity × BMI z-score	0.00 [0.00, 0.01]	.217	-0.01 [-0.01, 0.00]	.009
BMI z-score × Physical activity^2	0.00 [0.00, 0.00]	.025	0.00 [-0.01, 0.00]	.053
Sleep regularity (z)				
(Intercept)	0.13 [0.00, 0.25]	.044	0.14 [0.00, 0.29]	.055
Physical activity	0.23 [0.22, 0.25]	< .001	0.09 [0.08, 0.10]	< .001

Table 12 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
BMI z-score	-0.05 [-0.06, -0.04]	$< .001$	-0.07 [-0.08, -0.05]	$< .001$
Physical activity^2	-0.04 [-0.04, -0.04]	$< .001$	-0.02 [-0.02, -0.01]	$< .001$
Physical activity $\times$ BMI z-score	0.05 [0.04, 0.07]	$< .001$	0.03 [0.02, 0.03]	$< .001$
BMI z-score $\times$ Physical activity^2	-0.01 [-0.01, 0.00]	$< .001$	-0.01 [-0.01, 0.00]	.002

Note. Adjusted for SES, age, and sex. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 13

Sleep predicting physical activity controlling for SES, age, and sex

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.18 [-0.16, 0.53]	.300	-0.07 [-0.20, 0.06]	.277
Sleep duration	0.00 [0.00, 0.01]	.360	0.02 [0.02, 0.03]	< .001
BMI z-score	-0.07 [-0.08, -0.06]	< .001	-0.07 [-0.07, -0.06]	< .001
Sleep duration^2	-0.02 [-0.02, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep duration × BMI z-score	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.01]	.350
BMI z-score × Sleep duration^2	0.00 [0.00, 0.00]	.426	0.00 [0.00, 0.00]	.567
Sleep efficiency (z)				
(Intercept)	0.17 [-0.17, 0.51]	.331	-0.08 [-0.20, 0.05]	.250
Sleep efficiency	0.01 [0.00, 0.01]	.001	0.01 [0.00, 0.01]	.035
BMI z-score	-0.07 [-0.08, -0.06]	< .001	-0.07 [-0.07, -0.06]	< .001
Sleep efficiency^2	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep efficiency × BMI z-score	0.00 [-0.01, 0.00]	.075	0.00 [-0.01, 0.00]	.941
BMI z-score × Sleep efficiency^2	0.00 [0.00, 0.00]	.926	0.00 [0.00, 0.00]	.841
Sleep onset (z)				
(Intercept)	0.17 [-0.18, 0.53]	.331	-0.08 [-0.21, 0.05]	.219
Sleep onset	-0.05 [-0.06, -0.05]	< .001	-0.02 [-0.03, -0.01]	< .001
BMI z-score	-0.07 [-0.08, -0.06]	< .001	-0.07 [-0.07, -0.06]	< .001
Sleep onset^2	-0.01 [-0.01, 0.00]	.003	0.00 [0.00, 0.01]	.009
Sleep onset × BMI z-score	-0.01 [-0.02, -0.01]	< .001	-0.01 [-0.01, 0.00]	.002
BMI z-score × Sleep onset^2	0.00 [0.00, 0.01]	.016	0.00 [0.00, 0.00]	.191
Sleep regularity (z)				
(Intercept)	0.15 [-0.18, 0.48]	.379	-0.09 [-0.21, 0.04]	.167
Sleep regularity	0.10 [0.09, 0.10]	< .001	0.05 [0.05, 0.06]	< .001

Table 13 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
BMI z-score	-0.06 [-0.07, -0.05]	$< .001$	-0.06 [-0.07, -0.05]	$< .001$
Sleep regularity^2	0.00 [0.00, 0.01]	.253	0.00 [0.00, 0.00]	.381
Sleep regularity $\times$ BMI z-score	0.03 [0.02, 0.03]	$< .001$	0.02 [0.01, 0.02]	$< .001$
BMI z-score $\times$ Sleep regularity^2	0.00 [0.00, 0.01]	.119	0.00 [0.00, 0.00]	.837

Note. Adjusted for SES, age, and sex. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 14

Physical activity predicting sleep controlling for age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.00 [-0.17, 0.16]	.965	0.00 [-0.16, 0.15]	.986
Physical activity	0.07 [0.06, 0.09]	< .001	0.06 [0.04, 0.07]	< .001
Medium	0.01 [-0.01, 0.04]	.269	0.02 [-0.01, 0.04]	.169
High	0.01 [-0.01, 0.04]	.318	0.00 [-0.03, 0.02]	.888
Physical activity^2	-0.01 [-0.01, 0.00]	.037	0.01 [0.00, 0.01]	.156
Physical activity × Medium	0.00 [-0.02, 0.02]	.848	0.00 [-0.01, 0.02]	.804
Physical activity × High	0.01 [-0.01, 0.04]	.152	0.00 [-0.01, 0.02]	.703
Medium × Physical activity^2	0.00 [-0.01, 0.01]	.668	-0.01 [-0.02, 0.00]	.163
High × Physical activity^2	0.00 [-0.01, 0.00]	.511	0.00 [-0.01, 0.01]	.474
Sleep efficiency (z)				
(Intercept)	0.18 [0.02, 0.33]	.030	0.16 [0.00, 0.32]	.056
Physical activity	0.04 [0.02, 0.05]	< .001	0.00 [-0.02, 0.01]	.572
Medium	0.02 [0.00, 0.04]	.116	0.03 [0.00, 0.05]	.039
High	0.02 [0.00, 0.05]	.039	0.02 [0.00, 0.05]	.067
Physical activity^2	-0.01 [-0.02, -0.01]	< .001	0.00 [0.00, 0.01]	.355
Physical activity × Medium	0.00 [-0.02, 0.02]	.942	0.01 [-0.01, 0.02]	.219
Physical activity × High	0.02 [0.00, 0.05]	.019	0.02 [0.01, 0.04]	.009
Medium × Physical activity^2	0.00 [0.00, 0.01]	.407	-0.01 [-0.02, 0.00]	.180
High × Physical activity^2	0.00 [-0.01, 0.00]	.347	-0.01 [-0.02, 0.00]	.166
Sleep onset (z)				
(Intercept)	0.09 [-0.18, 0.36]	.529	0.08 [-0.18, 0.34]	.560
Physical activity	-0.10 [-0.11, -0.09]	< .001	-0.08 [-0.09, -0.07]	< .001
Medium	0.02 [0.00, 0.04]	.031	0.02 [0.00, 0.04]	.036

Table 14 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
High	0.01 [0.00, 0.03]	.085	0.02 [0.00, 0.04]	.018
Physical activity^2	0.01 [0.00, 0.01]	$<$ .001	0.00 [-0.01, 0.01]	.954
Physical activity $\times$ Medium	0.02 [0.00, 0.03]	.013	0.00 [-0.01, 0.01]	.788
Physical activity $\times$ High	0.04 [0.02, 0.05]	$<$ .001	0.02 [0.01, 0.03]	.001
Medium $\times$ Physical activity^2	0.00 [-0.01, 0.00]	.181	0.00 [-0.01, 0.01]	.541
High $\times$ Physical activity^2	-0.01 [-0.01, 0.00]	.010	-0.01 [-0.02, 0.00]	.037
Sleep regularity (z)				
(Intercept)	0.13 [0.01, 0.25]	.037	0.14 [0.00, 0.29]	.057
Physical activity	0.21 [0.19, 0.24]	$<$ .001	0.08 [0.06, 0.09]	$<$ .001
Medium	0.02 [0.00, 0.05]	.097	0.00 [-0.03, 0.03]	.970
High	0.03 [0.01, 0.06]	.012	0.02 [-0.01, 0.04]	.211
Physical activity^2	-0.03 [-0.04, -0.03]	$<$ .001	-0.02 [-0.02, -0.01]	$<$ .001
Physical activity $\times$ Medium	0.03 [0.01, 0.06]	.013	0.00 [-0.02, 0.02]	.776
Physical activity $\times$ High	0.02 [-0.01, 0.04]	.130	0.03 [0.01, 0.04]	.008
Medium $\times$ Physical activity^2	-0.01 [-0.02, 0.00]	.011	0.00 [-0.01, 0.01]	.876
High $\times$ Physical activity^2	0.00 [-0.01, 0.00]	.206	-0.01 [-0.02, 0.00]	.064

Note. Adjusted for age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 15

Sleep predicting physical activity controlling for age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.19 [-0.16, 0.53]	.292	-0.07 [-0.20, 0.06]	.276
Sleep duration	-0.01 [-0.02, 0.00]	.002	0.02 [0.01, 0.03]	< .001
Medium	-0.07 [-0.09, -0.04]	< .001	0.01 [-0.01, 0.03]	.466
High	-0.10 [-0.13, -0.08]	< .001	0.02 [0.00, 0.04]	.058
Sleep duration^2	-0.02 [-0.02, -0.01]	< .001	-0.01 [-0.02, -0.01]	< .001
Sleep duration × Medium	0.01 [0.00, 0.02]	.043	0.00 [-0.01, 0.01]	.899
Sleep duration × High	0.03 [0.02, 0.05]	< .001	0.01 [-0.01, 0.02]	.288
Medium × Sleep duration^2	0.01 [0.00, 0.01]	.089	0.00 [0.00, 0.01]	.383
High × Sleep duration^2	0.00 [-0.01, 0.01]	.552	0.00 [-0.01, 0.00]	.319
Sleep efficiency (z)				
(Intercept)	0.17 [-0.18, 0.51]	.340	-0.08 [-0.21, 0.05]	.230
Sleep efficiency	0.01 [0.00, 0.02]	.009	0.00 [-0.01, 0.01]	.530
Medium	-0.06 [-0.08, -0.04]	< .001	0.01 [-0.01, 0.03]	.218
High	-0.09 [-0.11, -0.07]	< .001	0.02 [0.01, 0.04]	.013
Sleep efficiency^2	-0.01 [-0.01, 0.00]	.016	-0.01 [-0.01, 0.00]	.017
Sleep efficiency × Medium	0.00 [-0.02, 0.01]	.659	0.01 [-0.01, 0.02]	.290
Sleep efficiency × High	-0.01 [-0.02, 0.00]	.197	0.00 [-0.01, 0.01]	.987
Medium × Sleep efficiency^2	0.00 [-0.01, 0.01]	.829	0.00 [-0.01, 0.01]	.718
High × Sleep efficiency^2	-0.01 [-0.01, 0.00]	.070	-0.01 [-0.01, 0.00]	.017
Sleep onset (z)				
(Intercept)	0.18 [-0.17, 0.53]	.315	-0.08 [-0.21, 0.05]	.235
Sleep onset	-0.05 [-0.07, -0.04]	< .001	-0.01 [-0.02, 0.00]	.038
Medium	-0.07 [-0.10, -0.05]	< .001	0.00 [-0.02, 0.02]	.652

Table 15 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
High	-0.11 [-0.13, -0.08]	< .001	0.01 [-0.01, 0.03]	.587
Sleep onset^2	-0.01 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.01]	.910
Sleep onset × Medium	0.00 [-0.01, 0.02]	.609	-0.01 [-0.02, 0.01]	.341
Sleep onset × High	0.00 [-0.02, 0.01]	.666	-0.02 [-0.03, -0.01]	.007
Medium × Sleep onset^2	0.01 [0.00, 0.02]	.014	0.00 [0.00, 0.01]	.269
High × Sleep onset^2	0.01 [0.00, 0.02]	.013	0.01 [0.00, 0.02]	.098
Sleep regularity (z)				
(Intercept)	0.15 [-0.19, 0.48]	.385	-0.09 [-0.21, 0.03]	.156
Sleep regularity	0.07 [0.06, 0.08]	< .001	0.04 [0.03, 0.05]	< .001
Medium	-0.06 [-0.08, -0.03]	< .001	0.01 [-0.01, 0.03]	.157
High	-0.09 [-0.11, -0.06]	< .001	0.02 [0.00, 0.04]	.015
Sleep regularity^2	0.00 [0.00, 0.01]	.510	0.00 [0.00, 0.01]	.711
Sleep regularity × Medium	0.02 [0.01, 0.04]	< .001	0.00 [-0.01, 0.01]	.677
Sleep regularity × High	0.03 [0.02, 0.04]	< .001	0.02 [0.01, 0.04]	< .001
Medium × Sleep regularity^2	0.00 [-0.01, 0.01]	.808	0.00 [-0.01, 0.00]	.522
High × Sleep regularity^2	0.00 [-0.01, 0.00]	.284	-0.01 [-0.02, 0.00]	.050

Note. Adjusted for age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 16

Physical activity predicting sleep controlling for SES, age, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.00 [-0.17, 0.16]	.989	0.00 [-0.16, 0.15]	.985
Physical activity	0.09 [0.08, 0.10]	< .001	0.06 [0.05, 0.07]	< .001
Male	-0.13 [-0.14, -0.11]	< .001	-0.12 [-0.14, -0.10]	< .001
Physical activity^2	-0.01 [-0.01, 0.00]	< .001	0.01 [0.00, 0.01]	.087
Physical activity × Male	-0.02 [-0.03, 0.00]	.026	0.00 [-0.01, 0.02]	.405
Male × Physical activity^2	0.00 [0.00, 0.01]	.118	0.00 [-0.01, 0.01]	.586
Sleep efficiency (z)				
(Intercept)	0.18 [0.03, 0.34]	.023	0.16 [0.00, 0.33]	.050
Physical activity	0.06 [0.04, 0.07]	< .001	-0.01 [-0.02, 0.00]	.084
Male	-0.18 [-0.20, -0.16]	< .001	-0.17 [-0.19, -0.15]	< .001
Physical activity^2	-0.02 [-0.02, -0.02]	< .001	0.00 [-0.01, 0.00]	.758
Physical activity × Male	-0.02 [-0.03, 0.00]	.051	0.03 [0.02, 0.04]	< .001
Male × Physical activity^2	0.01 [0.01, 0.02]	< .001	0.00 [-0.01, 0.01]	.838
Sleep onset (z)				
(Intercept)	0.09 [-0.18, 0.36]	.515	0.08 [-0.18, 0.34]	.569
Physical activity	-0.08 [-0.09, -0.07]	< .001	-0.08 [-0.09, -0.07]	< .001
Male	0.01 [0.00, 0.02]	.163	0.02 [0.01, 0.03]	.006
Physical activity^2	0.00 [0.00, 0.01]	.045	0.00 [0.00, 0.01]	.505
Physical activity × Male	0.01 [0.00, 0.02]	.072	0.02 [0.01, 0.02]	< .001
Male × Physical activity^2	0.00 [0.00, 0.00]	.926	-0.01 [-0.01, 0.00]	.055
Sleep regularity (z)				
(Intercept)	0.15 [0.02, 0.27]	.020	0.15 [0.00, 0.30]	.046
Physical activity	0.24 [0.22, 0.26]	< .001	0.07 [0.06, 0.08]	< .001

Table 16 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Male	-0.14 [-0.16, -0.12]	$< .001$	-0.13 [-0.15, -0.10]	$< .001$
Physical activity^2	-0.05 [-0.06, -0.05]	$< .001$	-0.02 [-0.03, -0.02]	$< .001$
Physical activity $\times$ Male	-0.02 [-0.03, 0.00]	.085	0.02 [0.01, 0.04]	$< .001$
Male $\times$ Physical activity^2	0.02 [0.01, 0.02]	$< .001$	0.01 [0.00, 0.02]	.160

Note. Adjusted for SES, age, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 17

Sleep predicting physical activity controlling for SES, age, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.18 [-0.16, 0.53]	.301	-0.07 [-0.20, 0.06]	.273
Sleep duration	-0.01 [-0.02, 0.00]	< .001	0.02 [0.02, 0.03]	< .001
Male	0.16 [0.14, 0.17]	< .001	0.23 [0.22, 0.24]	< .001
Sleep duration^2	-0.02 [-0.02, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep duration × Male	0.02 [0.02, 0.03]	< .001	0.00 [-0.01, 0.01]	.640
Male × Sleep duration^2	0.00 [0.00, 0.01]	.244	0.00 [-0.01, 0.00]	.883
Sleep efficiency (z)				
(Intercept)	0.17 [-0.18, 0.51]	.340	-0.07 [-0.20, 0.05]	.252
Sleep efficiency	0.02 [0.01, 0.03]	< .001	0.01 [0.00, 0.02]	.004
Male	0.16 [0.15, 0.18]	< .001	0.23 [0.21, 0.24]	< .001
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep efficiency × Male	-0.02 [-0.03, -0.01]	< .001	-0.01 [-0.02, 0.00]	.063
Male × Sleep efficiency^2	0.00 [-0.01, 0.00]	.254	0.00 [0.00, 0.01]	.509
Sleep onset (z)				
(Intercept)	0.18 [-0.18, 0.53]	.326	-0.08 [-0.21, 0.05]	.217
Sleep onset	-0.01 [-0.02, 0.00]	.007	-0.02 [-0.02, -0.01]	< .001
Male	0.14 [0.13, 0.16]	< .001	0.22 [0.21, 0.24]	< .001
Sleep onset^2	-0.01 [-0.01, 0.00]	< .001	0.00 [0.00, 0.01]	.199
Sleep onset × Male	-0.09 [-0.10, -0.07]	< .001	-0.01 [-0.02, 0.00]	.096
Male × Sleep onset^2	0.01 [0.00, 0.02]	.003	0.00 [0.00, 0.01]	.276
Sleep regularity (z)				
(Intercept)	0.15 [-0.18, 0.49]	.367	-0.09 [-0.21, 0.04]	.180
Sleep regularity	0.08 [0.07, 0.08]	< .001	0.05 [0.04, 0.06]	< .001

Table 17 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Male	0.16 [0.15, 0.18]	$< .001$	0.23 [0.22, 0.25]	$< .001$
Sleep regularity^2	0.00 [-0.01, 0.00]	.530	0.00 [-0.01, 0.00]	.193
Sleep regularity $\times$ Male	0.03 [0.02, 0.04]	$< .001$	0.00 [-0.01, 0.01]	.433
Male $\times$ Sleep regularity^2	0.00 [0.00, 0.01]	.118	0.00 [0.00, 0.01]	.673

Note. Adjusted for SES, age, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 18

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.00 [-0.16, 0.17]	.995	0.01 [-0.15, 0.16]	.941
Physical activity	0.08 [0.06, 0.10]	< .001	0.09 [0.08, 0.10]	< .001
Monday	-0.03 [-0.05, -0.01]	.002	-0.03 [-0.05, -0.01]	.009
Saturday	0.07 [0.06, 0.09]	< .001	0.07 [0.05, 0.09]	< .001
Sunday	0.05 [0.03, 0.07]	< .001	0.04 [0.02, 0.06]	< .001
Thursday	-0.03 [-0.05, -0.02]	< .001	-0.04 [-0.06, -0.02]	< .001
Tuesday	-0.05 [-0.07, -0.02]	< .001	-0.04 [-0.07, -0.02]	< .001
Wednesday	-0.06 [-0.08, -0.04]	< .001	-0.06 [-0.09, -0.04]	< .001
Physical activity^2	-0.01 [-0.01, 0.00]	.001	0.00 [-0.01, 0.01]	.534
Physical activity × Monday	0.00 [-0.02, 0.02]	.780	-0.02 [-0.04, -0.01]	.007
Physical activity × Saturday	0.04 [0.01, 0.06]	< .001	-0.06 [-0.08, -0.05]	< .001
Physical activity × Sunday	0.03 [0.01, 0.05]	.006	-0.07 [-0.09, -0.06]	< .001
Physical activity × Thursday	-0.02 [-0.04, 0.00]	.085	-0.01 [-0.02, 0.01]	.244
Physical activity × Tuesday	-0.01 [-0.03, 0.01]	.408	-0.01 [-0.02, 0.01]	.304
Physical activity × Wednesday	-0.02 [-0.04, 0.00]	.097	-0.01 [-0.02, 0.01]	.483
Monday × Physical activity^2	0.01 [0.00, 0.01]	.091	0.00 [-0.01, 0.01]	.917
Saturday × Physical activity^2	-0.01 [-0.02, 0.00]	.001	-0.01 [-0.02, 0.00]	.031
Sunday × Physical activity^2	-0.01 [-0.01, 0.00]	.111	-0.01 [-0.02, 0.00]	.233
Thursday × Physical activity^2	0.01 [0.00, 0.02]	.004	0.01 [0.00, 0.02]	.089
Tuesday × Physical activity^2	0.01 [0.00, 0.02]	.022	0.00 [-0.01, 0.02]	.559
Wednesday × Physical activity^2	0.01 [0.00, 0.02]	.001	0.01 [0.00, 0.02]	.058
Sleep efficiency (z)				
(Intercept)	0.19 [0.03, 0.35]	.018	0.19 [0.03, 0.36]	.022

Table 18 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity	0.04 [0.02, 0.05]	< .001	0.00 [-0.02, 0.01]	.618
Monday	-0.01 [-0.03, 0.01]	.191	-0.02 [-0.04, 0.00]	.104
Saturday	-0.07 [-0.08, -0.05]	< .001	-0.09 [-0.11, -0.07]	< .001
Sunday	-0.07 [-0.08, -0.05]	< .001	-0.10 [-0.12, -0.08]	< .001
Thursday	0.01 [0.00, 0.03]	.135	0.01 [-0.01, 0.03]	.225
Tuesday	0.01 [-0.01, 0.03]	.396	0.02 [-0.01, 0.04]	.211
Wednesday	0.02 [0.00, 0.04]	.065	0.02 [0.00, 0.05]	.090
Physical activity^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	.173
Physical activity × Monday	-0.01 [-0.03, 0.01]	.267	0.01 [-0.01, 0.02]	.273
Physical activity × Saturday	0.04 [0.02, 0.06]	< .001	-0.02 [-0.04, -0.01]	< .001
Physical activity × Sunday	0.03 [0.01, 0.05]	.010	-0.01 [-0.03, 0.00]	.137
Physical activity × Thursday	-0.02 [-0.04, 0.00]	.127	0.01 [0.00, 0.03]	.062
Physical activity × Tuesday	-0.03 [-0.05, -0.01]	.009	0.02 [0.00, 0.04]	.017
Physical activity × Wednesday	-0.01 [-0.03, 0.01]	.289	0.02 [0.00, 0.03]	.078
Monday × Physical activity^2	0.00 [-0.01, 0.00]	.408	0.00 [-0.01, 0.01]	.776
Saturday × Physical activity^2	-0.01 [-0.02, 0.00]	.020	0.02 [0.00, 0.03]	.007
Sunday × Physical activity^2	-0.01 [-0.01, 0.00]	.139	0.02 [0.00, 0.03]	.008
Thursday × Physical activity^2	0.00 [-0.01, 0.01]	.912	0.00 [-0.01, 0.01]	.696
Tuesday × Physical activity^2	0.00 [0.00, 0.01]	.360	-0.01 [-0.02, 0.01]	.285
Wednesday × Physical activity^2	0.00 [-0.01, 0.01]	.712	-0.01 [-0.02, 0.01]	.392
Sleep onset (z)				
(Intercept)	0.10 [-0.17, 0.37]	.463	0.09 [-0.17, 0.35]	.495
Physical activity	-0.05 [-0.06, -0.04]	< .001	-0.05 [-0.06, -0.04]	< .001
Monday	-0.08 [-0.10, -0.07]	< .001	-0.08 [-0.09, -0.06]	< .001

Table 18 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Saturday	0.09 [0.08, 0.10]	< .001	0.08 [0.06, 0.09]	< .001
Sunday	0.08 [0.07, 0.09]	< .001	0.06 [0.04, 0.07]	< .001
Thursday	-0.07 [-0.08, -0.05]	< .001	-0.06 [-0.08, -0.05]	< .001
Tuesday	-0.07 [-0.08, -0.05]	< .001	-0.06 [-0.08, -0.05]	< .001
Wednesday	-0.06 [-0.07, -0.04]	< .001	-0.04 [-0.06, -0.03]	< .001
Physical activity^2	0.00 [0.00, 0.01]	.062	0.00 [-0.01, 0.00]	.415
Physical activity × Monday	-0.07 [-0.08, -0.05]	< .001	-0.03 [-0.04, -0.02]	< .001
Physical activity × Saturday	0.04 [0.02, 0.05]	< .001	0.01 [0.00, 0.02]	.134
Physical activity × Sunday	0.01 [-0.01, 0.02]	.283	-0.03 [-0.05, -0.02]	< .001
Physical activity × Thursday	-0.06 [-0.08, -0.05]	< .001	-0.03 [-0.04, -0.02]	< .001
Physical activity × Tuesday	-0.07 [-0.09, -0.05]	< .001	-0.01 [-0.03, 0.00]	.011
Physical activity × Wednesday	-0.06 [-0.08, -0.04]	< .001	-0.02 [-0.03, 0.00]	.006
Monday × Physical activity^2	0.00 [0.00, 0.01]	.064	0.00 [0.00, 0.01]	.257
Saturday × Physical activity^2	0.00 [-0.01, 0.00]	.141	0.01 [0.00, 0.02]	.080
Sunday × Physical activity^2	-0.01 [-0.02, -0.01]	< .001	0.01 [0.00, 0.02]	.071
Thursday × Physical activity^2	0.01 [0.00, 0.01]	.008	0.00 [-0.01, 0.01]	.528
Tuesday × Physical activity^2	0.01 [0.00, 0.01]	.005	0.01 [0.00, 0.01]	.211
Wednesday × Physical activity^2	0.00 [0.00, 0.01]	.098	-0.01 [-0.01, 0.00]	.252
Sleep regularity (z)				
(Intercept)	0.09 [-0.04, 0.21]	.159	0.11 [-0.04, 0.26]	.149
Physical activity	0.19 [0.17, 0.22]	< .001	0.01 [-0.01, 0.03]	.223
Monday	0.04 [0.02, 0.06]	< .001	0.03 [0.01, 0.05]	.014
Saturday	-0.12 [-0.14, -0.09]	< .001	-0.13 [-0.16, -0.10]	< .001
Sunday	-0.13 [-0.16, -0.10]	< .001	-0.18 [-0.21, -0.15]	< .001

Table 18 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Thursday	0.18 [0.16, 0.20]	< .001	0.20 [0.17, 0.22]	< .001
Tuesday	0.21 [0.19, 0.23]	< .001	0.22 [0.20, 0.25]	< .001
Wednesday	0.20 [0.18, 0.22]	< .001	0.23 [0.20, 0.25]	< .001
Physical activity^2	-0.03 [-0.04, -0.03]	< .001	-0.01 [-0.02, 0.00]	.143
Physical activity × Monday	0.05 [0.03, 0.08]	< .001	0.11 [0.09, 0.12]	< .001
Physical activity × Saturday	0.01 [-0.01, 0.04]	.293	-0.04 [-0.05, -0.02]	< .001
Physical activity × Sunday	0.06 [0.03, 0.08]	< .001	-0.01 [-0.03, 0.00]	.101
Physical activity × Thursday	0.00 [-0.02, 0.03]	.773	0.09 [0.07, 0.11]	< .001
Physical activity × Tuesday	-0.02 [-0.04, 0.01]	.221	0.13 [0.11, 0.15]	< .001
Physical activity × Wednesday	-0.02 [-0.05, 0.01]	.181	0.09 [0.07, 0.11]	< .001
Monday × Physical activity^2	-0.01 [-0.02, -0.01]	.001	-0.03 [-0.04, -0.02]	< .001
Saturday × Physical activity^2	-0.01 [-0.02, 0.00]	.153	-0.01 [-0.03, 0.00]	.168
Sunday × Physical activity^2	-0.01 [-0.02, -0.01]	< .001	-0.01 [-0.02, 0.01]	.417
Thursday × Physical activity^2	0.00 [-0.01, 0.01]	.903	-0.02 [-0.04, -0.01]	.001
Tuesday × Physical activity^2	0.01 [0.00, 0.01]	.119	-0.02 [-0.04, -0.01]	.004
Wednesday × Physical activity^2	0.00 [-0.01, 0.01]	.856	-0.04 [-0.06, -0.03]	< .001

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 19

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.28 [-0.06, 0.63]	.107	-0.01 [-0.14, 0.12]	.860
Sleep duration	0.04 [0.03, 0.04]	< .001	0.07 [0.06, 0.08]	< .001
Monday	-0.20 [-0.22, -0.19]	< .001	-0.10 [-0.12, -0.09]	< .001
Saturday	-0.11 [-0.13, -0.10]	< .001	-0.10 [-0.11, -0.08]	< .001
Sunday	-0.20 [-0.22, -0.19]	< .001	-0.17 [-0.19, -0.16]	< .001
Thursday	-0.05 [-0.06, -0.03]	< .001	0.01 [-0.01, 0.02]	.504
Tuesday	-0.06 [-0.08, -0.05]	< .001	-0.02 [-0.04, -0.01]	.002
Wednesday	-0.06 [-0.08, -0.04]	< .001	0.00 [-0.01, 0.02]	.797
Sleep duration^2	-0.02 [-0.02, -0.01]	< .001	-0.01 [-0.02, 0.00]	.017
Sleep duration × Monday	-0.03 [-0.04, -0.02]	< .001	-0.06 [-0.07, -0.05]	< .001
Sleep duration × Saturday	-0.05 [-0.06, -0.03]	< .001	-0.06 [-0.07, -0.05]	< .001
Sleep duration × Sunday	-0.06 [-0.07, -0.04]	< .001	-0.08 [-0.09, -0.07]	< .001
Sleep duration × Thursday	-0.01 [-0.03, 0.00]	.043	0.00 [-0.01, 0.01]	.921
Sleep duration × Tuesday	-0.02 [-0.03, 0.00]	.018	-0.02 [-0.03, 0.00]	.007
Sleep duration × Wednesday	-0.02 [-0.03, 0.00]	.008	-0.02 [-0.03, 0.00]	.018
Monday × Sleep duration^2	0.01 [0.00, 0.01]	.242	0.00 [-0.01, 0.01]	.523
Saturday × Sleep duration^2	0.00 [-0.01, 0.01]	.599	0.00 [-0.01, 0.01]	.719
Sunday × Sleep duration^2	0.00 [-0.01, 0.01]	.750	0.00 [-0.01, 0.01]	.882
Thursday × Sleep duration^2	0.00 [-0.01, 0.01]	.733	0.00 [-0.01, 0.01]	.953
Tuesday × Sleep duration^2	0.00 [-0.01, 0.01]	.423	0.00 [-0.01, 0.01]	.524
Wednesday × Sleep duration^2	0.00 [-0.01, 0.01]	.434	0.00 [-0.01, 0.01]	.731
Sleep efficiency (z)				
(Intercept)	0.28 [-0.07, 0.62]	.115	-0.01 [-0.14, 0.12]	.861

Table 19 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency	-0.01 [-0.02, 0.00]	.021	-0.03 [-0.04, -0.02]	< .001
Monday	-0.21 [-0.22, -0.19]	< .001	-0.10 [-0.12, -0.09]	< .001
Saturday	-0.13 [-0.14, -0.11]	< .001	-0.11 [-0.12, -0.09]	< .001
Sunday	-0.21 [-0.22, -0.20]	< .001	-0.18 [-0.19, -0.17]	< .001
Thursday	-0.05 [-0.07, -0.04]	< .001	0.00 [-0.02, 0.01]	.973
Tuesday	-0.06 [-0.07, -0.05]	< .001	-0.02 [-0.03, 0.00]	.011
Wednesday	-0.06 [-0.07, -0.04]	< .001	0.00 [-0.02, 0.01]	.564
Sleep efficiency^2	-0.01 [-0.02, -0.01]	< .001	-0.01 [-0.02, -0.01]	< .001
Sleep efficiency × Monday	0.02 [0.00, 0.03]	.014	0.05 [0.04, 0.07]	< .001
Sleep efficiency × Saturday	0.02 [0.00, 0.03]	.015	0.04 [0.02, 0.05]	< .001
Sleep efficiency × Sunday	0.05 [0.03, 0.06]	< .001	0.09 [0.07, 0.10]	< .001
Sleep efficiency × Thursday	0.02 [0.01, 0.04]	.007	0.02 [0.00, 0.03]	.042
Sleep efficiency × Tuesday	0.01 [0.00, 0.03]	.088	0.01 [0.00, 0.03]	.046
Sleep efficiency × Wednesday	0.01 [-0.01, 0.02]	.243	0.02 [0.00, 0.03]	.035
Monday × Sleep efficiency^2	0.01 [0.00, 0.02]	.014	0.01 [0.00, 0.01]	.087
Saturday × Sleep efficiency^2	0.01 [0.01, 0.02]	< .001	0.01 [0.00, 0.02]	.027
Sunday × Sleep efficiency^2	0.01 [0.00, 0.02]	.011	0.01 [0.00, 0.02]	.010
Thursday × Sleep efficiency^2	0.01 [0.00, 0.02]	.153	0.00 [0.00, 0.01]	.341
Tuesday × Sleep efficiency^2	0.00 [-0.01, 0.01]	.932	0.00 [-0.01, 0.01]	.597
Wednesday × Sleep efficiency^2	0.00 [-0.01, 0.01]	.581	0.00 [0.00, 0.01]	.249
Sleep onset (z)				
(Intercept)	0.27 [-0.08, 0.62]	.137	-0.04 [-0.17, 0.09]	.522
Sleep onset	-0.08 [-0.09, -0.07]	< .001	-0.07 [-0.08, -0.06]	< .001
Monday	-0.20 [-0.21, -0.18]	< .001	-0.08 [-0.09, -0.06]	< .001

Table 19 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Saturday	-0.11 [-0.12, -0.09]	< .001	-0.08 [-0.09, -0.06]	< .001
Sunday	-0.18 [-0.20, -0.17]	< .001	-0.14 [-0.15, -0.13]	< .001
Thursday	-0.04 [-0.06, -0.03]	< .001	0.01 [-0.01, 0.03]	.216
Tuesday	-0.05 [-0.07, -0.04]	< .001	-0.01 [-0.03, 0.00]	.066
Wednesday	-0.05 [-0.07, -0.04]	< .001	0.01 [0.00, 0.03]	.133
Sleep onset^2	0.00 [-0.01, 0.00]	.329	0.02 [0.01, 0.02]	< .001
Sleep onset × Monday	0.03 [0.02, 0.04]	< .001	0.08 [0.07, 0.10]	< .001
Sleep onset × Saturday	0.05 [0.04, 0.06]	< .001	0.07 [0.06, 0.09]	< .001
Sleep onset × Sunday	0.09 [0.08, 0.11]	< .001	0.15 [0.13, 0.16]	< .001
Sleep onset × Thursday	0.02 [0.01, 0.04]	.001	0.01 [0.00, 0.02]	.106
Sleep onset × Tuesday	0.03 [0.02, 0.05]	< .001	0.04 [0.02, 0.05]	< .001
Sleep onset × Wednesday	0.03 [0.02, 0.05]	< .001	0.03 [0.01, 0.04]	< .001
Monday × Sleep onset^2	0.01 [0.00, 0.02]	.014	-0.02 [-0.03, -0.01]	< .001
Saturday × Sleep onset^2	0.00 [-0.01, 0.01]	.918	-0.02 [-0.03, -0.01]	< .001
Sunday × Sleep onset^2	-0.01 [-0.02, -0.01]	.002	-0.03 [-0.04, -0.02]	< .001
Thursday × Sleep onset^2	0.00 [-0.01, 0.01]	.617	0.00 [-0.01, 0.00]	.335
Tuesday × Sleep onset^2	-0.01 [-0.01, 0.00]	.272	0.00 [-0.01, 0.01]	.505
Wednesday × Sleep onset^2	0.00 [-0.01, 0.01]	.570	-0.01 [-0.02, 0.00]	.041
Sleep regularity (z)				
(Intercept)	0.24 [-0.10, 0.57]	.171	-0.02 [-0.15, 0.11]	.760
Sleep regularity	0.10 [0.09, 0.11]	< .001	0.03 [0.02, 0.04]	< .001
Monday	-0.16 [-0.17, -0.14]	< .001	-0.10 [-0.11, -0.08]	< .001
Saturday	-0.08 [-0.09, -0.06]	< .001	-0.09 [-0.11, -0.08]	< .001
Sunday	-0.16 [-0.17, -0.14]	< .001	-0.17 [-0.18, -0.15]	< .001

Table 19 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Thursday	-0.04 [-0.05, -0.02]	$< .001$	0.00 [-0.02, 0.01]	.590
Tuesday	-0.05 [-0.06, -0.03]	$< .001$	-0.02 [-0.04, -0.01]	.003
Wednesday	-0.05 [-0.07, -0.03]	$< .001$	-0.01 [-0.03, 0.01]	.171
Sleep regularity^2	0.01 [0.00, 0.01]	.144	-0.01 [-0.02, 0.00]	.005
Sleep regularity × Monday	-0.04 [-0.06, -0.03]	$< .001$	0.00 [-0.01, 0.02]	.542
Sleep regularity × Saturday	-0.02 [-0.03, 0.00]	.020	0.01 [0.00, 0.03]	.070
Sleep regularity × Sunday	-0.01 [-0.02, 0.01]	.436	0.03 [0.01, 0.04]	$< .001$
Sleep regularity × Thursday	-0.02 [-0.04, -0.01]	.002	0.00 [-0.01, 0.02]	.794
Sleep regularity × Tuesday	-0.03 [-0.04, -0.01]	$< .001$	-0.01 [-0.02, 0.01]	.214
Sleep regularity × Wednesday	-0.03 [-0.04, -0.01]	$< .001$	0.01 [-0.01, 0.02]	.361
Monday × Sleep regularity^2	-0.01 [-0.02, 0.00]	.028	0.01 [0.00, 0.02]	.017
Saturday × Sleep regularity^2	-0.02 [-0.03, -0.01]	.001	0.00 [-0.01, 0.01]	.748
Sunday × Sleep regularity^2	-0.01 [-0.02, 0.00]	.004	0.01 [0.00, 0.02]	.041
Thursday × Sleep regularity^2	0.00 [-0.01, 0.01]	.494	0.01 [0.00, 0.02]	.143
Tuesday × Sleep regularity^2	0.00 [-0.01, 0.01]	.511	0.01 [0.00, 0.02]	.071
Wednesday × Sleep regularity^2	0.00 [-0.02, 0.01]	.395	0.01 [0.00, 0.02]	.050

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 20

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.03 [-0.14, 0.19]	.729	0.02 [-0.13, 0.18]	.791
Physical activity	0.10 [0.08, 0.12]	< .001	0.06 [0.05, 0.07]	< .001
Spring	-0.05 [-0.07, -0.03]	< .001	-0.03 [-0.06, -0.01]	.006
Summer	-0.12 [-0.14, -0.09]	< .001	-0.10 [-0.12, -0.07]	< .001
Winter	0.05 [0.03, 0.07]	< .001	0.06 [0.04, 0.08]	< .001
Physical activity^2	-0.01 [-0.01, 0.00]	< .001	0.01 [0.00, 0.02]	.020
Physical activity × Spring	-0.03 [-0.05, -0.01]	.010	-0.01 [-0.02, 0.01]	.484
Physical activity × Summer	-0.03 [-0.05, -0.01]	.015	-0.01 [-0.03, 0.01]	.327
Physical activity × Winter	-0.02 [-0.04, 0.00]	.037	0.01 [-0.01, 0.03]	.187
Spring × Physical activity^2	0.01 [0.00, 0.01]	.110	-0.01 [-0.02, 0.00]	.127
Summer × Physical activity^2	0.00 [-0.01, 0.01]	.581	-0.01 [-0.02, 0.00]	.054
Winter × Physical activity^2	0.00 [0.00, 0.01]	.257	-0.01 [-0.02, 0.00]	.221
Sleep efficiency (z)				
(Intercept)	0.18 [0.02, 0.33]	.029	0.16 [-0.01, 0.32]	.063
Physical activity	0.06 [0.05, 0.08]	< .001	0.02 [0.01, 0.03]	.005
Spring	0.00 [-0.02, 0.02]	.972	0.01 [-0.02, 0.03]	.532
Summer	-0.02 [-0.05, 0.00]	.082	-0.01 [-0.03, 0.02]	.548
Winter	0.03 [0.00, 0.05]	.027	0.04 [0.02, 0.06]	.001
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.01]	.805
Physical activity × Spring	-0.04 [-0.06, -0.02]	< .001	-0.02 [-0.04, -0.01]	.002
Physical activity × Summer	-0.02 [-0.05, 0.00]	.087	-0.01 [-0.03, 0.00]	.135
Physical activity × Winter	-0.01 [-0.03, 0.01]	.211	0.00 [-0.02, 0.01]	.618
Spring × Physical activity^2	0.01 [0.00, 0.02]	.007	0.00 [-0.01, 0.01]	.741

Table 20 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Summer × Physical activity^2	0.01 [0.00, 0.02]	.072	0.00 [-0.02, 0.01]	.533
Winter × Physical activity^2	0.01 [0.00, 0.01]	.039	-0.01 [-0.02, 0.00]	.164
Sleep onset (z)				
(Intercept)	0.08 [-0.19, 0.35]	.558	0.07 [-0.19, 0.33]	.609
Physical activity	-0.07 [-0.08, -0.06]	< .001	-0.06 [-0.07, -0.05]	< .001
Spring	0.01 [-0.01, 0.03]	.238	0.01 [0.00, 0.03]	.120
Summer	0.04 [0.02, 0.05]	< .001	0.03 [0.01, 0.05]	.006
Winter	-0.01 [-0.03, 0.01]	.177	0.00 [-0.02, 0.01]	.696
Physical activity^2	0.00 [-0.01, 0.00]	.118	-0.01 [-0.01, 0.00]	.079
Physical activity × Spring	-0.03 [-0.04, -0.01]	< .001	-0.02 [-0.03, 0.00]	.006
Physical activity × Summer	0.00 [-0.02, 0.02]	.870	0.00 [-0.01, 0.02]	.698
Physical activity × Winter	-0.01 [-0.03, 0.00]	.113	-0.02 [-0.03, 0.00]	.006
Spring × Physical activity^2	0.01 [0.01, 0.02]	< .001	0.00 [0.00, 0.01]	.226
Summer × Physical activity^2	0.01 [0.00, 0.02]	< .001	0.02 [0.01, 0.03]	< .001
Winter × Physical activity^2	0.01 [0.00, 0.01]	.005	0.00 [-0.01, 0.01]	.912
Sleep regularity (z)				
(Intercept)	0.16 [0.04, 0.28]	.011	0.18 [0.04, 0.33]	.015
Physical activity	0.21 [0.18, 0.23]	< .001	0.10 [0.09, 0.12]	< .001
Spring	-0.06 [-0.08, -0.03]	< .001	-0.08 [-0.11, -0.06]	< .001
Summer	0.02 [-0.01, 0.04]	.238	0.00 [-0.03, 0.03]	.877
Winter	-0.03 [-0.06, 0.00]	.023	-0.04 [-0.07, -0.01]	.008
Physical activity^2	-0.03 [-0.04, -0.02]	< .001	-0.02 [-0.03, -0.01]	< .001
Physical activity × Spring	0.04 [0.02, 0.07]	.002	-0.04 [-0.06, -0.02]	< .001
Physical activity × Summer	0.03 [0.00, 0.06]	.046	-0.02 [-0.04, 0.00]	.087

Table 20 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity × Winter	0.02 [0.00, 0.05]	.054	0.00 [-0.02, 0.02]	.704
Spring × Physical activity^2	-0.02 [-0.03, -0.01]	< .001	0.01 [0.00, 0.02]	.079
Summer × Physical activity^2	-0.01 [-0.02, 0.00]	.171	0.02 [0.00, 0.03]	.014
Winter × Physical activity^2	-0.01 [-0.01, 0.00]	.059	0.00 [-0.02, 0.01]	.544

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 21

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.18 [-0.17, 0.53]	.308	-0.06 [-0.18, 0.07]	.404
Sleep duration	0.02 [0.01, 0.03]	< .001	0.04 [0.03, 0.05]	< .001
Spring	0.01 [-0.01, 0.04]	.169	-0.01 [-0.03, 0.01]	.204
Summer	0.00 [-0.03, 0.02]	.711	-0.04 [-0.06, -0.02]	< .001
Winter	-0.02 [-0.05, 0.00]	.019	-0.03 [-0.05, -0.01]	.005
Sleep duration^2	-0.02 [-0.02, -0.01]	< .001	-0.01 [-0.01, 0.00]	.001
Sleep duration × Spring	-0.03 [-0.05, -0.02]	< .001	-0.03 [-0.04, -0.02]	< .001
Sleep duration × Summer	-0.04 [-0.06, -0.03]	< .001	-0.03 [-0.05, -0.02]	< .001
Sleep duration × Winter	0.00 [-0.01, 0.02]	.461	-0.01 [-0.02, 0.01]	.385
Spring × Sleep duration^2	0.00 [-0.01, 0.00]	.275	0.00 [-0.01, 0.01]	.861
Summer × Sleep duration^2	-0.01 [-0.02, 0.00]	.093	-0.01 [-0.02, 0.00]	.020
Winter × Sleep duration^2	0.00 [-0.01, 0.01]	.579	0.00 [-0.01, 0.01]	.806
Sleep efficiency (z)				
(Intercept)	0.17 [-0.17, 0.52]	.321	-0.05 [-0.18, 0.08]	.435
Sleep efficiency	0.02 [0.01, 0.03]	.002	0.01 [0.00, 0.02]	.022
Spring	0.01 [-0.01, 0.03]	.503	-0.02 [-0.04, 0.00]	.049
Summer	-0.02 [-0.04, 0.01]	.188	-0.06 [-0.08, -0.04]	< .001
Winter	-0.03 [-0.05, -0.01]	.008	-0.03 [-0.05, -0.01]	< .001
Sleep efficiency^2	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.02, -0.01]	< .001
Sleep efficiency × Spring	-0.01 [-0.03, 0.00]	.054	-0.01 [-0.02, 0.00]	.108
Sleep efficiency × Summer	0.00 [-0.01, 0.02]	.571	0.00 [-0.01, 0.02]	.835
Sleep efficiency × Winter	-0.01 [-0.03, 0.00]	.070	-0.01 [-0.02, 0.00]	.108
Spring × Sleep efficiency^2	0.00 [0.00, 0.01]	.645	0.00 [0.00, 0.01]	.310

Table 21 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Summer × Sleep efficiency^2	0.01 [0.00, 0.01]	.128	0.01 [0.00, 0.01]	.063
Winter × Sleep efficiency^2	0.00 [0.00, 0.01]	.469	0.00 [0.00, 0.01]	.671
Sleep onset (z)				
(Intercept)	0.17 [-0.18, 0.53]	.334	-0.06 [-0.19, 0.07]	.361
Sleep onset	-0.05 [-0.07, -0.04]	< .001	-0.02 [-0.03, -0.01]	< .001
Spring	0.01 [-0.01, 0.04]	.202	-0.02 [-0.04, 0.00]	.123
Summer	-0.01 [-0.03, 0.02]	.475	-0.05 [-0.07, -0.03]	< .001
Winter	-0.03 [-0.05, -0.01]	.014	-0.03 [-0.06, -0.01]	< .001
Sleep onset^2	0.00 [-0.01, 0.00]	.217	0.00 [0.00, 0.01]	.397
Sleep onset × Spring	0.00 [-0.01, 0.02]	.841	0.00 [-0.01, 0.01]	.929
Sleep onset × Summer	0.03 [0.01, 0.05]	< .001	0.02 [0.01, 0.04]	.008
Sleep onset × Winter	-0.02 [-0.04, -0.01]	.006	-0.01 [-0.02, 0.01]	.309
Spring × Sleep onset^2	0.00 [-0.01, 0.01]	.434	0.00 [-0.01, 0.01]	.742
Summer × Sleep onset^2	0.00 [-0.01, 0.01]	.611	0.00 [-0.01, 0.01]	.595
Winter × Sleep onset^2	0.00 [-0.01, 0.01]	.640	0.01 [0.00, 0.01]	.241
Sleep regularity (z)				
(Intercept)	0.15 [-0.18, 0.49]	.376	-0.07 [-0.19, 0.06]	.277
Sleep regularity	0.09 [0.08, 0.10]	< .001	0.06 [0.05, 0.07]	< .001
Spring	0.02 [0.00, 0.04]	.102	-0.01 [-0.03, 0.01]	.478
Summer	-0.01 [-0.03, 0.01]	.353	-0.05 [-0.07, -0.02]	< .001
Winter	-0.02 [-0.04, 0.00]	.026	-0.03 [-0.05, -0.01]	.007
Sleep regularity^2	0.00 [-0.01, 0.00]	.467	0.00 [-0.01, 0.01]	.783
Sleep regularity × Spring	0.01 [0.00, 0.03]	.093	-0.02 [-0.03, -0.01]	.005
Sleep regularity × Summer	-0.02 [-0.03, 0.00]	.023	-0.03 [-0.04, -0.01]	< .001

Table 21 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep regularity × Winter	0.01 [0.00, 0.03]	.096	0.00 [-0.02, 0.01]	.563
Spring × Sleep regularity^2	0.00 [-0.01, 0.01]	.727	0.00 [-0.01, 0.00]	.436
Summer × Sleep regularity^2	0.00 [0.00, 0.01]	.341	0.00 [-0.01, 0.00]	.332
Winter × Sleep regularity^2	0.01 [0.00, 0.02]	.082	0.00 [-0.01, 0.01]	.645

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 22

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.20 [0.02, 0.37]	.033	0.21 [0.04, 0.39]	.015
Physical activity	0.01 [-0.03, 0.06]	.482	0.02 [-0.04, 0.08]	.566
Europe	-0.22 [-0.28, -0.16]	< .001	-0.23 [-0.30, -0.17]	< .001
Africa	-0.21 [-0.28, -0.15]	< .001	-0.24 [-0.31, -0.17]	< .001
Asia	-0.48 [-0.56, -0.40]	< .001	-0.54 [-0.61, -0.46]	< .001
North America	-0.12 [-0.19, -0.05]	.001	-0.13 [-0.21, -0.06]	< .001
South America	-0.45 [-0.52, -0.38]	< .001	-0.48 [-0.56, -0.41]	< .001
Physical activity^2	0.00 [0.00, 0.01]	.165	0.03 [0.01, 0.05]	.002
Physical activity × Europe	0.06 [0.02, 0.10]	.004	0.04 [-0.02, 0.09]	.177
Physical activity × Africa	0.07 [0.01, 0.13]	.015	0.02 [-0.06, 0.10]	.605
Physical activity × Asia	0.08 [0.00, 0.16]	.040	0.06 [-0.02, 0.13]	.121
Physical activity × North America	0.07 [0.01, 0.12]	.019	0.02 [-0.05, 0.09]	.572
Physical activity × South America	0.11 [0.06, 0.16]	< .001	0.04 [-0.02, 0.11]	.165
Europe × Physical activity^2	-0.01 [-0.02, 0.00]	.059	-0.03 [-0.06, -0.01]	.003
Africa × Physical activity^2	-0.03 [-0.07, 0.01]	.172	-0.02 [-0.06, 0.02]	.320
Asia × Physical activity^2	-0.05 [-0.11, 0.01]	.087	-0.03 [-0.08, 0.02]	.214
North America × Physical activity^2	0.00 [-0.02, 0.02]	.906	-0.02 [-0.05, 0.02]	.317
South America × Physical activity^2	-0.01 [-0.04, 0.01]	.165	-0.04 [-0.06, -0.01]	.019
Sleep efficiency (z)				
(Intercept)	0.28 [0.11, 0.45]	.001	0.31 [0.14, 0.48]	< .001
Physical activity	-0.01 [-0.04, 0.01]	.327	0.00 [-0.03, 0.04]	.888
Europe	-0.14 [-0.21, -0.08]	< .001	-0.18 [-0.24, -0.11]	< .001
Africa	0.00 [-0.07, 0.07]	.927	-0.05 [-0.12, 0.03]	.206

Table 22 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Asia	-0.21 [-0.29, -0.13]	< .001	-0.35 [-0.42, -0.27]	< .001
North America	0.02 [-0.06, 0.09]	.623	-0.09 [-0.16, -0.01]	.020
South America	-0.15 [-0.22, -0.08]	< .001	-0.29 [-0.37, -0.21]	< .001
Physical activity^2	0.00 [0.00, 0.01]	.935	-0.02 [-0.03, 0.00]	.022
Physical activity × Europe	0.03 [0.00, 0.06]	.035	0.00 [-0.03, 0.03]	.966
Physical activity × Africa	0.06 [0.01, 0.11]	.013	-0.02 [-0.08, 0.04]	.577
Physical activity × Asia	0.05 [-0.02, 0.13]	.166	0.07 [0.01, 0.13]	.014
Physical activity × North America	0.14 [0.09, 0.19]	< .001	0.07 [0.01, 0.12]	.018
Physical activity × South America	0.18 [0.14, 0.22]	< .001	0.07 [0.03, 0.11]	< .001
Europe × Physical activity^2	-0.01 [-0.01, 0.00]	.010	0.02 [0.01, 0.04]	.007
Africa × Physical activity^2	-0.05 [-0.09, 0.00]	.033	0.02 [-0.01, 0.06]	.204
Asia × Physical activity^2	-0.16 [-0.23, -0.09]	< .001	-0.04 [-0.08, 0.01]	.091
North America × Physical activity^2	-0.03 [-0.04, -0.01]	< .001	-0.01 [-0.05, 0.02]	.426
South America × Physical activity^2	-0.09 [-0.11, -0.07]	< .001	-0.03 [-0.06, -0.01]	.013
Sleep onset (z)				
(Intercept)	0.01 [-0.27, 0.28]	.965	0.03 [-0.24, 0.29]	.848
Physical activity	-0.08 [-0.10, -0.06]	< .001	-0.07 [-0.10, -0.04]	< .001
Europe	0.09 [0.04, 0.14]	< .001	0.07 [0.02, 0.12]	.007
Africa	0.07 [0.02, 0.12]	.009	0.00 [-0.05, 0.06]	.931
Asia	0.22 [0.16, 0.28]	< .001	0.15 [0.10, 0.21]	< .001
North America	0.08 [0.02, 0.13]	.006	-0.02 [-0.08, 0.03]	.446
South America	0.13 [0.08, 0.18]	< .001	0.08 [0.02, 0.13]	.007
Physical activity^2	0.00 [0.00, 0.01]	.211	-0.03 [-0.05, -0.02]	< .001
Physical activity × Europe	-0.02 [-0.04, 0.00]	.097	0.00 [-0.03, 0.03]	.847

Table 22 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity × Africa	0.06 [0.03, 0.10]	< .001	0.03 [-0.02, 0.07]	.289
Physical activity × Asia	0.02 [-0.03, 0.08]	.381	0.02 [-0.02, 0.07]	.282
Physical activity × North America	0.10 [0.06, 0.13]	< .001	0.08 [0.03, 0.12]	< .001
Physical activity × South America	0.03 [0.00, 0.05]	.077	0.01 [-0.02, 0.05]	.475
Europe × Physical activity^2	0.01 [0.00, 0.01]	.003	0.04 [0.02, 0.05]	< .001
Africa × Physical activity^2	-0.02 [-0.05, 0.01]	.245	0.03 [0.00, 0.05]	.042
Asia × Physical activity^2	-0.04 [-0.09, 0.00]	.044	0.01 [-0.03, 0.04]	.721
North America × Physical activity^2	-0.02 [-0.03, 0.00]	.014	0.00 [-0.03, 0.02]	.943
South America × Physical activity^2	-0.02 [-0.04, -0.01]	.003	0.02 [0.00, 0.04]	.037
Sleep regularity (z)				
(Intercept)	-0.03 [-0.17, 0.10]	.636	0.02 [-0.14, 0.17]	.846
Physical activity	0.27 [0.24, 0.30]	< .001	0.22 [0.17, 0.27]	< .001
Europe	0.24 [0.18, 0.31]	< .001	0.20 [0.13, 0.26]	< .001
Africa	0.45 [0.38, 0.53]	< .001	0.35 [0.28, 0.43]	< .001
Asia	0.48 [0.40, 0.57]	< .001	0.40 [0.32, 0.48]	< .001
North America	0.18 [0.10, 0.26]	< .001	0.08 [0.00, 0.16]	.045
South America	0.13 [0.06, 0.20]	< .001	0.00 [-0.07, 0.07]	.979
Physical activity^2	-0.04 [-0.04, -0.03]	< .001	-0.11 [-0.13, -0.09]	< .001
Physical activity × Europe	-0.05 [-0.09, -0.01]	.014	-0.15 [-0.21, -0.10]	< .001
Physical activity × Africa	-0.21 [-0.27, -0.15]	< .001	0.10 [0.02, 0.18]	.015
Physical activity × Asia	-0.25 [-0.34, -0.16]	< .001	0.02 [-0.05, 0.10]	.535
Physical activity × North America	0.01 [-0.05, 0.06]	.840	0.02 [-0.06, 0.09]	.678
Physical activity × South America	0.01 [-0.03, 0.05]	.645	0.03 [-0.03, 0.09]	.268
Europe × Physical activity^2	-0.01 [-0.01, 0.00]	.087	0.11 [0.08, 0.13]	< .001

Table 22 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Africa × Physical activity^2	-0.18 [-0.25, -0.11]	< .001	-0.05 [-0.09, 0.00]	.043
Asia × Physical activity^2	-0.19 [-0.28, -0.10]	< .001	0.02 [-0.04, 0.07]	.564
North America × Physical activity^2	-0.02 [-0.04, -0.01]	.011	-0.01 [-0.06, 0.03]	.528
South America × Physical activity^2	-0.09 [-0.12, -0.07]	< .001	-0.01 [-0.04, 0.02]	.608

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 23

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.35 [0.01, 0.69]	.041	0.07 [-0.05, 0.20]	.250
Sleep duration	0.11 [0.09, 0.12]	< .001	0.06 [0.04, 0.07]	< .001
Europe	-0.18 [-0.24, -0.12]	< .001	-0.16 [-0.21, -0.10]	< .001
Africa	-0.15 [-0.21, -0.08]	< .001	-0.04 [-0.10, 0.02]	.218
Asia	-0.43 [-0.50, -0.36]	< .001	-0.45 [-0.51, -0.38]	< .001
North America	-0.34 [-0.41, -0.27]	< .001	-0.26 [-0.32, -0.20]	< .001
South America	-0.24 [-0.30, -0.17]	< .001	-0.26 [-0.32, -0.20]	< .001
Sleep duration^2	-0.05 [-0.06, -0.04]	< .001	-0.02 [-0.03, -0.01]	< .001
Sleep duration × Europe	-0.12 [-0.14, -0.10]	< .001	-0.03 [-0.05, -0.02]	< .001
Sleep duration × Africa	-0.06 [-0.10, -0.03]	< .001	-0.06 [-0.10, -0.03]	< .001
Sleep duration × Asia	-0.10 [-0.14, -0.06]	< .001	0.01 [-0.03, 0.05]	.675
Sleep duration × North America	-0.04 [-0.07, -0.01]	.005	-0.02 [-0.05, 0.01]	.161
Sleep duration × South America	-0.14 [-0.16, -0.12]	< .001	-0.09 [-0.11, -0.07]	< .001
Europe × Sleep duration^2	0.03 [0.02, 0.04]	< .001	0.01 [0.00, 0.02]	.084
Africa × Sleep duration^2	0.02 [0.00, 0.05]	.055	-0.02 [-0.04, 0.00]	.063
Asia × Sleep duration^2	0.05 [0.02, 0.08]	.003	-0.04 [-0.08, -0.01]	.005
North America × Sleep duration^2	0.03 [0.01, 0.05]	< .001	0.01 [-0.01, 0.03]	.290
South America × Sleep duration^2	0.03 [0.02, 0.05]	< .001	-0.01 [-0.02, 0.01]	.371
Sleep efficiency (z)				
(Intercept)	0.38 [0.04, 0.71]	.028	0.10 [-0.03, 0.22]	.121
Sleep efficiency	-0.05 [-0.07, -0.02]	< .001	-0.01 [-0.03, 0.02]	.551
Europe	-0.24 [-0.29, -0.18]	< .001	-0.18 [-0.23, -0.13]	< .001
Africa	-0.19 [-0.25, -0.13]	< .001	-0.14 [-0.20, -0.08]	< .001

Table 23 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Asia	-0.47 [-0.54, -0.40]	< .001	-0.49 [-0.55, -0.43]	< .001
North America	-0.35 [-0.41, -0.29]	< .001	-0.29 [-0.34, -0.23]	< .001
South America	-0.30 [-0.36, -0.24]	< .001	-0.34 [-0.40, -0.29]	< .001
Sleep efficiency^2	-0.04 [-0.05, -0.03]	< .001	-0.03 [-0.04, -0.02]	< .001
Sleep efficiency × Europe	0.06 [0.04, 0.09]	< .001	0.02 [-0.01, 0.04]	.174
Sleep efficiency × Africa	0.04 [0.00, 0.08]	.036	-0.01 [-0.05, 0.02]	.531
Sleep efficiency × Asia	0.06 [0.02, 0.11]	.004	0.02 [-0.02, 0.06]	.332
Sleep efficiency × North America	0.05 [0.01, 0.08]	.015	0.01 [-0.02, 0.05]	.535
Sleep efficiency × South America	0.05 [0.02, 0.09]	< .001	0.01 [-0.02, 0.03]	.664
Europe × Sleep efficiency^2	0.03 [0.02, 0.04]	< .001	0.02 [0.00, 0.03]	.006
Africa × Sleep efficiency^2	0.03 [0.02, 0.05]	< .001	0.02 [0.01, 0.04]	.002
Asia × Sleep efficiency^2	0.05 [0.03, 0.06]	< .001	0.02 [0.00, 0.04]	.014
North America × Sleep efficiency^2	0.03 [0.02, 0.05]	< .001	0.03 [0.01, 0.04]	< .001
South America × Sleep efficiency^2	0.04 [0.02, 0.05]	< .001	0.02 [0.01, 0.04]	< .001
Sleep onset (z)				
(Intercept)	0.19 [-0.15, 0.53]	.282	-0.01 [-0.14, 0.11]	.870
Sleep onset	-0.20 [-0.23, -0.18]	< .001	-0.10 [-0.12, -0.08]	< .001
Europe	-0.01 [-0.07, 0.06]	.869	-0.06 [-0.12, 0.00]	.048
Africa	0.09 [0.00, 0.19]	.060	-0.05 [-0.14, 0.04]	.316
Asia	-0.20 [-0.29, -0.12]	< .001	-0.36 [-0.44, -0.28]	< .001
North America	-0.05 [-0.14, 0.03]	.190	-0.17 [-0.24, -0.09]	< .001
South America	-0.04 [-0.11, 0.03]	.275	-0.19 [-0.25, -0.12]	< .001
Sleep onset^2	0.02 [0.01, 0.03]	< .001	0.01 [0.00, 0.02]	.010
Sleep onset × Europe	0.17 [0.14, 0.19]	< .001	0.10 [0.07, 0.12]	< .001

Table 23 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep onset × Africa	0.17 [0.02, 0.32]	.029	-0.02 [-0.17, 0.13]	.761
Sleep onset × Asia	0.15 [0.02, 0.28]	.019	0.01 [-0.12, 0.13]	.930
Sleep onset × North America	0.26 [0.19, 0.32]	< .001	0.09 [0.03, 0.15]	.004
Sleep onset × South America	0.16 [0.13, 0.19]	< .001	0.10 [0.07, 0.13]	< .001
Europe × Sleep onset^2	-0.03 [-0.04, -0.02]	< .001	-0.01 [-0.02, 0.00]	.129
Africa × Sleep onset^2	-0.05 [-0.14, 0.03]	.235	-0.06 [-0.15, 0.02]	.134
Asia × Sleep onset^2	-0.06 [-0.14, 0.02]	.170	-0.09 [-0.17, -0.01]	.032
North America × Sleep onset^2	0.01 [-0.03, 0.04]	.707	0.00 [-0.03, 0.04]	.965
South America × Sleep onset^2	-0.05 [-0.06, -0.04]	< .001	-0.02 [-0.03, 0.00]	.013
Sleep regularity (z)				
(Intercept)	0.39 [0.06, 0.72]	.022	0.09 [-0.03, 0.21]	.161
Sleep regularity	0.17 [0.15, 0.18]	< .001	0.07 [0.05, 0.08]	< .001
Europe	-0.29 [-0.35, -0.24]	< .001	-0.20 [-0.25, -0.15]	< .001
Africa	-0.22 [-0.29, -0.16]	< .001	-0.14 [-0.19, -0.08]	< .001
Asia	-0.51 [-0.58, -0.45]	< .001	-0.51 [-0.57, -0.44]	< .001
North America	-0.36 [-0.42, -0.30]	< .001	-0.25 [-0.30, -0.19]	< .001
South America	-0.32 [-0.38, -0.26]	< .001	-0.31 [-0.37, -0.26]	< .001
Sleep regularity^2	0.01 [0.00, 0.02]	.025	0.00 [-0.01, 0.01]	.625
Sleep regularity × Europe	-0.10 [-0.11, -0.08]	< .001	-0.03 [-0.05, -0.02]	< .001
Sleep regularity × Africa	-0.09 [-0.12, -0.06]	< .001	0.04 [0.01, 0.06]	.006
Sleep regularity × Asia	-0.10 [-0.13, -0.07]	< .001	0.07 [0.04, 0.10]	< .001
Sleep regularity × North America	-0.06 [-0.09, -0.03]	< .001	0.03 [0.00, 0.06]	.074
Sleep regularity × South America	-0.02 [-0.04, 0.00]	.062	0.05 [0.03, 0.07]	< .001
Europe × Sleep regularity^2	-0.01 [-0.02, 0.00]	.061	0.00 [-0.01, 0.01]	.468

Table 23 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Africa × Sleep regularity^2	-0.03 [-0.05, -0.02]	< .001	0.00 [-0.02, 0.02]	.954
Asia × Sleep regularity^2	-0.03 [-0.05, -0.01]	.015	-0.01 [-0.03, 0.01]	.518
North America × Sleep regularity^2	-0.02 [-0.03, 0.00]	.046	-0.01 [-0.03, 0.00]	.086
South America × Sleep regularity^2	0.00 [-0.01, 0.02]	.572	0.01 [0.00, 0.02]	.211

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 24

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.34 [0.17, 0.51]	< .001	0.33 [0.17, 0.49]	< .001
Physical activity	0.14 [0.10, 0.19]	< .001	0.12 [0.09, 0.16]	< .001
Daylight hours	-0.03 [-0.03, -0.02]	< .001	-0.03 [-0.03, -0.02]	< .001
Physical activity^2	-0.01 [-0.03, 0.00]	.117	0.01 [-0.01, 0.03]	.519
Physical activity × Daylight hours	-0.01 [-0.01, 0.00]	.005	-0.01 [-0.01, 0.00]	< .001
Daylight hours × Physical activity^2	0.00 [0.00, 0.00]	.468	0.00 [0.00, 0.00]	.729
Sleep efficiency (z)				
(Intercept)	0.24 [0.07, 0.40]	.005	0.22 [0.05, 0.39]	.011
Physical activity	0.11 [0.07, 0.16]	< .001	0.04 [0.01, 0.08]	.007
Daylight hours	0.00 [-0.01, 0.00]	.011	0.00 [-0.01, 0.00]	.021
Physical activity^2	-0.02 [-0.04, -0.01]	.002	-0.01 [-0.03, 0.01]	.176
Physical activity × Daylight hours	-0.01 [-0.01, 0.00]	.002	0.00 [-0.01, 0.00]	.019
Daylight hours × Physical activity^2	0.00 [0.00, 0.00]	.137	0.00 [0.00, 0.00]	.203
Sleep onset (z)				
(Intercept)	-0.02 [-0.29, 0.25]	.882	-0.01 [-0.27, 0.25]	.950
Physical activity	-0.09 [-0.13, -0.06]	< .001	-0.09 [-0.12, -0.07]	< .001
Daylight hours	0.01 [0.01, 0.01]	< .001	0.01 [0.00, 0.01]	< .001
Physical activity^2	-0.01 [-0.02, 0.00]	.008	-0.03 [-0.04, -0.01]	< .001
Physical activity × Daylight hours	0.00 [0.00, 0.00]	.460	0.00 [0.00, 0.00]	.047
Daylight hours × Physical activity^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.001
Sleep regularity (z)				
(Intercept)	0.05 [-0.09, 0.18]	.492	0.07 [-0.09, 0.23]	.388
Physical activity	0.25 [0.19, 0.31]	< .001	0.17 [0.13, 0.21]	< .001

Table 24 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Daylight hours	0.01 [0.00, 0.01]	.001	0.01 [0.00, 0.01]	.010
Physical activity^2	-0.02 [-0.04, 0.00]	.015	-0.04 [-0.07, -0.02]	.001
Physical activity × Daylight hours	0.00 [-0.01, 0.00]	.502	-0.01 [-0.01, 0.00]	$<$ .001
Daylight hours × Physical activity^2	0.00 [0.00, 0.00]	.040	0.00 [0.00, 0.00]	.074

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 25

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.01 [-0.34, 0.36]	.943	-0.09 [-0.22, 0.04]	.192
Sleep duration	0.09 [0.07, 0.11]	< .001	0.08 [0.06, 0.11]	< .001
Daylight hours	0.01 [0.01, 0.02]	< .001	0.00 [0.00, 0.00]	.378
Sleep duration^2	0.00 [-0.01, 0.02]	.660	0.01 [-0.01, 0.02]	.394
Sleep duration × Daylight hours	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.01, 0.00]	< .001
Daylight hours × Sleep duration^2	0.00 [0.00, 0.00]	.004	0.00 [0.00, 0.00]	.015
Sleep efficiency (z)				
(Intercept)	0.02 [-0.33, 0.37]	.903	-0.06 [-0.19, 0.08]	.402
Sleep efficiency	0.02 [-0.01, 0.05]	.117	0.01 [-0.02, 0.03]	.559
Daylight hours	0.01 [0.01, 0.02]	< .001	0.00 [0.00, 0.00]	.341
Sleep efficiency^2	-0.02 [-0.04, -0.01]	< .001	-0.03 [-0.05, -0.02]	< .001
Sleep efficiency × Daylight hours	0.00 [0.00, 0.00]	.351	0.00 [0.00, 0.00]	.855
Daylight hours × Sleep efficiency^2	0.00 [0.00, 0.00]	.022	0.00 [0.00, 0.00]	.001
Sleep onset (z)				
(Intercept)	0.01 [-0.34, 0.37]	.947	-0.09 [-0.23, 0.05]	.191
Sleep onset	-0.08 [-0.11, -0.04]	< .001	-0.03 [-0.06, 0.00]	.026
Daylight hours	0.01 [0.01, 0.02]	< .001	0.00 [0.00, 0.00]	.728
Sleep onset^2	-0.01 [-0.02, 0.01]	.469	0.01 [-0.01, 0.03]	.271
Sleep onset × Daylight hours	0.00 [0.00, 0.00]	.184	0.00 [0.00, 0.00]	.350
Daylight hours × Sleep onset^2	0.00 [0.00, 0.00]	.820	0.00 [0.00, 0.00]	.539
Sleep regularity (z)				
(Intercept)	0.00 [-0.34, 0.34]	.980	-0.09 [-0.22, 0.04]	.182
Sleep regularity	0.12 [0.09, 0.14]	< .001	0.11 [0.08, 0.13]	< .001

Table 25 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Daylight hours	0.01 [0.01, 0.02]	$< .001$	0.00 [0.00, 0.00]	.885
Sleep regularity^2	0.01 [-0.01, 0.02]	.485	0.00 [-0.01, 0.02]	.882
Sleep regularity $\times$ Daylight hours	0.00 [0.00, 0.00]	.022	0.00 [-0.01, 0.00]	$< .001$
Daylight hours $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.536	0.00 [0.00, 0.00]	.673

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 26

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.50 [0.24, 0.75]	< .001	0.41 [0.15, 0.67]	.002
Physical activity	0.11 [0.09, 0.12]	< .001	0.08 [0.06, 0.09]	< .001
Wrist	-0.63 [-0.91, -0.34]	< .001	-0.52 [-0.81, -0.23]	< .001
Physical activity^2	-0.03 [-0.04, -0.01]	.001	0.00 [-0.01, 0.01]	.630
Physical activity × Wrist	-0.05 [-0.06, -0.03]	< .001	-0.03 [-0.05, -0.01]	.004
Wrist × Physical activity^2	0.03 [0.01, 0.04]	.002	0.00 [-0.01, 0.01]	.827
Sleep efficiency (z)				
(Intercept)	-0.39 [-0.55, -0.23]	< .001	-0.48 [-0.65, -0.32]	< .001
Physical activity	0.13 [0.11, 0.14]	< .001	0.09 [0.08, 0.11]	< .001
Wrist	0.72 [0.55, 0.90]	< .001	0.81 [0.64, 0.99]	< .001
Physical activity^2	-0.09 [-0.11, -0.07]	< .001	-0.04 [-0.05, -0.03]	< .001
Physical activity × Wrist	-0.15 [-0.17, -0.13]	< .001	-0.11 [-0.13, -0.09]	< .001
Wrist × Physical activity^2	0.09 [0.07, 0.11]	< .001	0.04 [0.03, 0.05]	< .001
Sleep onset (z)				
(Intercept)	-0.80 [-1.18, -0.43]	< .001	-0.77 [-1.15, -0.40]	< .001
Physical activity	-0.02 [-0.03, 0.00]	.007	-0.03 [-0.04, -0.02]	< .001
Wrist	1.13 [0.71, 1.56]	< .001	1.08 [0.65, 1.50]	< .001
Physical activity^2	-0.02 [-0.03, -0.01]	< .001	-0.02 [-0.03, -0.01]	< .001
Physical activity × Wrist	-0.11 [-0.12, -0.10]	< .001	-0.06 [-0.07, -0.05]	< .001
Wrist × Physical activity^2	0.03 [0.02, 0.04]	< .001	0.02 [0.01, 0.03]	< .001
Sleep regularity (z)				
(Intercept)	0.01 [-0.25, 0.27]	.913	-0.28 [-0.56, -0.01]	.045
Physical activity	0.20 [0.18, 0.22]	< .001	0.29 [0.27, 0.31]	< .001

Table 26 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Wrist	0.16 [-0.13, 0.45]	.271	0.52 [0.22, 0.82]	$< .001$
Physical activity^2	-0.17 [-0.19, -0.14]	$< .001$	-0.13 [-0.14, -0.11]	$< .001$
Physical activity $\times$ Wrist	0.02 [0.00, 0.04]	.117	-0.25 [-0.27, -0.23]	$< .001$
Wrist $\times$ Physical activity^2	0.13 [0.11, 0.16]	$< .001$	0.11 [0.10, 0.12]	$< .001$

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 27

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-1.03 [-1.48, -0.58]	< .001	-0.39 [-0.62, -0.15]	.001
Sleep duration	0.06 [0.05, 0.07]	< .001	0.06 [0.05, 0.07]	< .001
Wrist	1.52 [1.02, 2.03]	< .001	0.40 [0.13, 0.66]	.003
Sleep duration^2	-0.02 [-0.03, -0.02]	< .001	-0.03 [-0.03, -0.02]	< .001
Sleep duration × Wrist	-0.08 [-0.09, -0.07]	< .001	-0.05 [-0.06, -0.03]	< .001
Wrist × Sleep duration^2	0.00 [-0.01, 0.01]	.740	0.02 [0.01, 0.02]	< .001
Sleep efficiency (z)				
(Intercept)	-1.00 [-1.45, -0.55]	< .001	-0.36 [-0.60, -0.12]	.003
Sleep efficiency	0.02 [0.01, 0.03]	.002	0.02 [0.01, 0.02]	.002
Wrist	1.48 [0.97, 1.98]	< .001	0.36 [0.09, 0.63]	.009
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency × Wrist	-0.01 [-0.02, 0.00]	.102	-0.02 [-0.03, 0.00]	.010
Wrist × Sleep efficiency^2	0.00 [0.00, 0.01]	.289	0.00 [0.00, 0.01]	.175
Sleep onset (z)				
(Intercept)	-1.02 [-1.46, -0.58]	< .001	-0.40 [-0.64, -0.16]	< .001
Sleep onset	-0.04 [-0.07, -0.01]	.016	-0.07 [-0.10, -0.04]	< .001
Wrist	1.52 [1.02, 2.01]	< .001	0.40 [0.13, 0.66]	.003
Sleep onset^2	-0.02 [-0.04, 0.00]	.042	-0.03 [-0.05, -0.01]	.002
Sleep onset × Wrist	-0.04 [-0.07, 0.00]	.028	0.05 [0.01, 0.08]	.004
Wrist × Sleep onset^2	0.02 [0.00, 0.04]	.021	0.03 [0.02, 0.05]	< .001
Sleep regularity (z)				
(Intercept)	-0.98 [-1.42, -0.54]	< .001	-0.36 [-0.60, -0.12]	.003
Sleep regularity	0.12 [0.12, 0.13]	< .001	0.13 [0.12, 0.14]	< .001

Table 27 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Wrist	1.44 [0.94, 1.93]	$< .001$	0.35 [0.08, 0.62]	.011
Sleep regularity^2	-0.01 [-0.01, 0.00]	.014	0.00 [0.00, 0.01]	.524
Sleep regularity $\times$ Wrist	-0.05 [-0.06, -0.04]	$< .001$	-0.11 [-0.12, -0.10]	$< .001$
Wrist $\times$ Sleep regularity^2	0.01 [0.01, 0.02]	$< .001$	0.00 [-0.01, 0.01]	.713

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 28

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.02 [-0.15, 0.18]	.830	0.02 [-0.14, 0.17]	.832
Physical activity	0.08 [0.06, 0.10]	< .001	0.06 [0.04, 0.07]	< .001
Most active hour	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.016
Physical activity^2	-0.01 [-0.01, 0.00]	< .001	0.00 [0.00, 0.01]	.286
Physical activity × Most active hour	0.00 [0.00, 0.00]	.945	0.00 [0.00, 0.00]	.681
Most active hour × Physical activity^2	0.00 [0.00, 0.00]	.187	0.00 [0.00, 0.00]	.852
Sleep efficiency (z)				
(Intercept)	0.17 [0.02, 0.33]	.031	0.17 [0.00, 0.33]	.044
Physical activity	0.01 [-0.01, 0.03]	.216	0.00 [-0.01, 0.02]	.651
Most active hour	0.00 [0.00, 0.00]	.892	0.00 [0.00, 0.00]	.588
Physical activity^2	0.00 [-0.01, 0.00]	.075	0.00 [-0.01, 0.01]	.786
Physical activity × Most active hour	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.470
Most active hour × Physical activity^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.605
Sleep onset (z)				
(Intercept)	0.03 [-0.24, 0.31]	.810	0.03 [-0.23, 0.29]	.810
Physical activity	-0.10 [-0.11, -0.09]	< .001	-0.06 [-0.06, -0.05]	< .001
Most active hour	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Physical activity^2	0.01 [0.01, 0.01]	< .001	-0.01 [-0.01, 0.00]	.076
Physical activity × Most active hour	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	< .001
Most active hour × Physical activity^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.085
Sleep regularity (z)				
(Intercept)	0.15 [0.03, 0.28]	.016	0.19 [0.04, 0.34]	.011
Physical activity	0.18 [0.15, 0.20]	< .001	0.05 [0.04, 0.07]	< .001

Table 28 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Most active hour	0.00 [0.00, 0.00]	.009	0.00 [-0.01, 0.00]	$<$ .001
Physical activity^2	-0.03 [-0.03, -0.02]	$<$ .001	-0.03 [-0.05, -0.02]	$<$ .001
Physical activity $\times$ Most active hour	0.00 [0.00, 0.01]	$<$ .001	0.00 [0.00, 0.00]	$<$ .001
Most active hour $\times$ Physical activity^2	0.00 [0.00, 0.00]	$<$ .001	0.00 [0.00, 0.00]	$<$ .001

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor.

Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 29

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.14 [-0.21, 0.48]	.434	-0.10 [-0.23, 0.03]	.136
Sleep duration	0.04 [0.03, 0.05]	< .001	0.05 [0.04, 0.06]	< .001
Most active hour	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	< .001
Sleep duration^2	-0.01 [-0.01, 0.00]	.022	0.00 [-0.01, 0.00]	.502
Sleep duration × Most active hour	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	< .001
Most active hour × Sleep duration^2	0.00 [0.00, 0.00]	.006	0.00 [0.00, 0.00]	.011
Sleep efficiency (z)				
(Intercept)	0.14 [-0.20, 0.48]	.425	-0.09 [-0.22, 0.04]	.172
Sleep efficiency	-0.01 [-0.03, 0.00]	.016	-0.02 [-0.03, -0.01]	.002
Most active hour	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.004
Sleep efficiency^2	-0.01 [-0.02, -0.01]	< .001	-0.01 [-0.02, -0.01]	< .001
Sleep efficiency × Most active hour	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	< .001
Most active hour × Sleep efficiency^2	0.00 [0.00, 0.00]	.043	0.00 [0.00, 0.00]	.030
Sleep onset (z)				
(Intercept)	0.12 [-0.23, 0.47]	.514	-0.11 [-0.24, 0.02]	.087
Sleep onset	-0.10 [-0.11, -0.09]	< .001	-0.05 [-0.07, -0.04]	< .001
Most active hour	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Sleep onset^2	0.01 [0.00, 0.01]	.071	0.01 [0.00, 0.02]	.017
Sleep onset × Most active hour	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	< .001
Most active hour × Sleep onset^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.117
Sleep regularity (z)				
(Intercept)	0.11 [-0.22, 0.45]	.514	-0.10 [-0.23, 0.03]	.118
Sleep regularity	0.07 [0.06, 0.08]	< .001	0.02 [0.01, 0.03]	< .001

Table 29 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Most active hour	0.00 [0.00, 0.00]	$< .001$	0.00 [0.00, 0.00]	.018
Sleep regularity^2	0.00 [0.00, 0.01]	.278	-0.01 [-0.02, 0.00]	.019
Sleep regularity $\times$ Most active hour	0.00 [0.00, 0.00]	$< .001$	0.00 [0.00, 0.00]	$< .001$
Most active hour $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.282	0.00 [0.00, 0.00]	.028

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 30

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.07 [-0.10, 0.23]	.439	0.07 [-0.09, 0.23]	.384
Physical activity	0.08 [0.06, 0.09]	< .001	0.06 [0.05, 0.07]	< .001
Non-white	-0.16 [-0.19, -0.12]	< .001	-0.19 [-0.22, -0.15]	< .001
Unclear	-0.12 [-0.15, -0.09]	< .001	-0.11 [-0.15, -0.08]	< .001
Physical activity^2	0.00 [-0.01, 0.00]	.108	0.00 [-0.01, 0.00]	.692
Physical activity × Non-white	0.05 [0.02, 0.08]	< .001	0.02 [0.00, 0.04]	.061
Physical activity × Unclear	-0.01 [-0.03, 0.02]	.574	-0.02 [-0.04, 0.00]	.025
Non-white × Physical activity^2	0.00 [-0.01, 0.01]	.989	0.00 [-0.01, 0.02]	.740
Unclear × Physical activity^2	0.00 [-0.01, 0.00]	.548	0.01 [0.00, 0.02]	.017
Sleep efficiency (z)				
(Intercept)	0.19 [0.03, 0.35]	.021	0.17 [0.01, 0.33]	.037
Physical activity	0.04 [0.03, 0.05]	< .001	0.01 [0.00, 0.02]	.115
Non-white	-0.07 [-0.10, -0.03]	< .001	-0.12 [-0.15, -0.09]	< .001
Unclear	-0.01 [-0.05, 0.02]	.451	0.00 [-0.03, 0.04]	.853
Physical activity^2	-0.01 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.00]	.794
Physical activity × Non-white	0.07 [0.04, 0.10]	< .001	0.05 [0.03, 0.07]	< .001
Physical activity × Unclear	-0.01 [-0.03, 0.01]	.305	-0.02 [-0.03, 0.00]	.038
Non-white × Physical activity^2	-0.03 [-0.04, -0.02]	< .001	-0.01 [-0.03, 0.00]	.172
Unclear × Physical activity^2	0.00 [0.00, 0.01]	.059	0.00 [-0.01, 0.01]	.948
Sleep onset (z)				
(Intercept)	0.06 [-0.22, 0.33]	.693	0.04 [-0.22, 0.31]	.737
Physical activity	-0.09 [-0.10, -0.08]	< .001	-0.07 [-0.07, -0.06]	< .001
Non-white	0.08 [0.05, 0.10]	< .001	0.06 [0.03, 0.08]	< .001

Table 30 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Unclear	0.06 [0.04, 0.09]	< .001	0.06 [0.04, 0.09]	< .001
Physical activity^2	0.00 [0.00, 0.01]	.046	0.00 [0.00, 0.01]	.327
Physical activity × Non-white	0.03 [0.01, 0.05]	.003	0.00 [-0.02, 0.01]	.691
Physical activity × Unclear	0.02 [0.00, 0.03]	.011	-0.01 [-0.02, 0.00]	.020
Non-white × Physical activity^2	-0.02 [-0.03, -0.01]	< .001	0.00 [-0.02, 0.01]	.544
Unclear × Physical activity^2	0.00 [0.00, 0.01]	.440	-0.01 [-0.01, 0.00]	.038
Sleep regularity (z)				
(Intercept)	0.18 [0.05, 0.30]	.007	0.17 [0.02, 0.33]	.027
Physical activity	0.22 [0.20, 0.24]	< .001	0.08 [0.07, 0.10]	< .001
Non-white	0.05 [0.02, 0.09]	.002	0.05 [0.02, 0.09]	.002
Unclear	-0.09 [-0.12, -0.05]	< .001	-0.05 [-0.09, -0.02]	.003
Physical activity^2	-0.04 [-0.04, -0.03]	< .001	0.00 [-0.01, 0.01]	.757
Physical activity × Non-white	-0.02 [-0.05, 0.01]	.278	0.13 [0.10, 0.16]	< .001
Physical activity × Unclear	0.03 [0.01, 0.05]	.007	-0.01 [-0.03, 0.00]	.128
Non-white × Physical activity^2	-0.03 [-0.05, -0.01]	< .001	-0.08 [-0.10, -0.06]	< .001
Unclear × Physical activity^2	0.00 [-0.01, 0.00]	.452	-0.03 [-0.04, -0.02]	< .001

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 31

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.19 [-0.15, 0.53]	.282	-0.04 [-0.17, 0.08]	.488
Sleep duration	0.00 [0.00, 0.01]	.595	0.03 [0.03, 0.04]	< .001
Non-white	-0.11 [-0.14, -0.08]	< .001	-0.05 [-0.08, -0.03]	< .001
Unclear	0.00 [-0.03, 0.03]	.877	-0.04 [-0.07, -0.02]	.001
Sleep duration^2	-0.01 [-0.01, -0.01]	< .001	0.00 [-0.01, 0.00]	.057
Sleep duration × Non-white	0.00 [-0.02, 0.01]	.903	-0.03 [-0.05, -0.02]	< .001
Sleep duration × Unclear	0.00 [-0.01, 0.01]	.960	-0.02 [-0.03, -0.01]	< .001
Non-white × Sleep duration^2	0.00 [-0.01, 0.01]	.787	-0.02 [-0.03, -0.01]	.001
Unclear × Sleep duration^2	-0.02 [-0.02, -0.01]	< .001	-0.01 [-0.02, -0.01]	< .001
Sleep efficiency (z)				
(Intercept)	0.19 [-0.15, 0.53]	.281	-0.03 [-0.15, 0.10]	.648
Sleep efficiency	0.02 [0.01, 0.02]	< .001	0.01 [0.01, 0.02]	< .001
Non-white	-0.11 [-0.14, -0.09]	< .001	-0.09 [-0.11, -0.06]	< .001
Unclear	-0.02 [-0.05, 0.01]	.180	-0.08 [-0.11, -0.05]	< .001
Sleep efficiency^2	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep efficiency × Non-white	-0.01 [-0.03, 0.01]	.212	-0.01 [-0.03, 0.00]	.098
Sleep efficiency × Unclear	-0.02 [-0.03, -0.01]	.005	-0.02 [-0.03, -0.01]	.002
Non-white × Sleep efficiency^2	0.01 [0.00, 0.01]	.117	0.00 [0.00, 0.01]	.373
Unclear × Sleep efficiency^2	0.00 [-0.01, 0.00]	.593	0.00 [0.00, 0.01]	.273
Sleep onset (z)				
(Intercept)	0.20 [-0.15, 0.55]	.267	-0.04 [-0.17, 0.09]	.526
Sleep onset	-0.05 [-0.05, -0.04]	< .001	-0.02 [-0.02, -0.01]	< .001
Non-white	-0.08 [-0.12, -0.05]	< .001	-0.06 [-0.09, -0.03]	< .001

Table 31 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Unclear	-0.05 [-0.08, -0.01]	.007	-0.08 [-0.11, -0.05]	$< .001$
Sleep onset^2	-0.01 [-0.01, 0.00]	.011	0.01 [0.00, 0.01]	$< .001$
Sleep onset × Non-white	0.03 [0.01, 0.05]	.006	0.02 [0.00, 0.04]	.016
Sleep onset × Unclear	-0.03 [-0.05, -0.02]	$< .001$	-0.02 [-0.03, -0.01]	.005
Non-white × Sleep onset^2	0.00 [-0.01, 0.01]	.903	0.00 [-0.02, 0.01]	.446
Unclear × Sleep onset^2	0.00 [0.00, 0.01]	.181	-0.01 [-0.01, 0.00]	.056
Sleep regularity (z)				
(Intercept)	0.15 [-0.18, 0.49]	.370	-0.05 [-0.17, 0.07]	.415
Sleep regularity	0.07 [0.06, 0.08]	$< .001$	0.04 [0.04, 0.05]	$< .001$
Non-white	-0.09 [-0.12, -0.06]	$< .001$	-0.07 [-0.09, -0.04]	$< .001$
Unclear	0.00 [-0.03, 0.03]	.902	-0.06 [-0.09, -0.04]	$< .001$
Sleep regularity^2	0.00 [0.00, 0.01]	.557	0.00 [-0.01, 0.00]	.745
Sleep regularity × Non-white	0.03 [0.01, 0.05]	$< .001$	0.05 [0.04, 0.07]	$< .001$
Sleep regularity × Unclear	0.05 [0.04, 0.06]	$< .001$	0.01 [0.00, 0.02]	.053
Non-white × Sleep regularity^2	-0.01 [-0.02, 0.00]	.020	-0.01 [-0.02, 0.00]	.158
Unclear × Sleep regularity^2	0.00 [-0.01, 0.01]	.524	0.00 [-0.01, 0.01]	.862

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

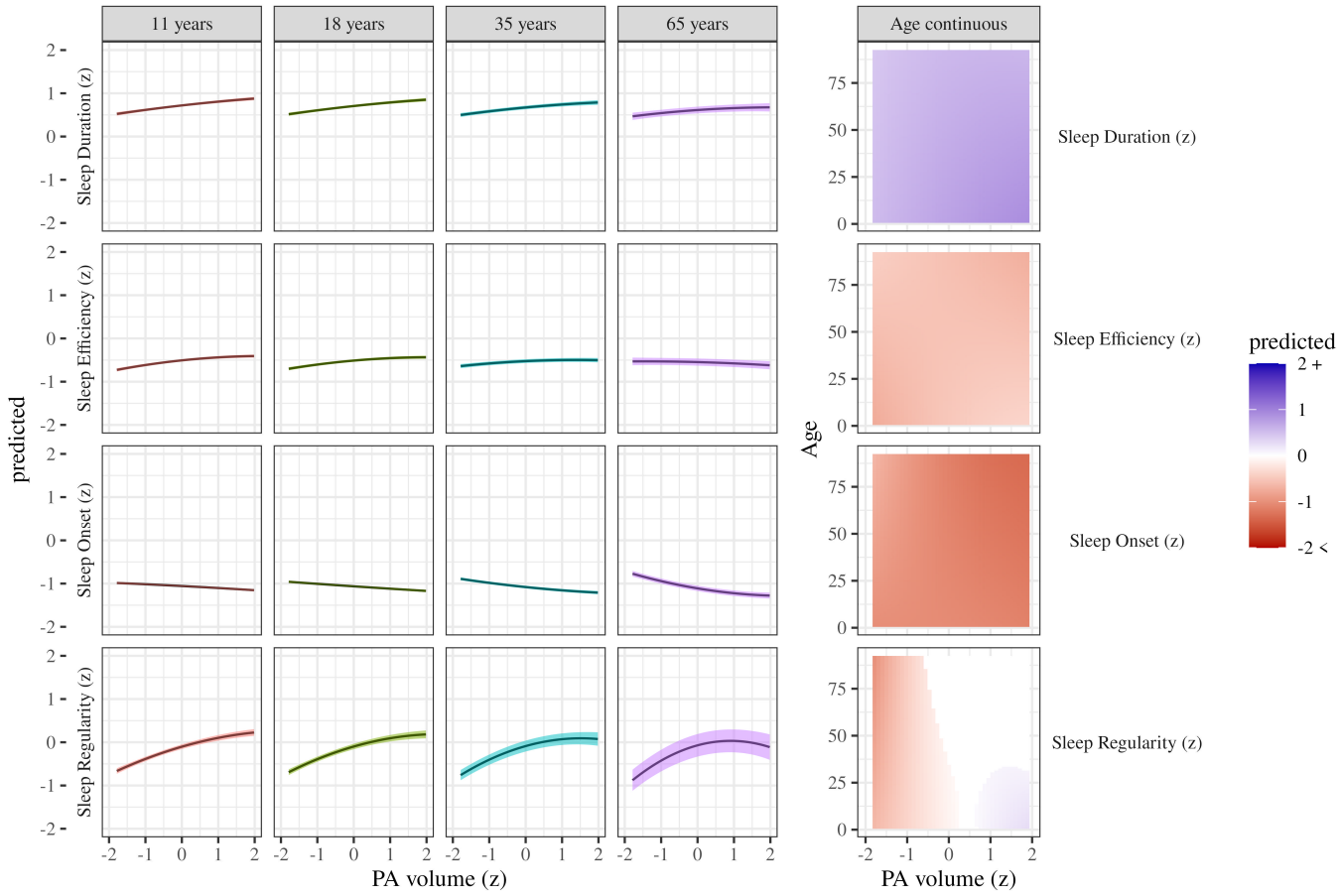

Figure 1. Sleep metrics on Physical activity volume

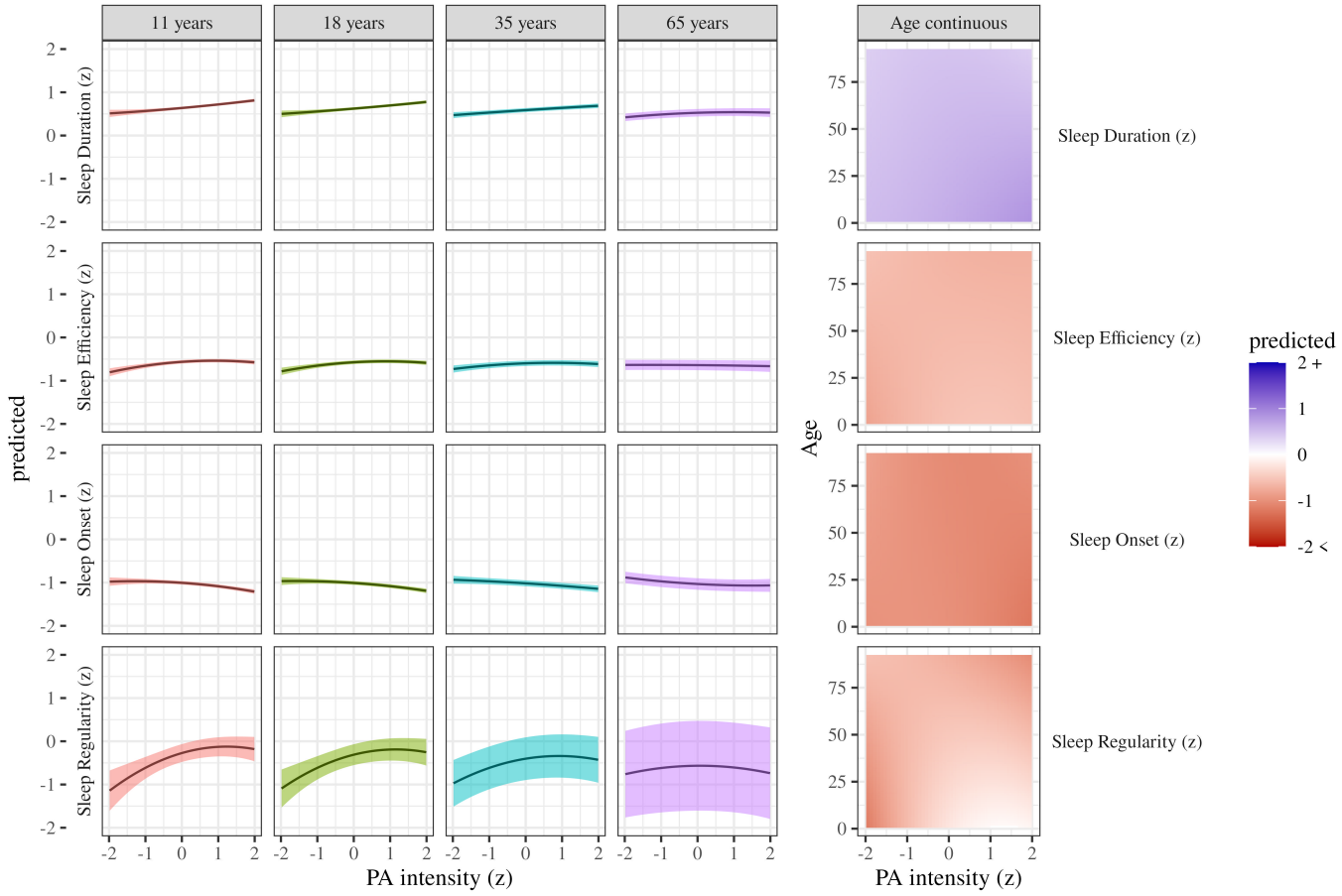

Figure 2. Sleep metrics on Physical activity intensity

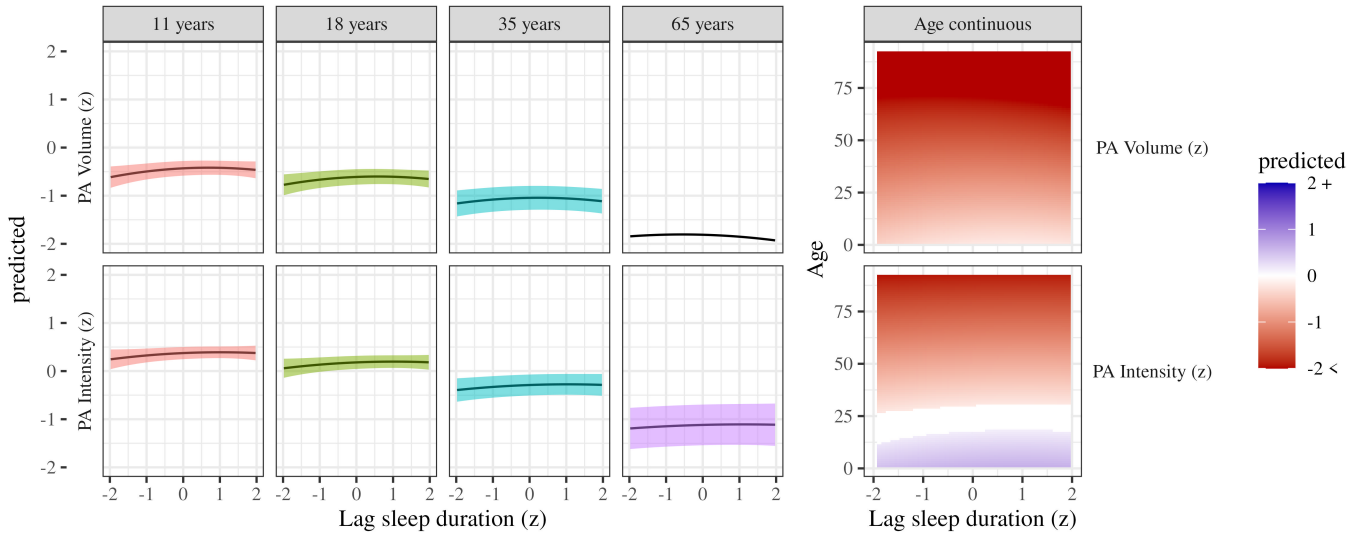

Figure 3. Physical activity by sleep duration

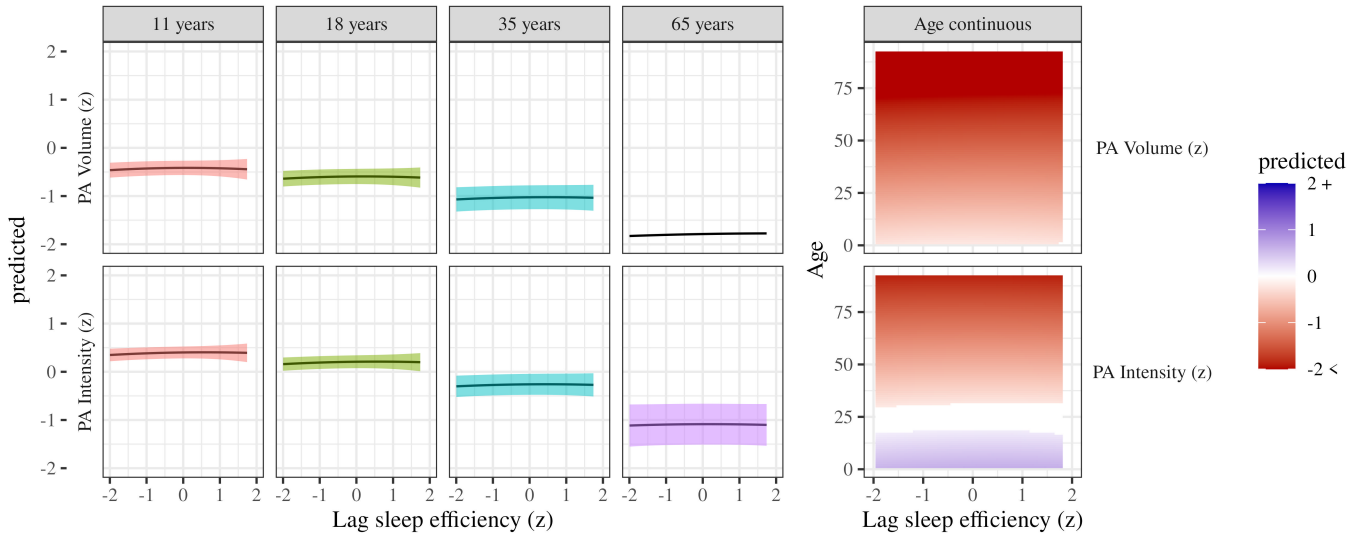

Figure 4. Physical activity by sleep efficiency

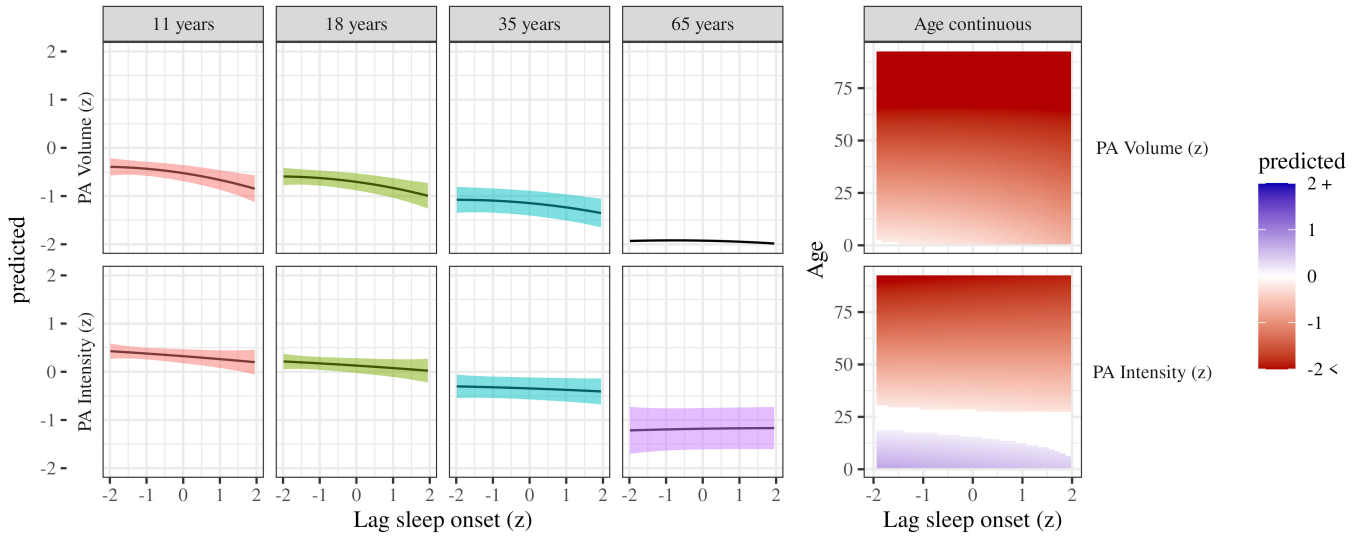

Figure 5. Physical activity by sleep onset

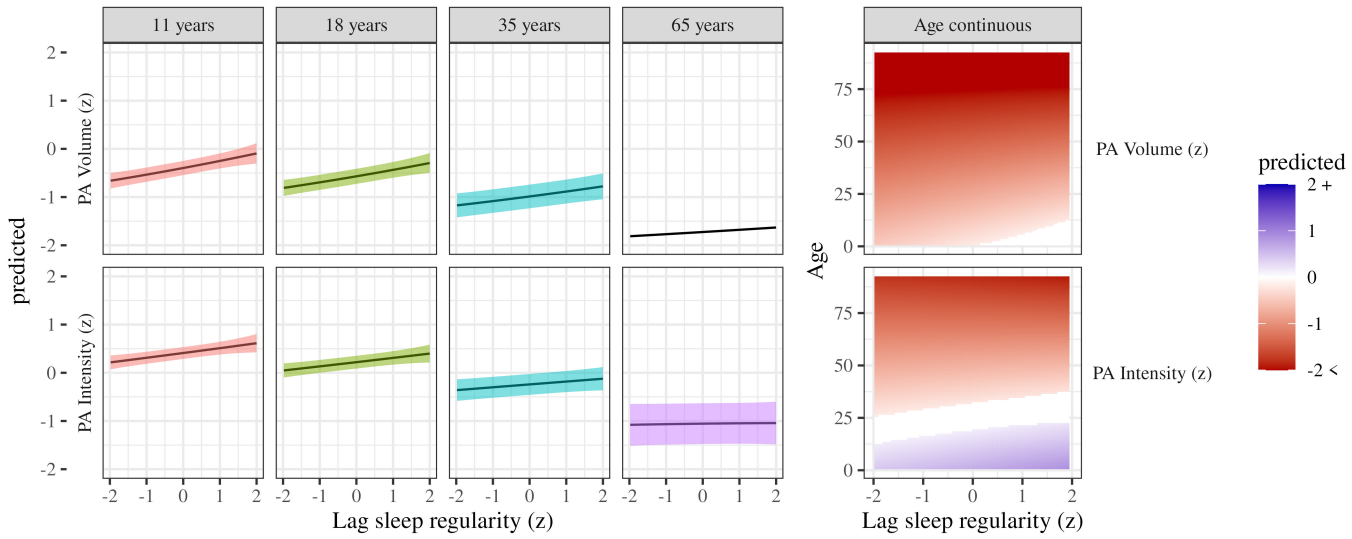

Figure 6. Physical activity by sleep regularity

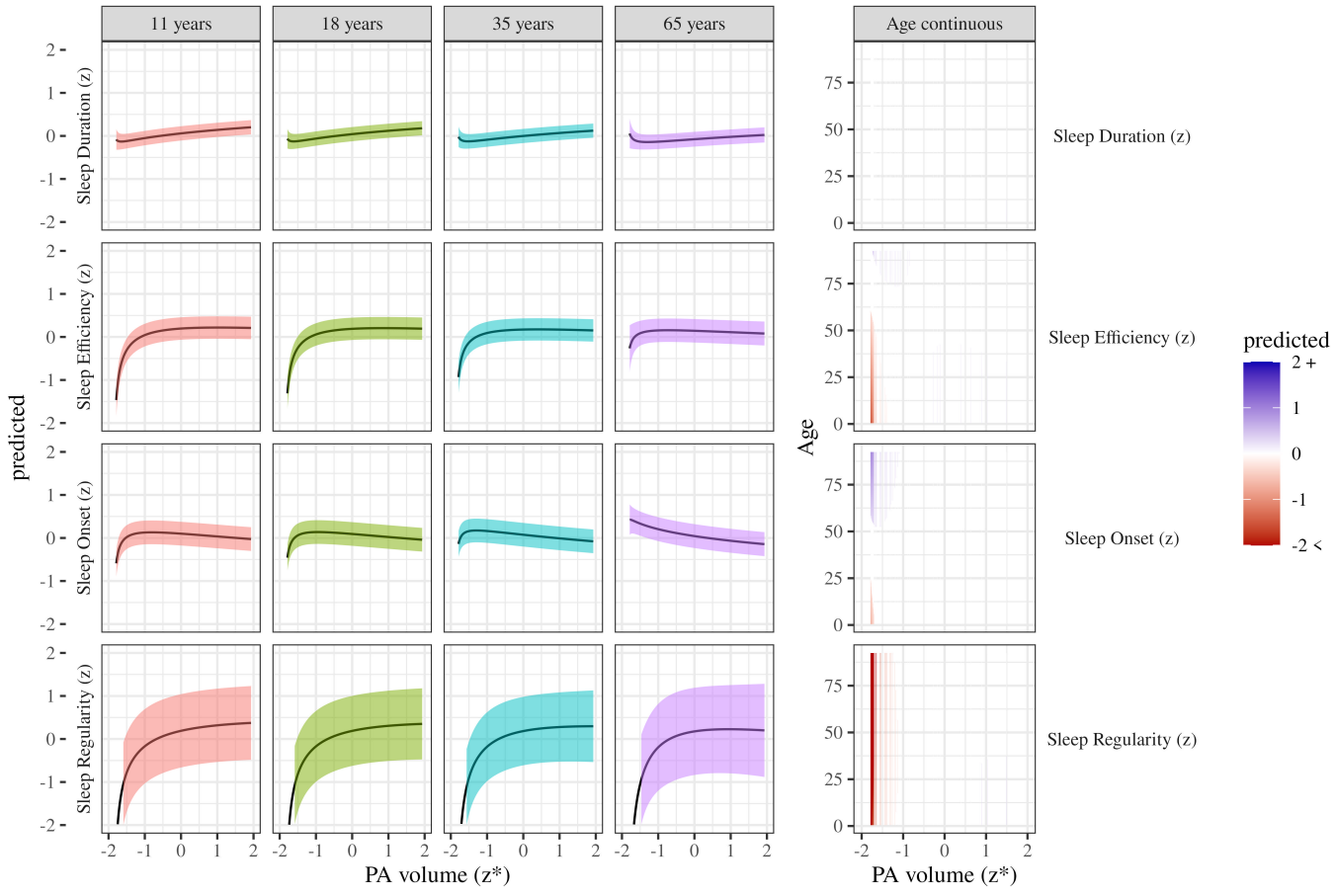

Figure 7. Sleep metrics on Physical activity volume

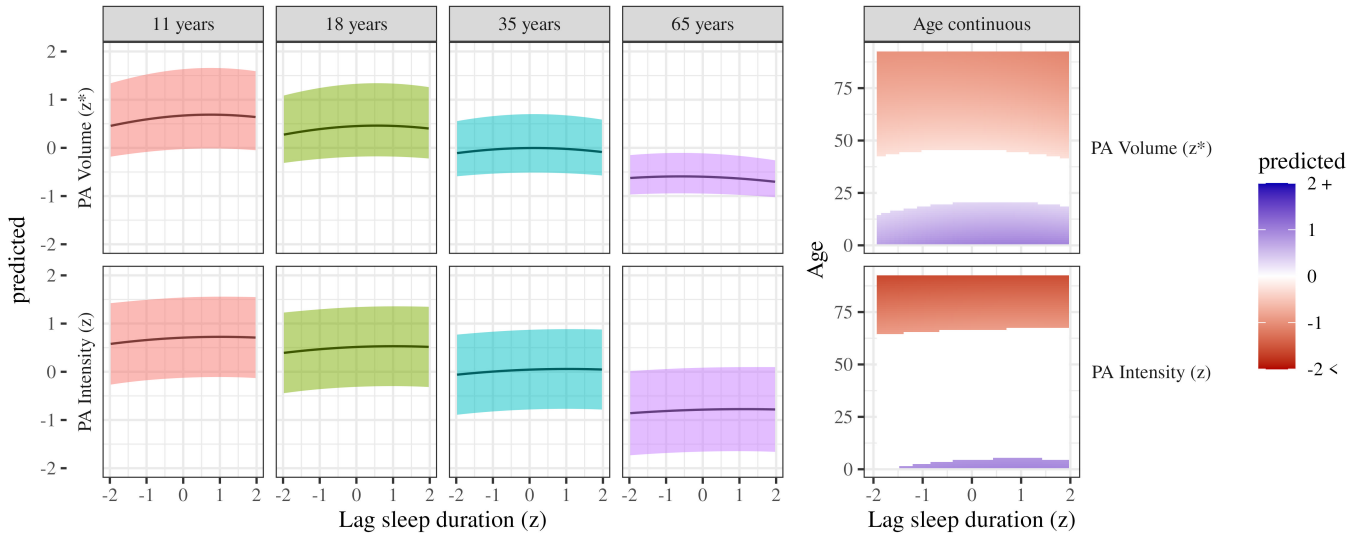

Figure 8. Physical activity by sleep duration

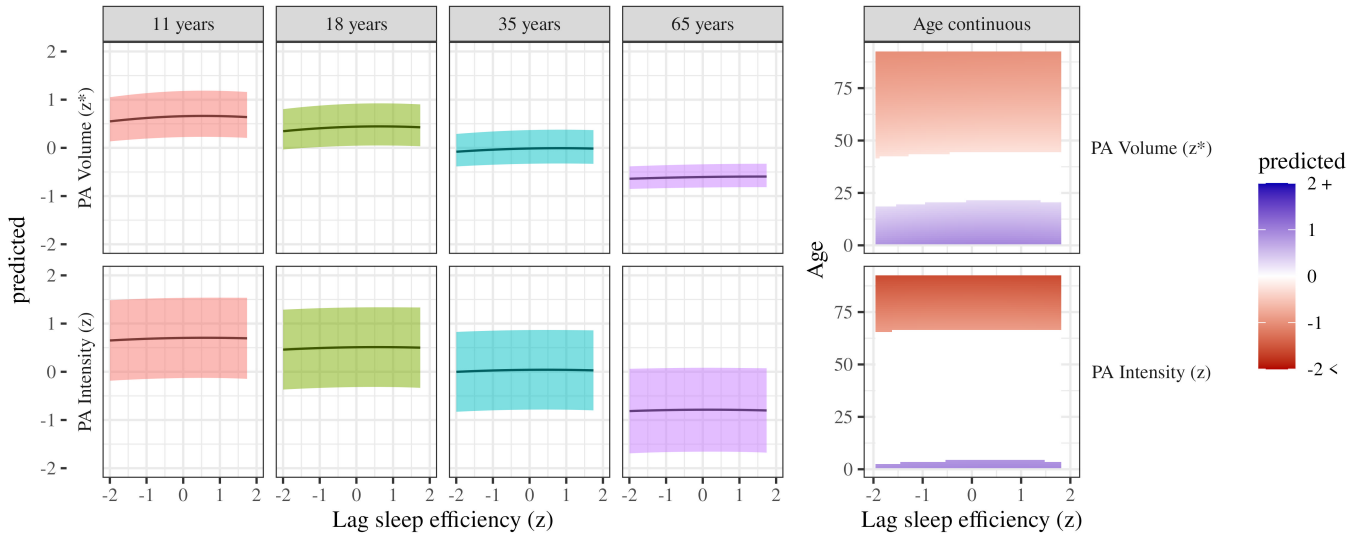

Figure 9. Physical activity by sleep efficiency

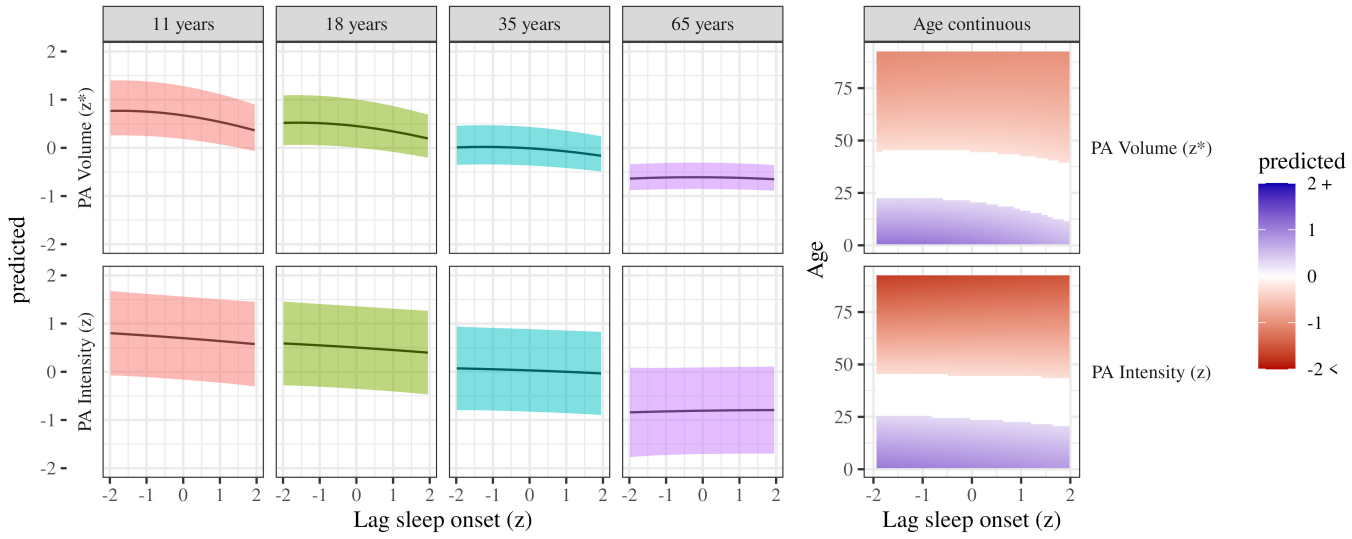

Figure 10. Physical activity by sleep onset

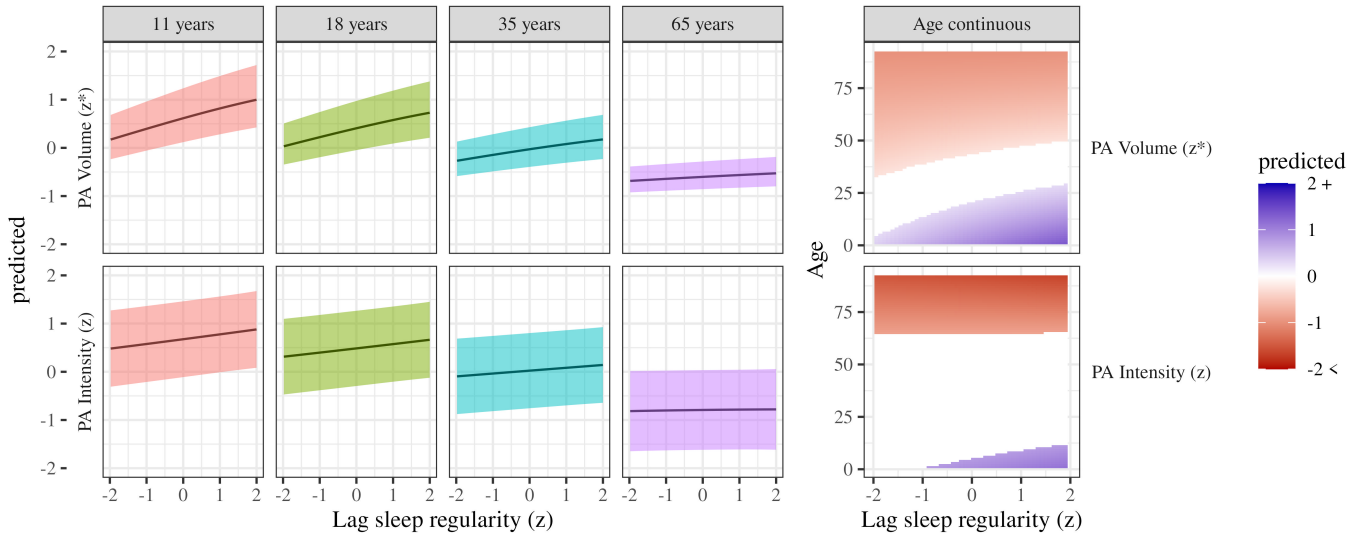

Figure 11. Physical activity by sleep regularity

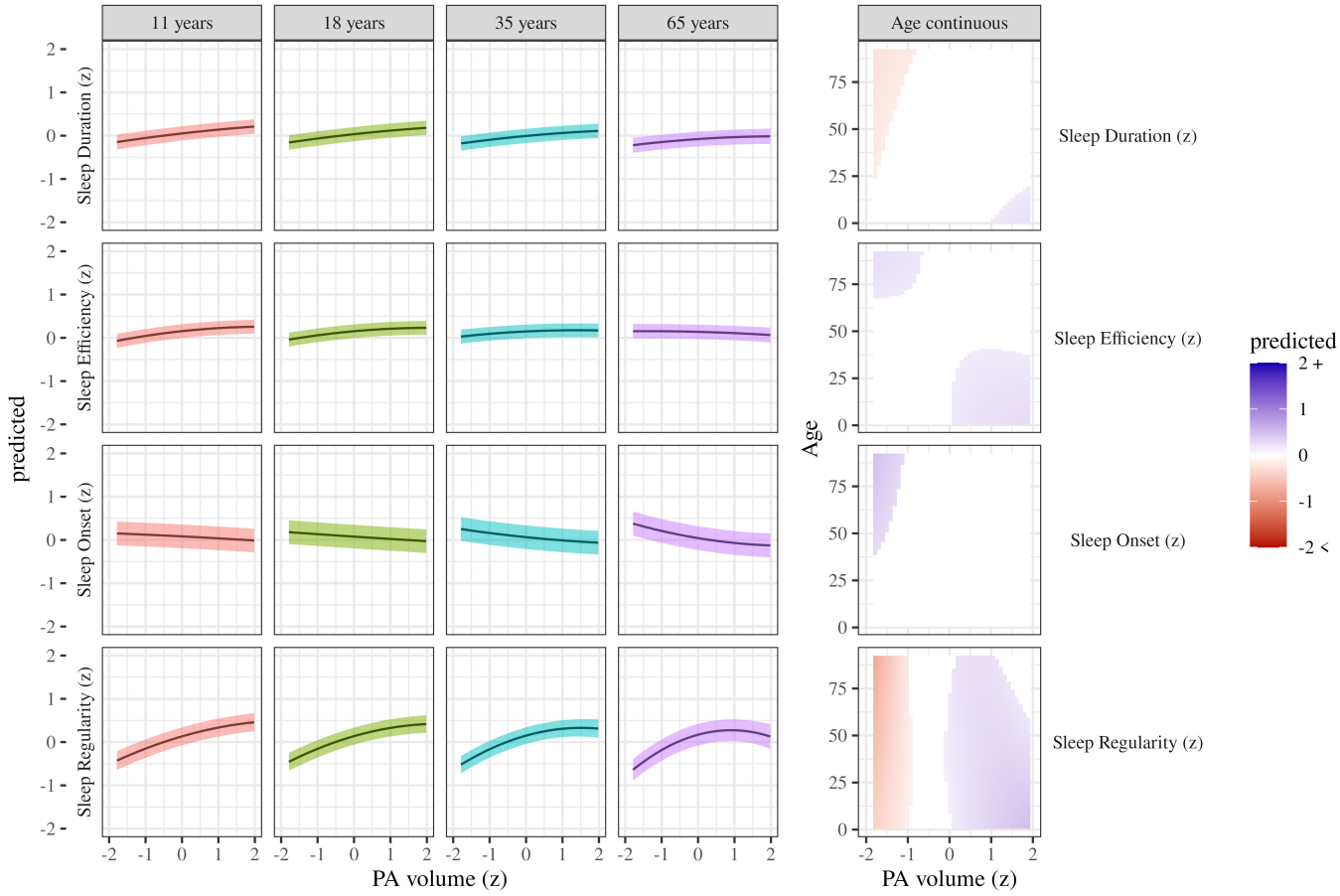

Figure 12. Sleep metrics on physical activity volume (all wear days)

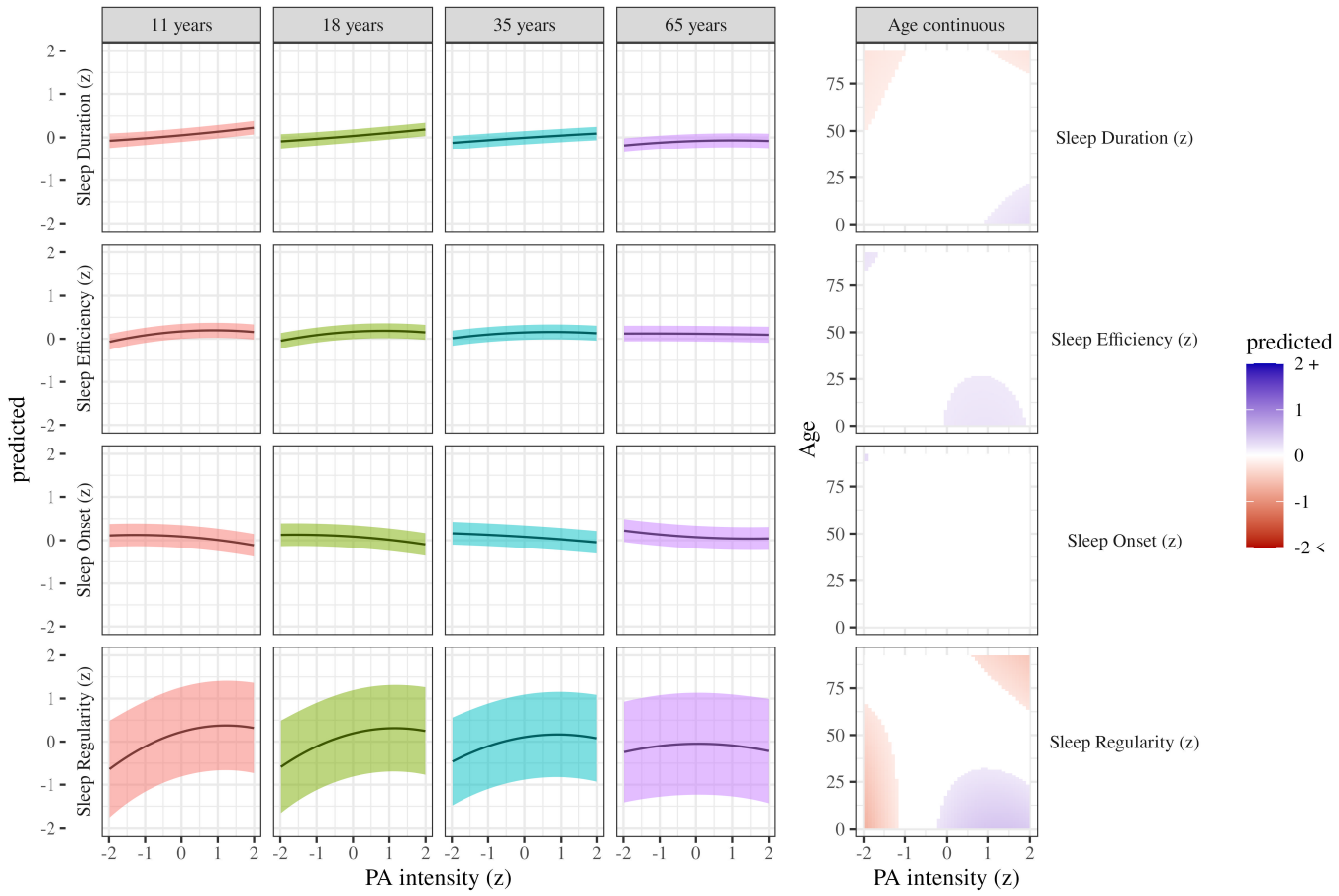

Figure 13. Sleep metrics on physical activity intensity (all wear days)

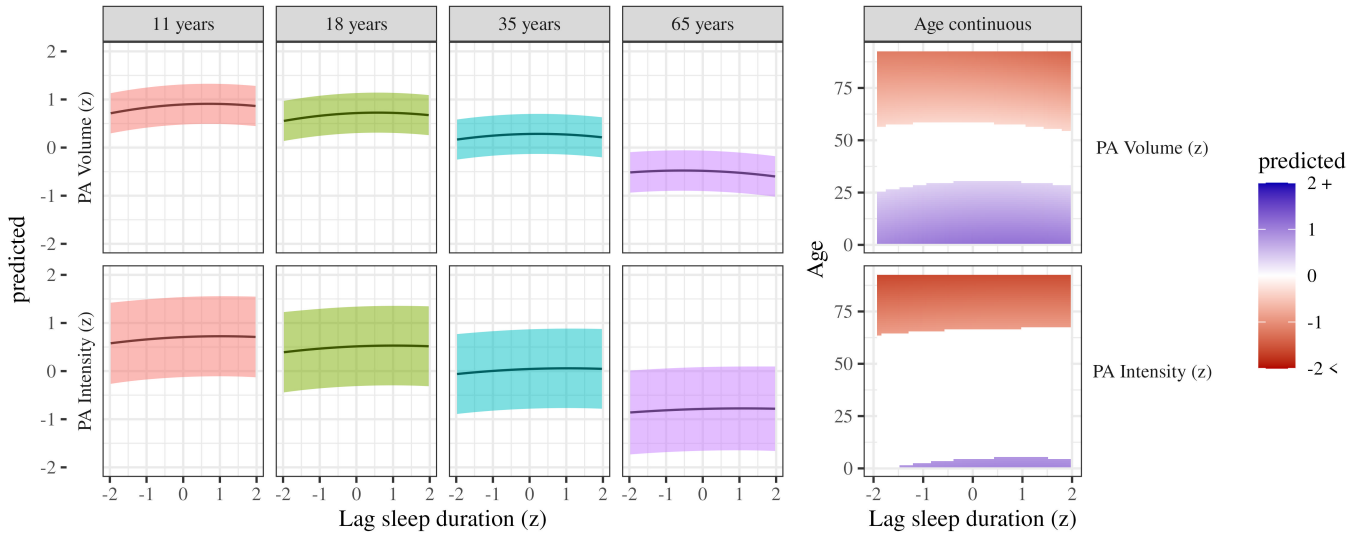

Figure 14. Physical activity by previous night sleep duration (all wear days)

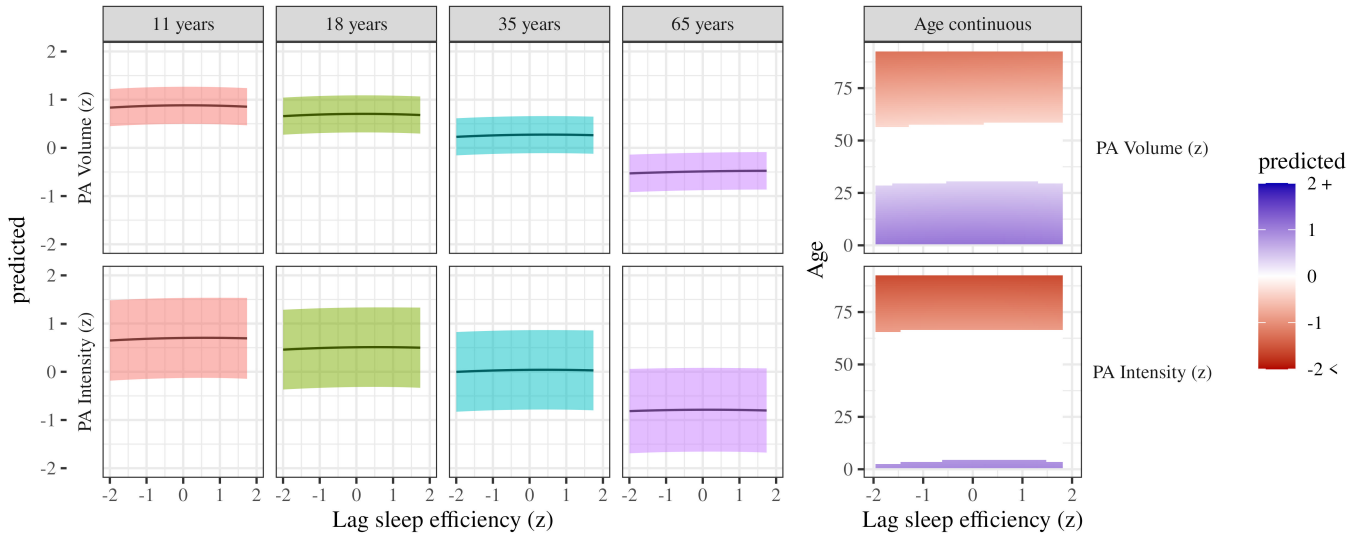

Figure 15. Physical activity by previous night sleep efficiency (all wear days)

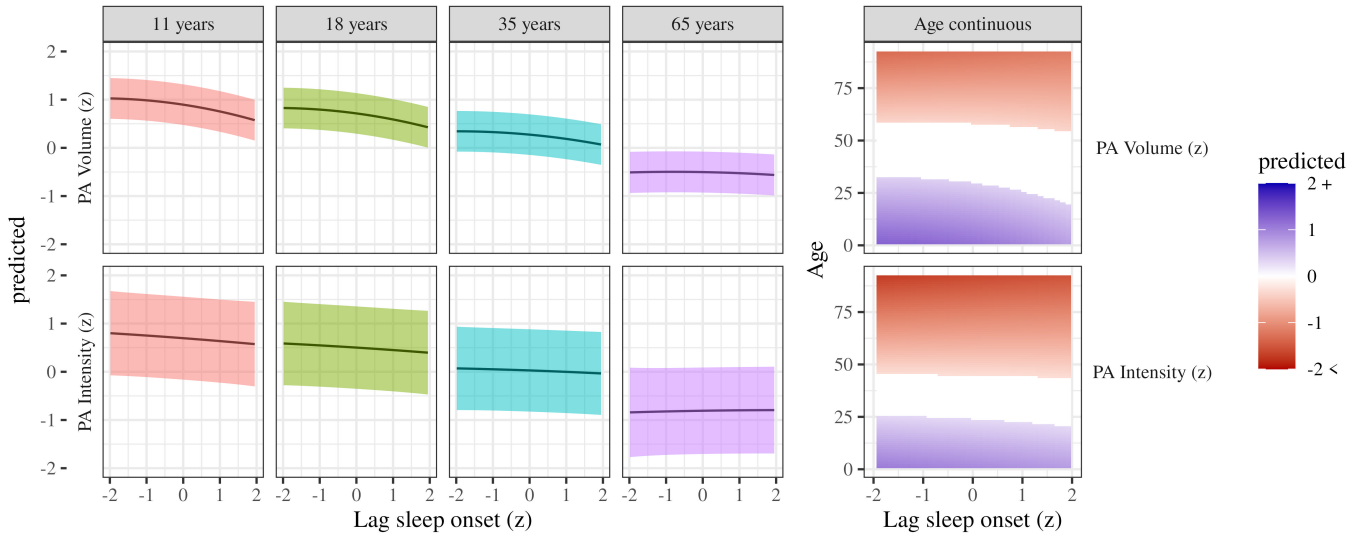

Figure 16. Physical activity by previous night sleep onset (all wear days)

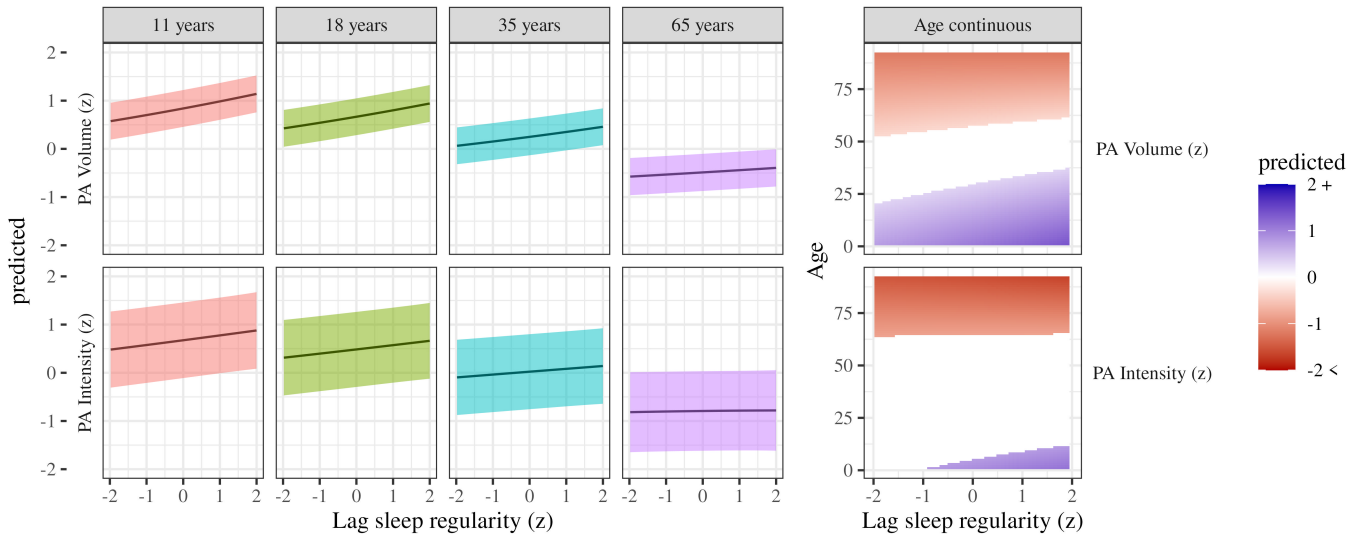

Figure 17. Physical activity by previous night sleep regularity (all wear days)

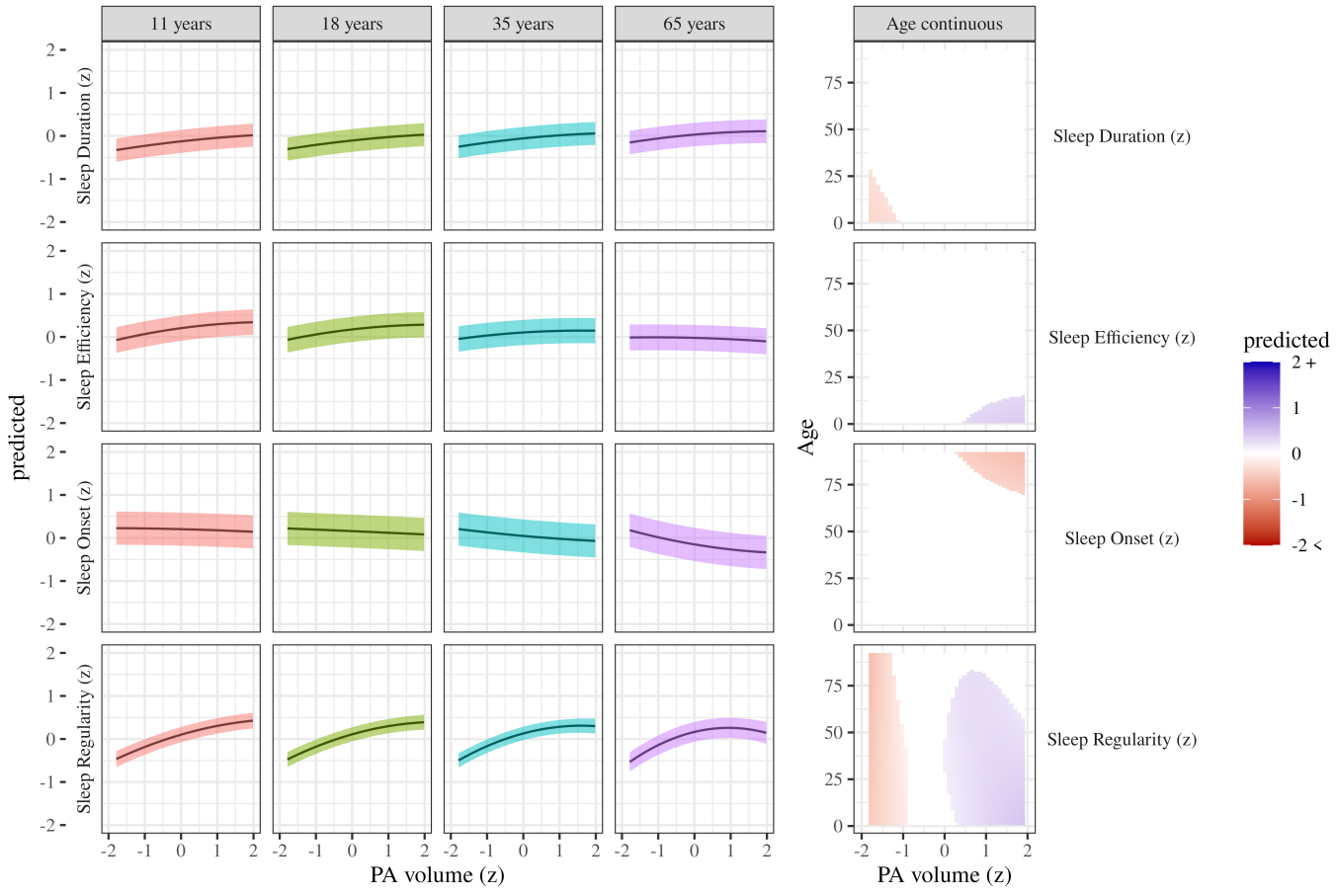

Figure 18. Sleep metrics on physical activity volume (at least seven valid days)

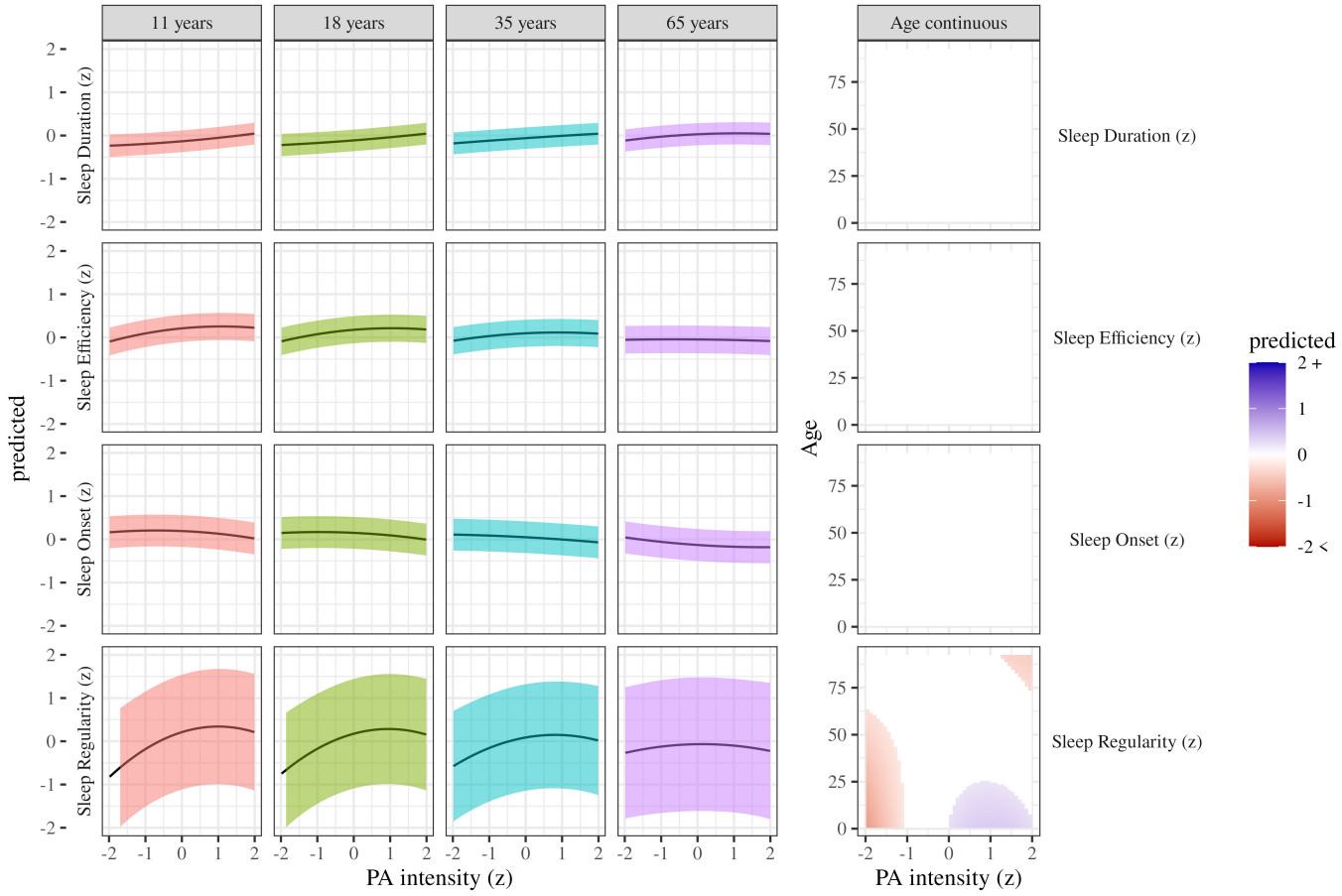

Figure 19. Sleep metrics on physical activity intensity (at least seven valid days)

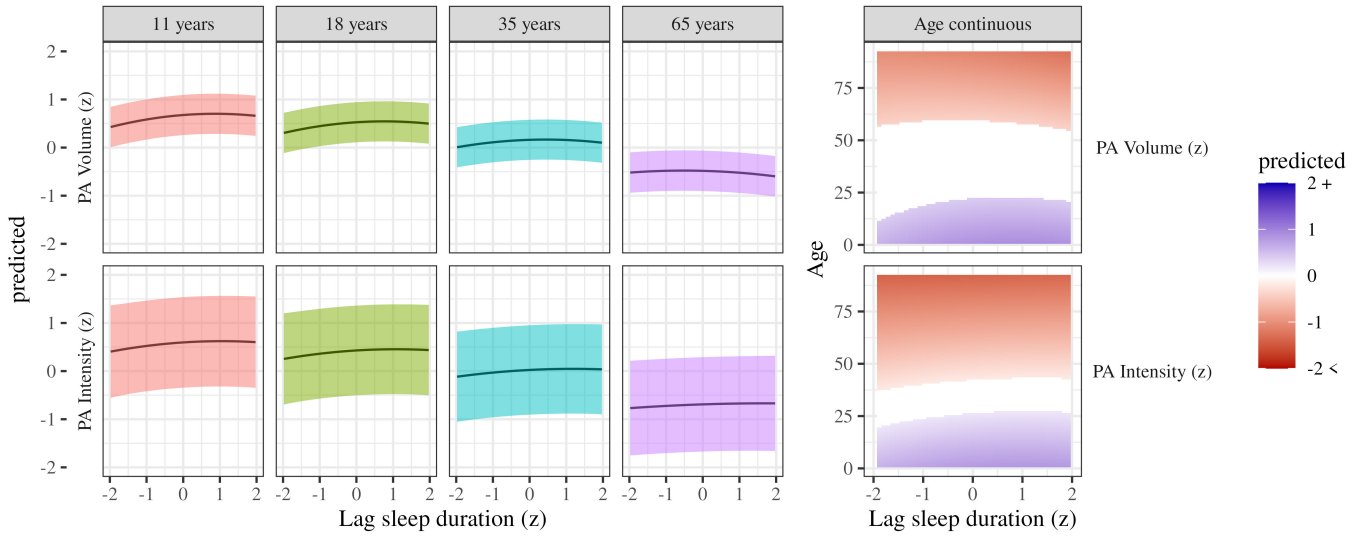

Figure 20. Physical activity by previous night sleep duration (at least seven valid days)

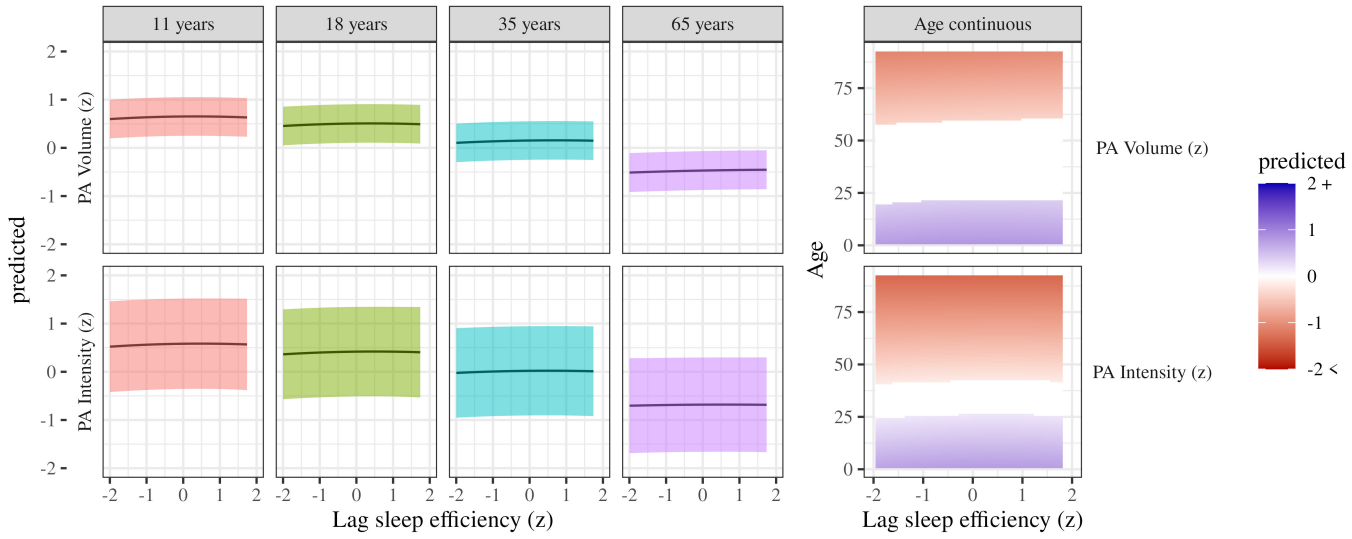

Figure 21. Physical activity by previous night sleep efficiency (at least seven valid days)

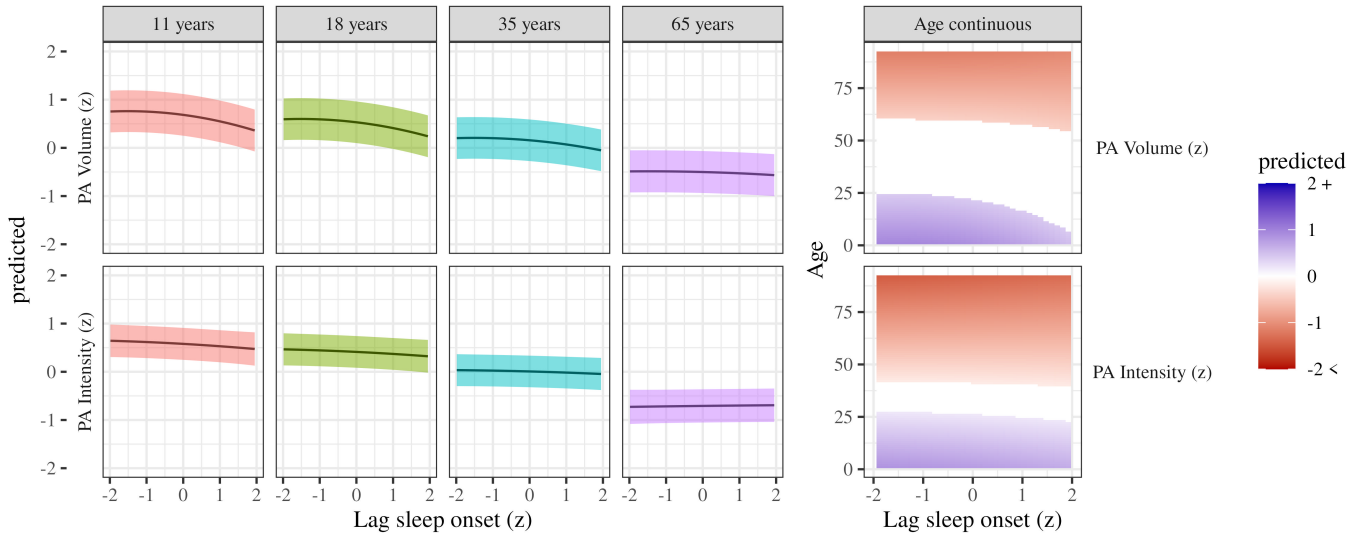

Figure 22. Physical activity by previous night sleep onset (at least seven valid days)

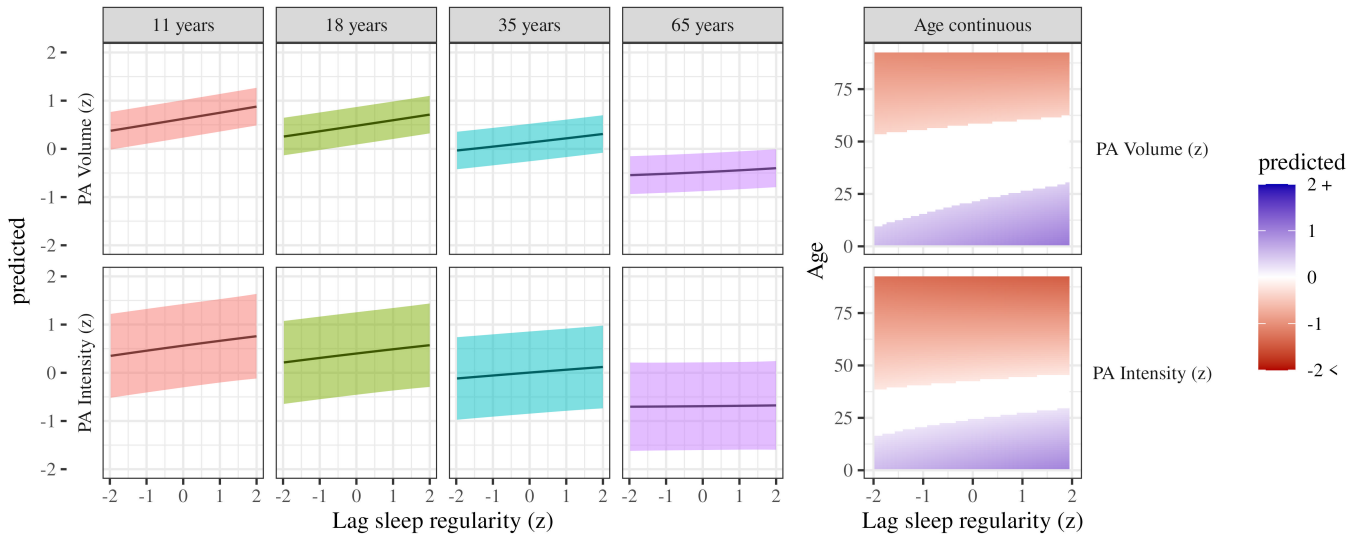

Figure 23. Physical activity by previous night sleep regularity (at least seven valid days)

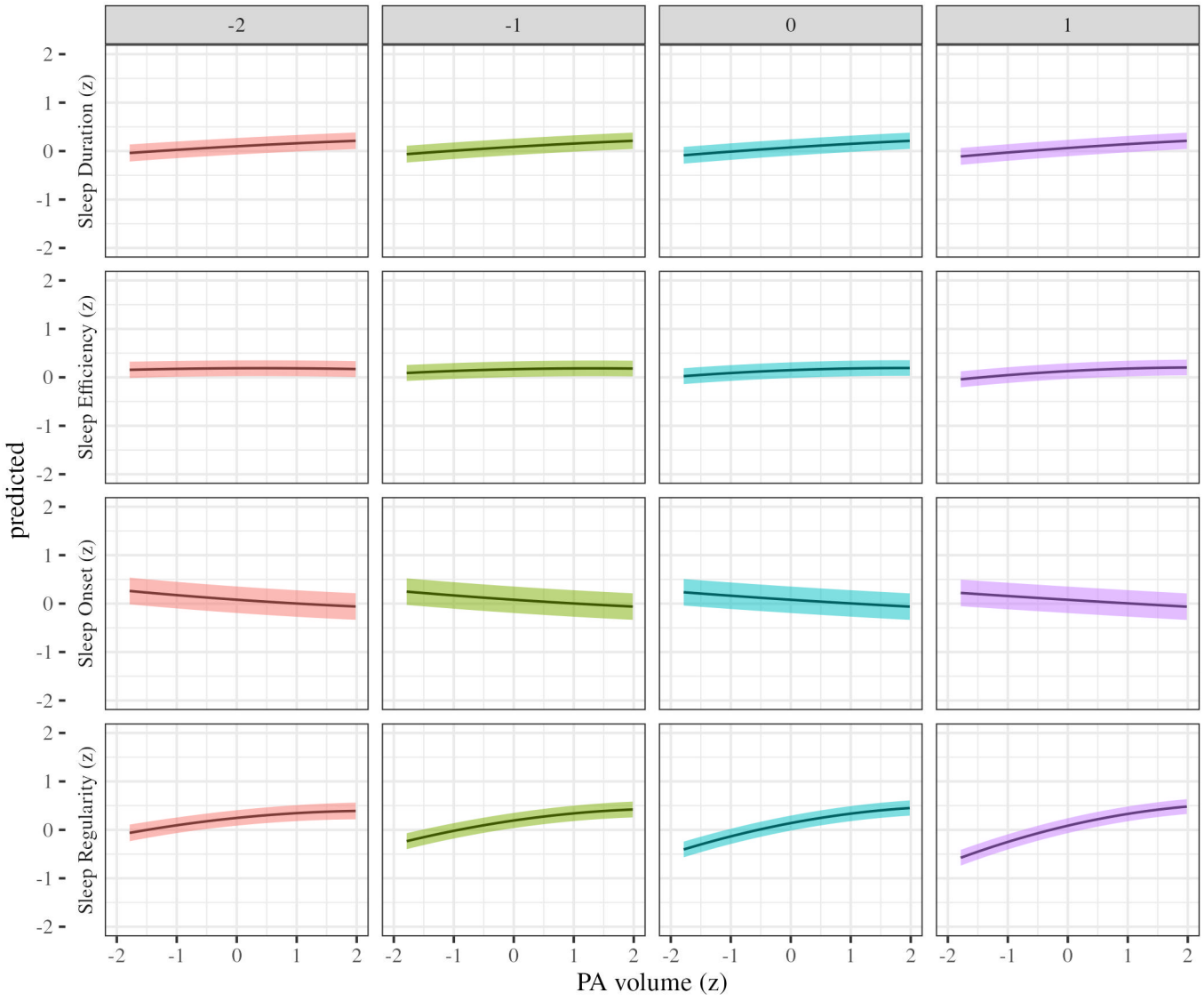

Figure 24. Sleep metrics on Physical activity volume by BMI z-score

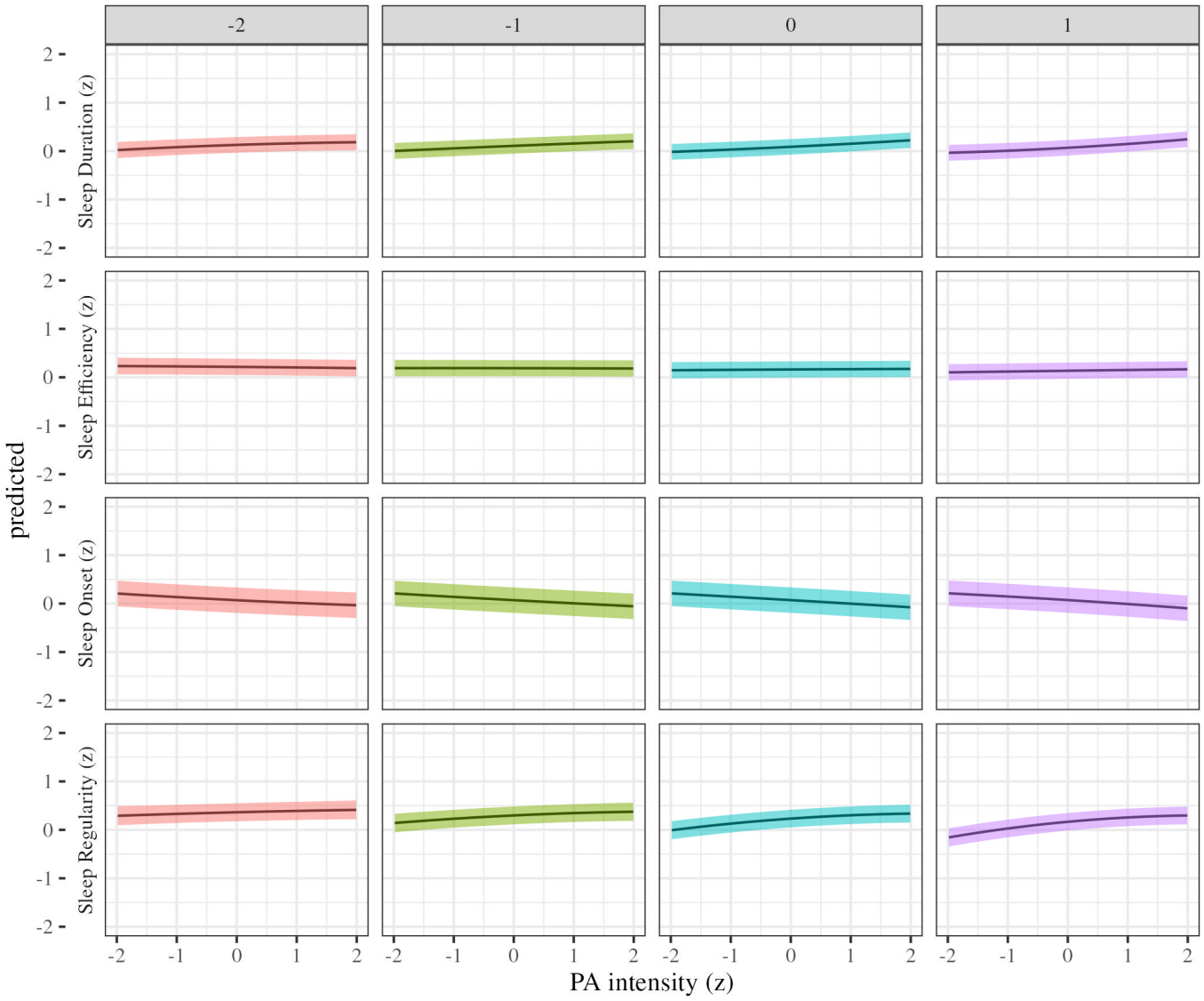

Figure 25. Sleep metrics on Physical activity intensity moderated by BMI z-score

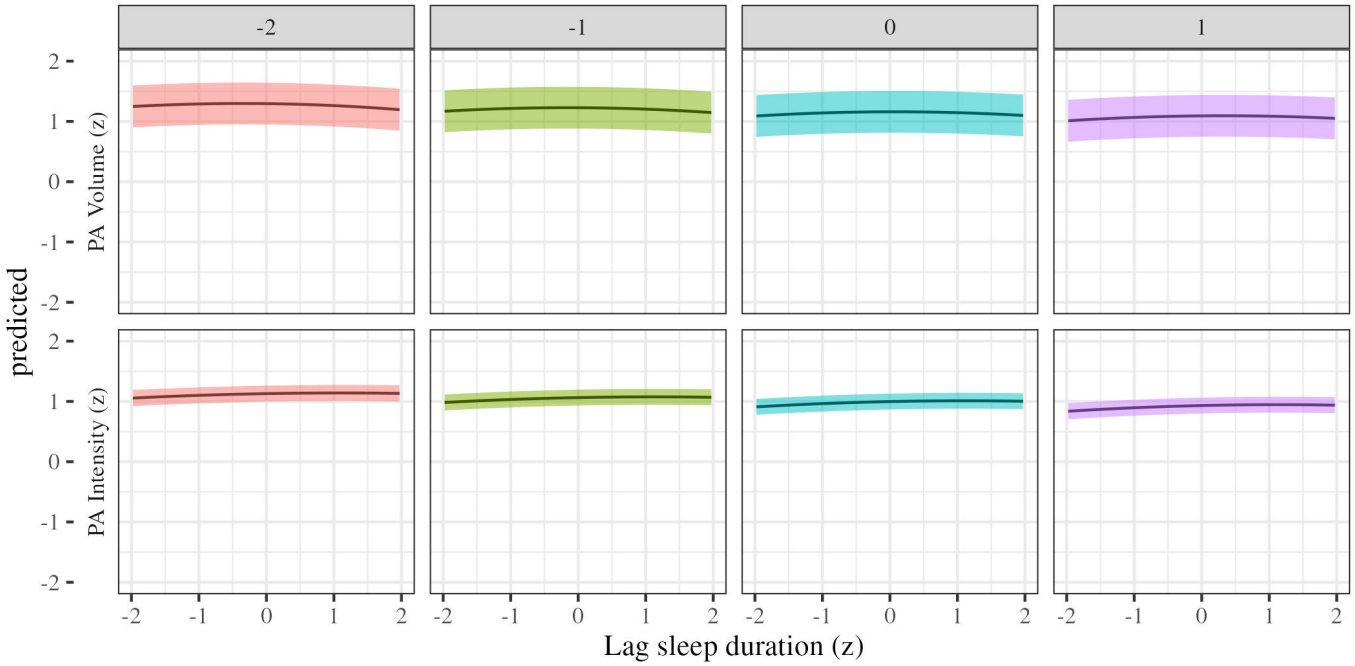

Figure 26. Physical activity by sleep duration moderated by BMI z-score

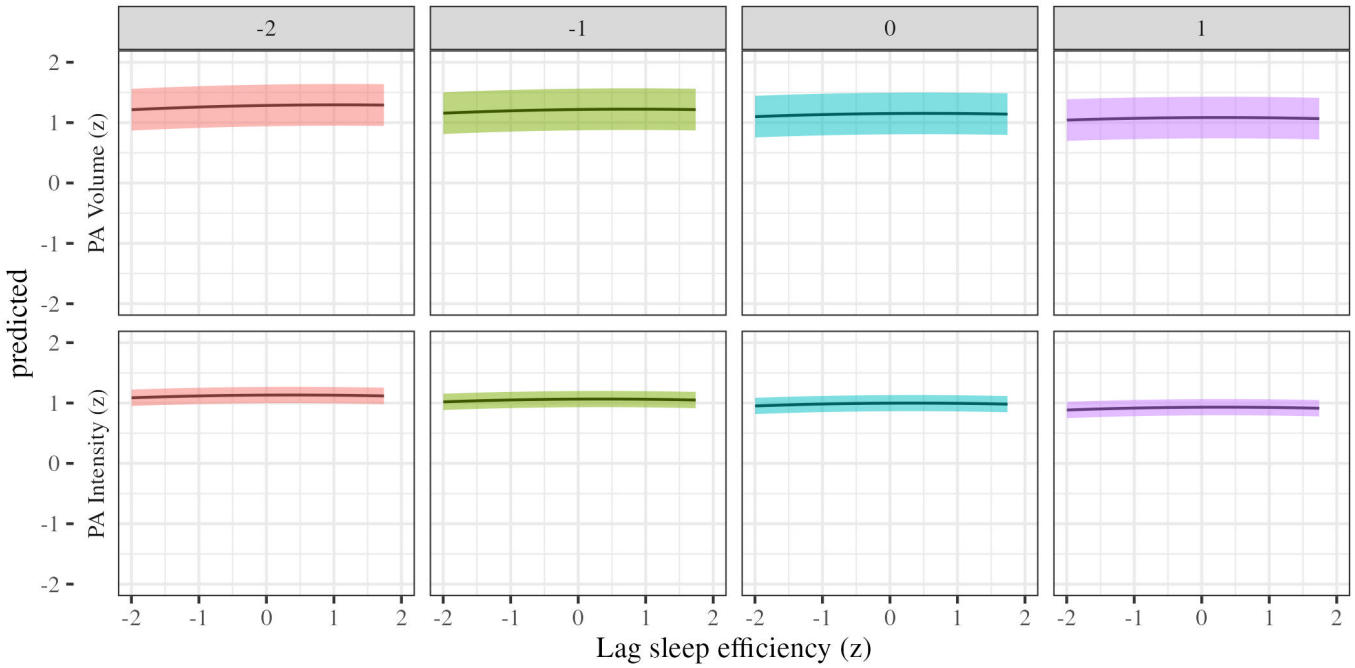

Figure 27. Physical activity by sleep efficiency moderated by BMI z-score

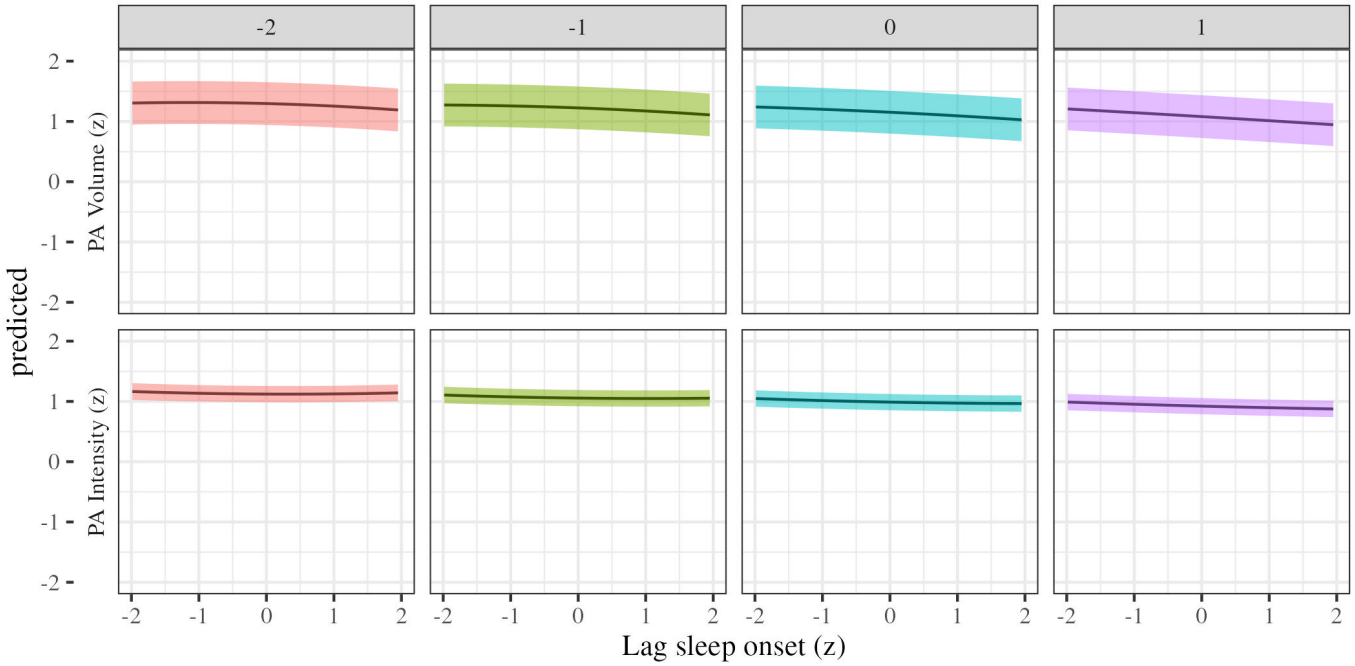

Figure 28. Physical activity by sleep onset moderated by BMI z-score

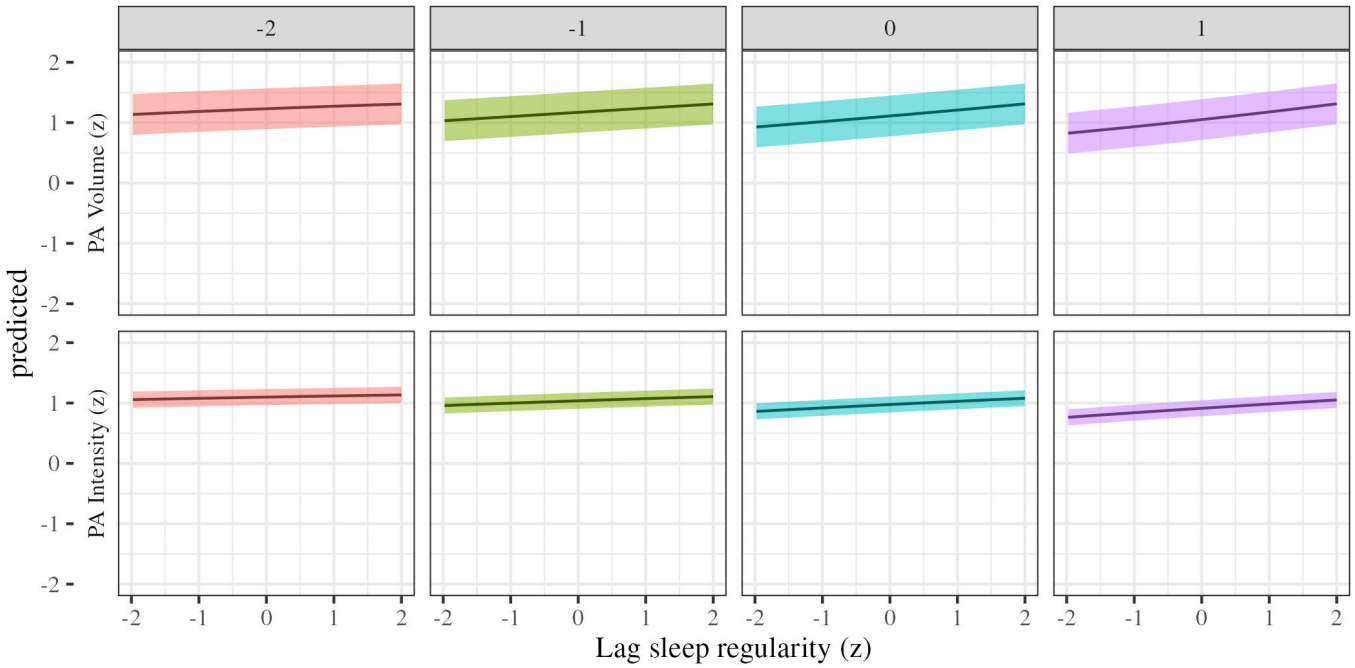

Figure 29. Physical activity by sleep regularity moderated by BMI z-score

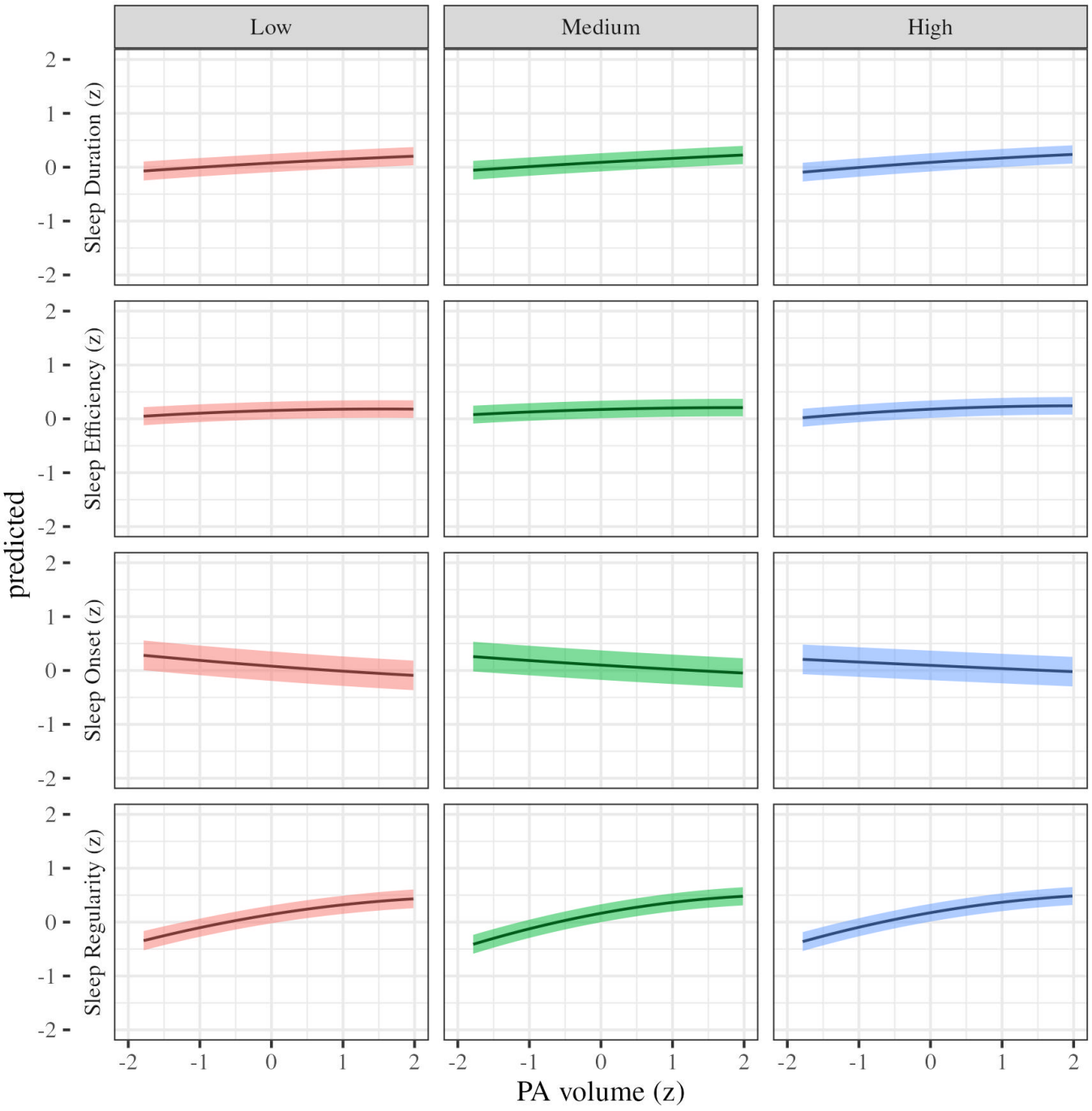

Figure 30. Sleep metrics on Physical activity volume by SES

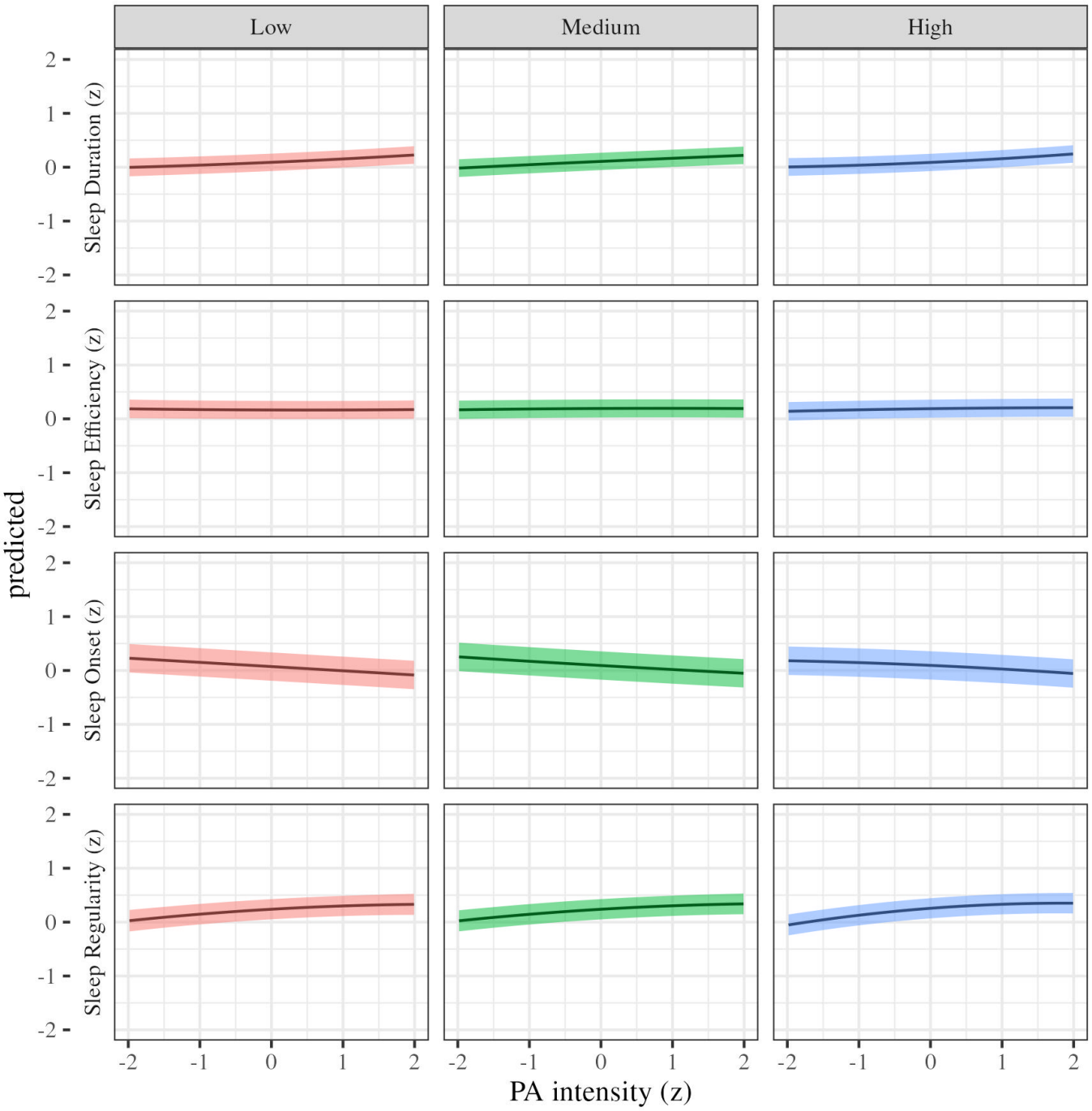

Figure 31. Sleep metrics on Physical activity intensity moderated by SES

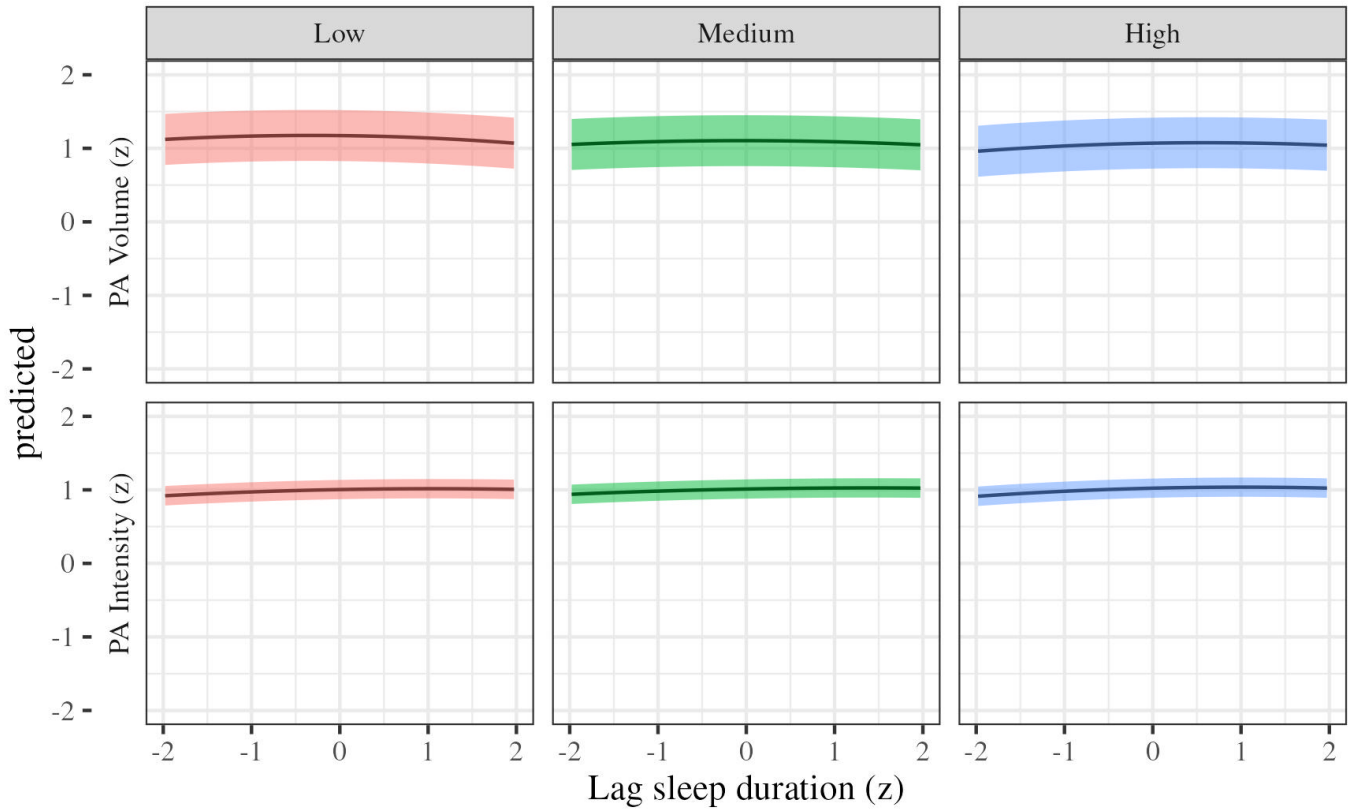

Figure 32. Physical activity by sleep duration moderated by SES

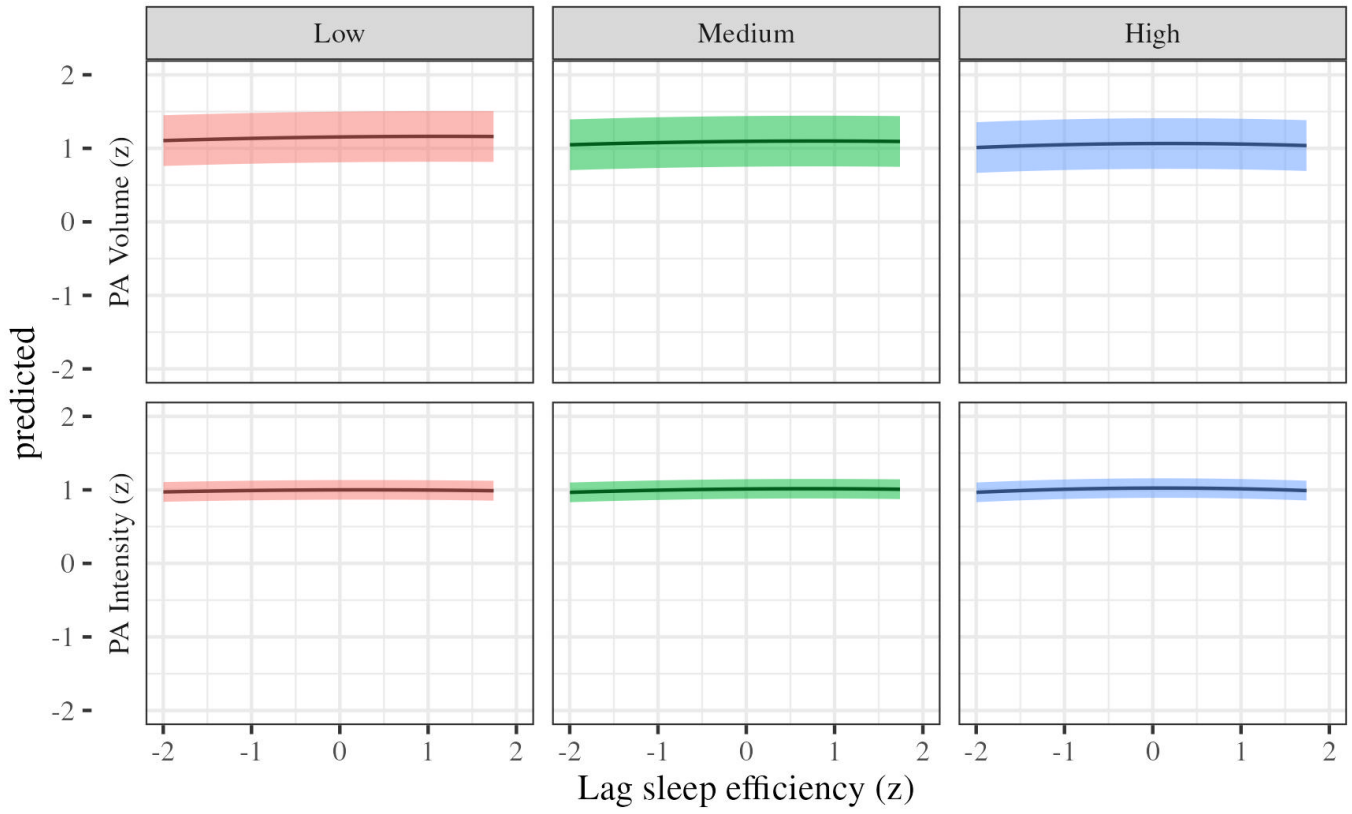

Figure 33. Physical activity by sleep efficiency moderated by SES

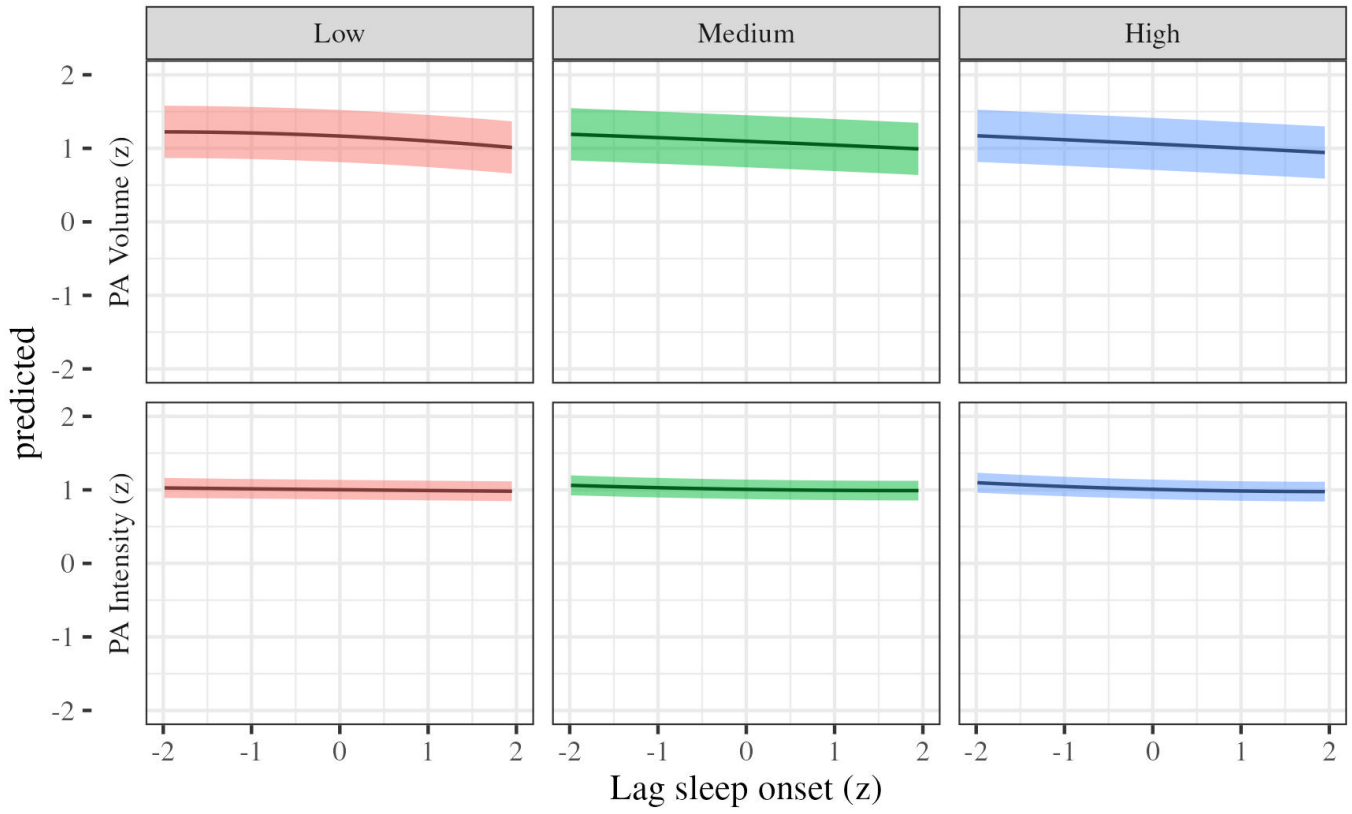

Figure 34. Physical activity by sleep onset moderated by SES

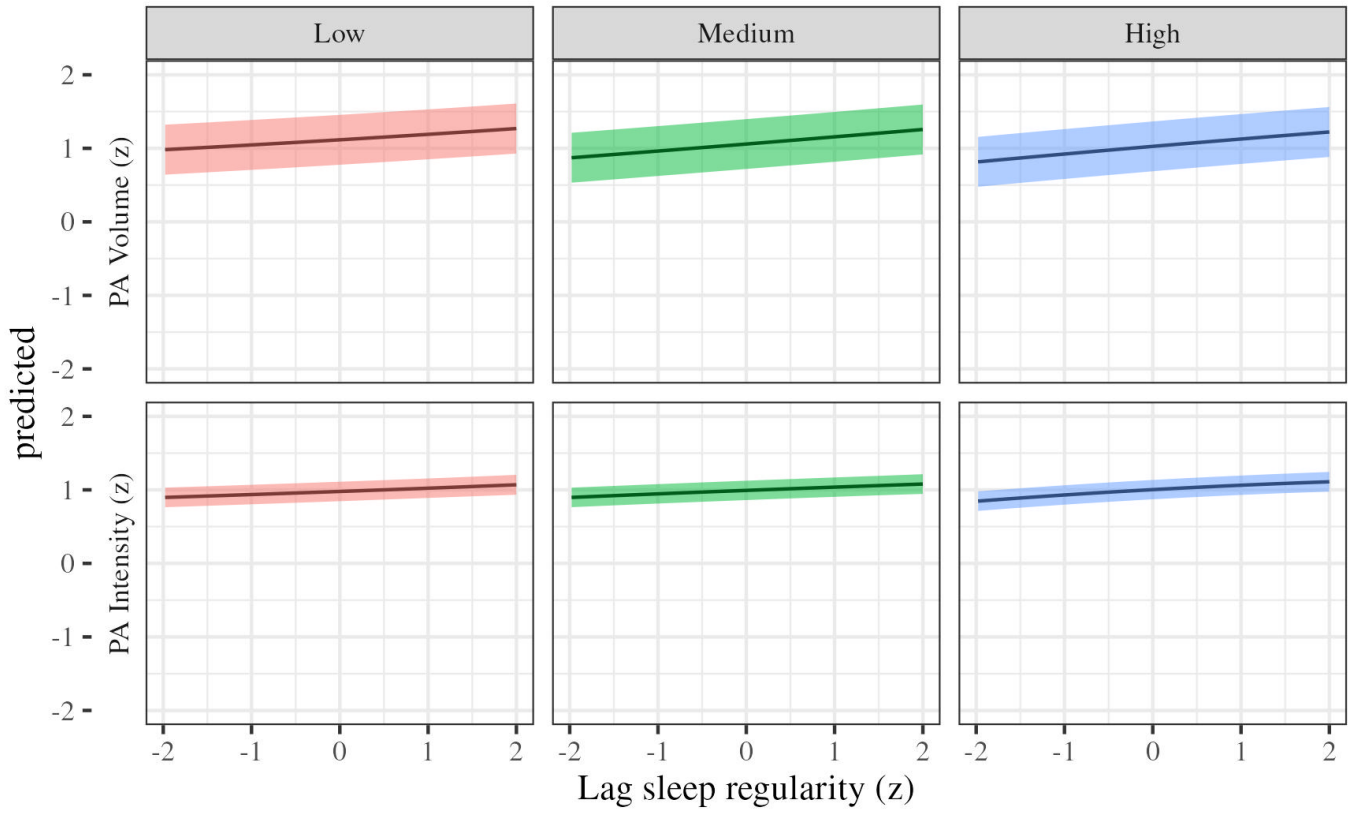

Figure 35. Physical activity by sleep regularity moderated by SES

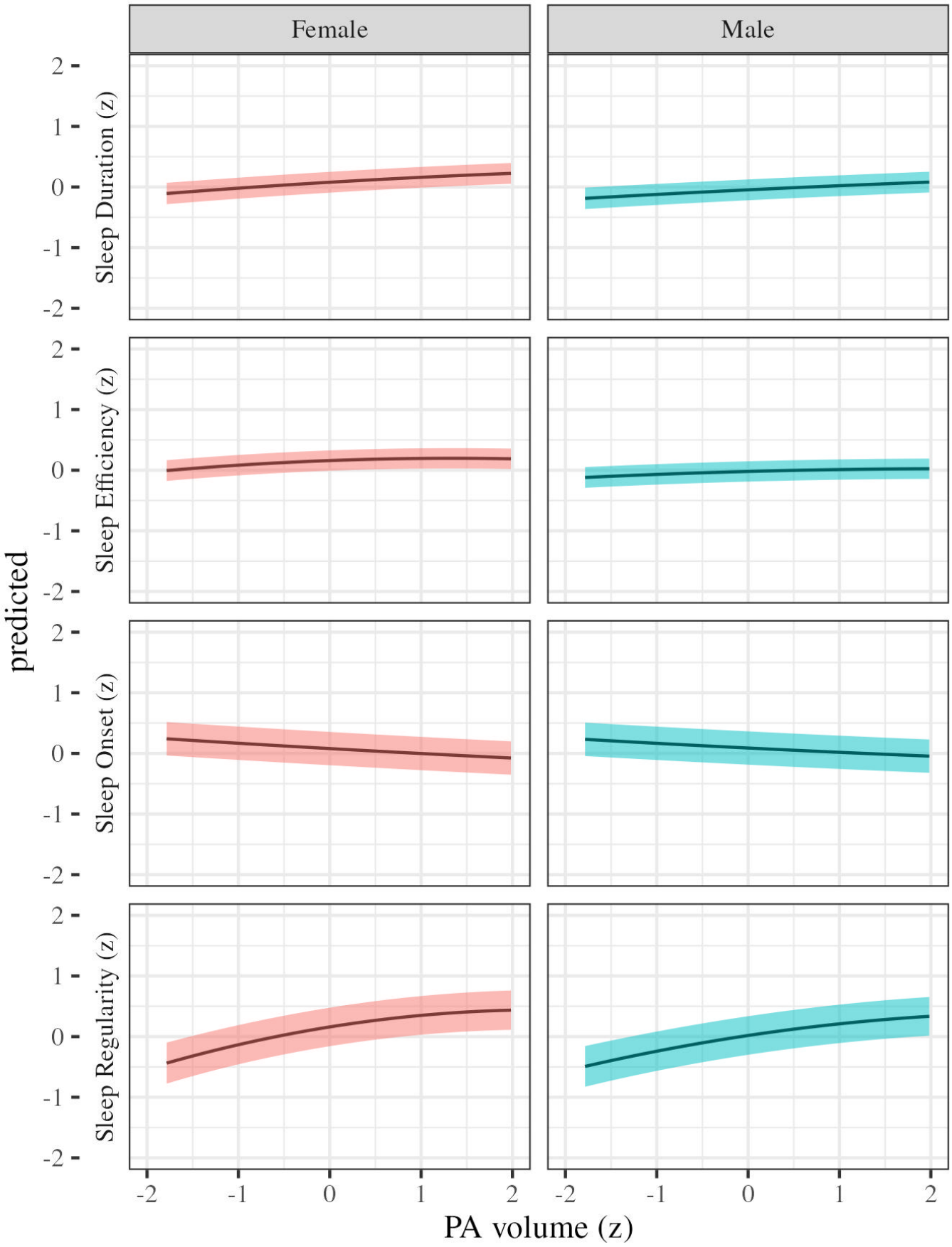

Figure 36. Sleep metrics on Physical activity volume by sex

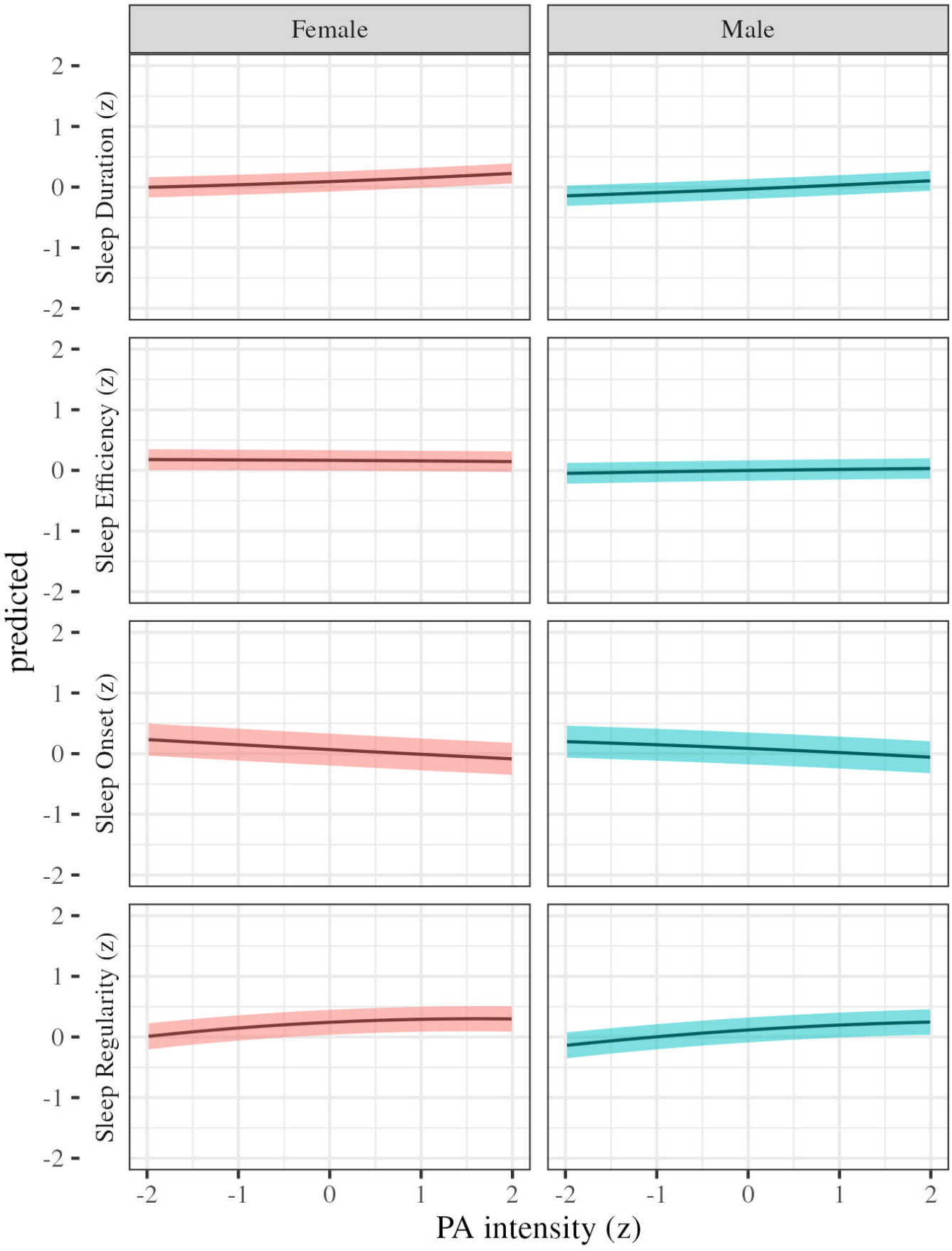

Figure 37. Sleep metrics on Physical activity intensity moderated by sex

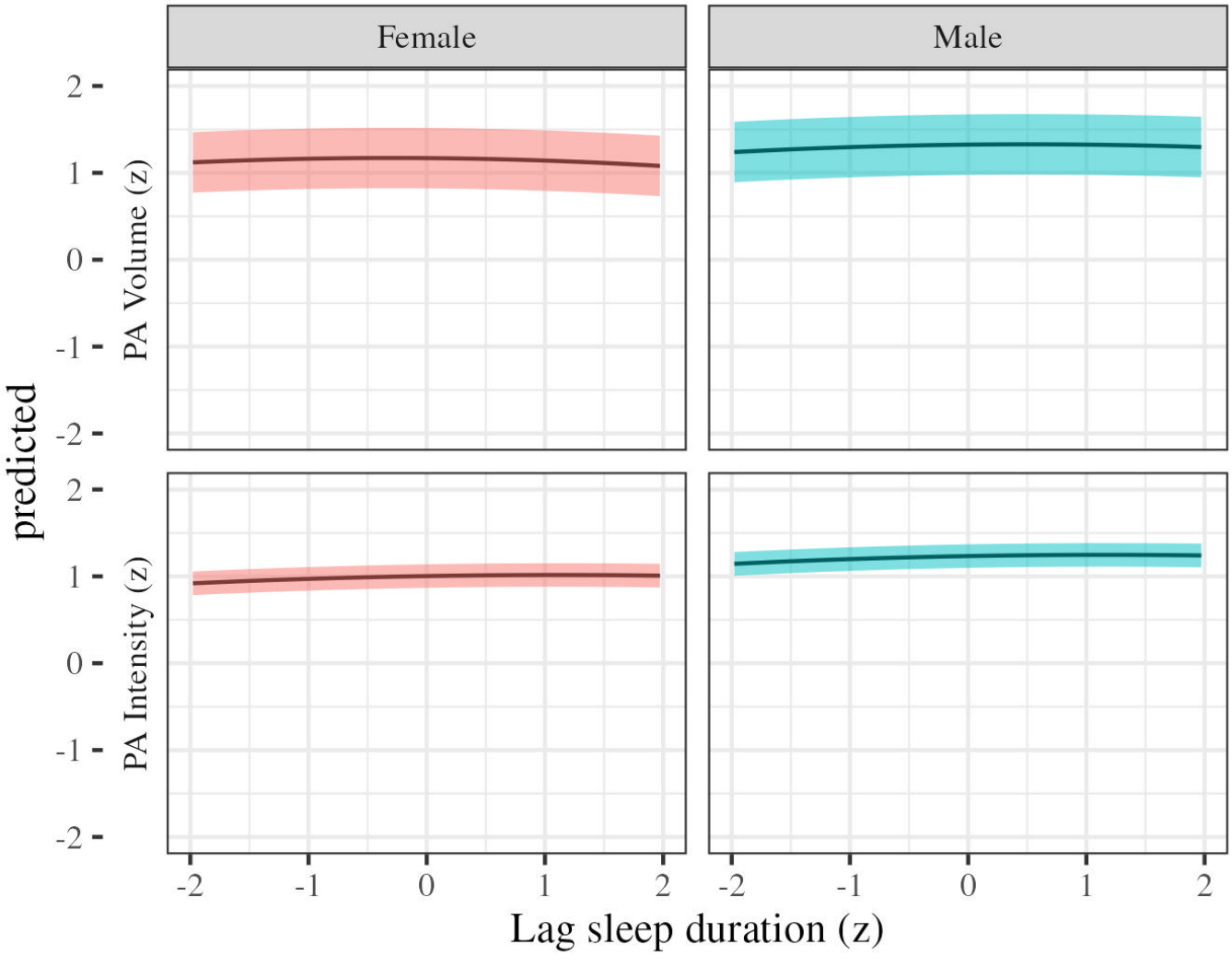

Figure 38. Physical activity by sleep duration moderated by sex

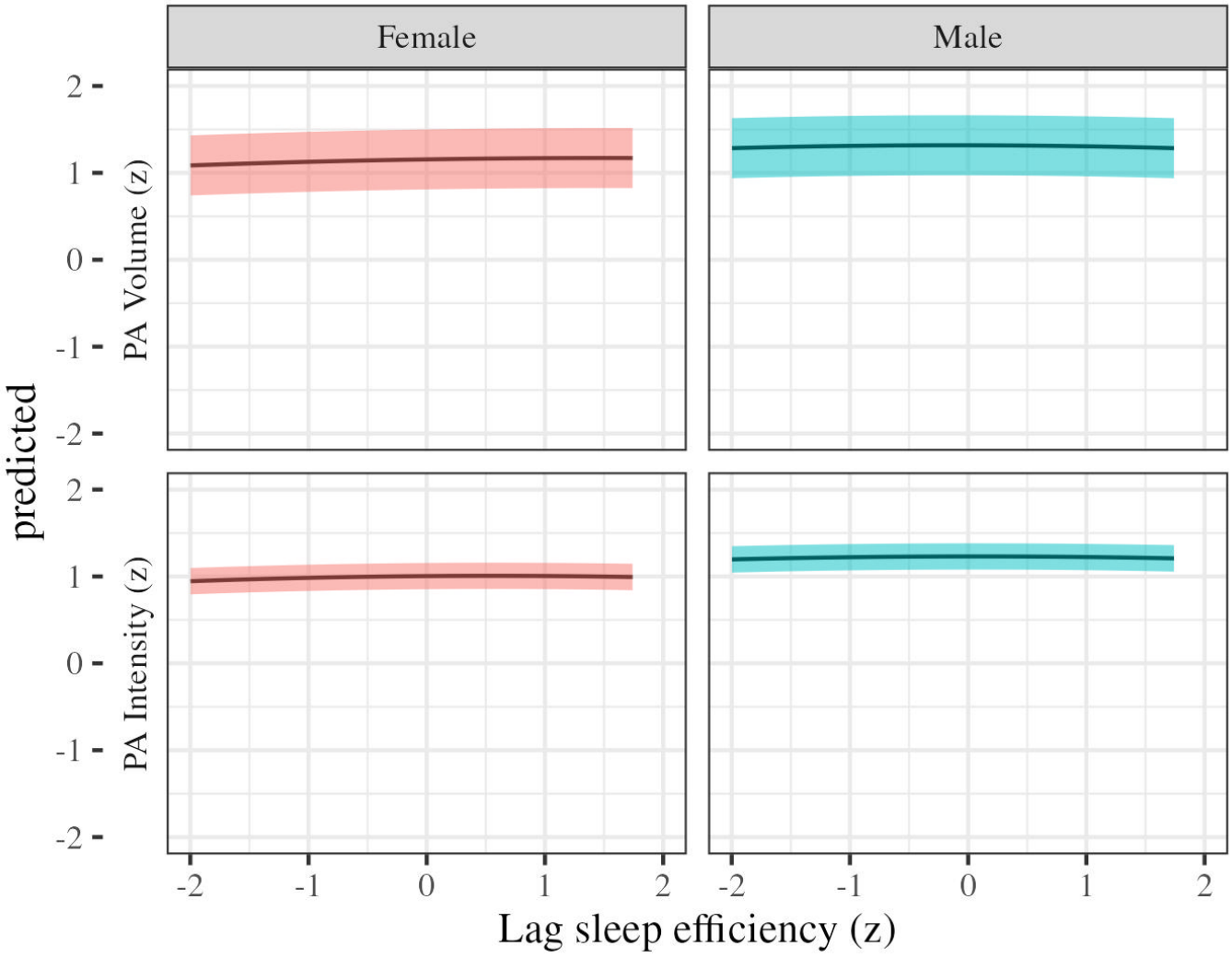

Figure 39. Physical activity by sleep efficiency moderated by sex

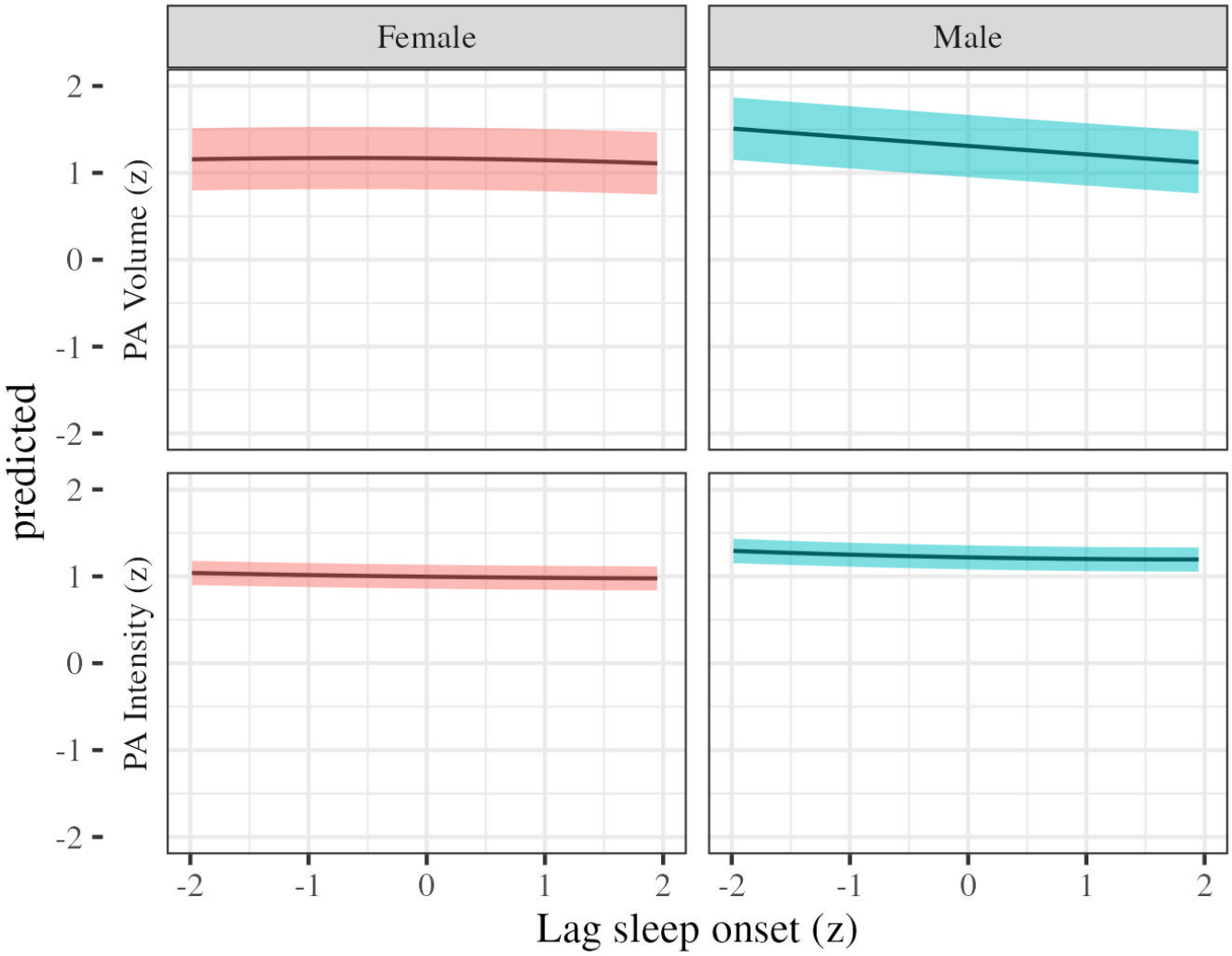

Figure 40. Physical activity by sleep onset moderated by sex

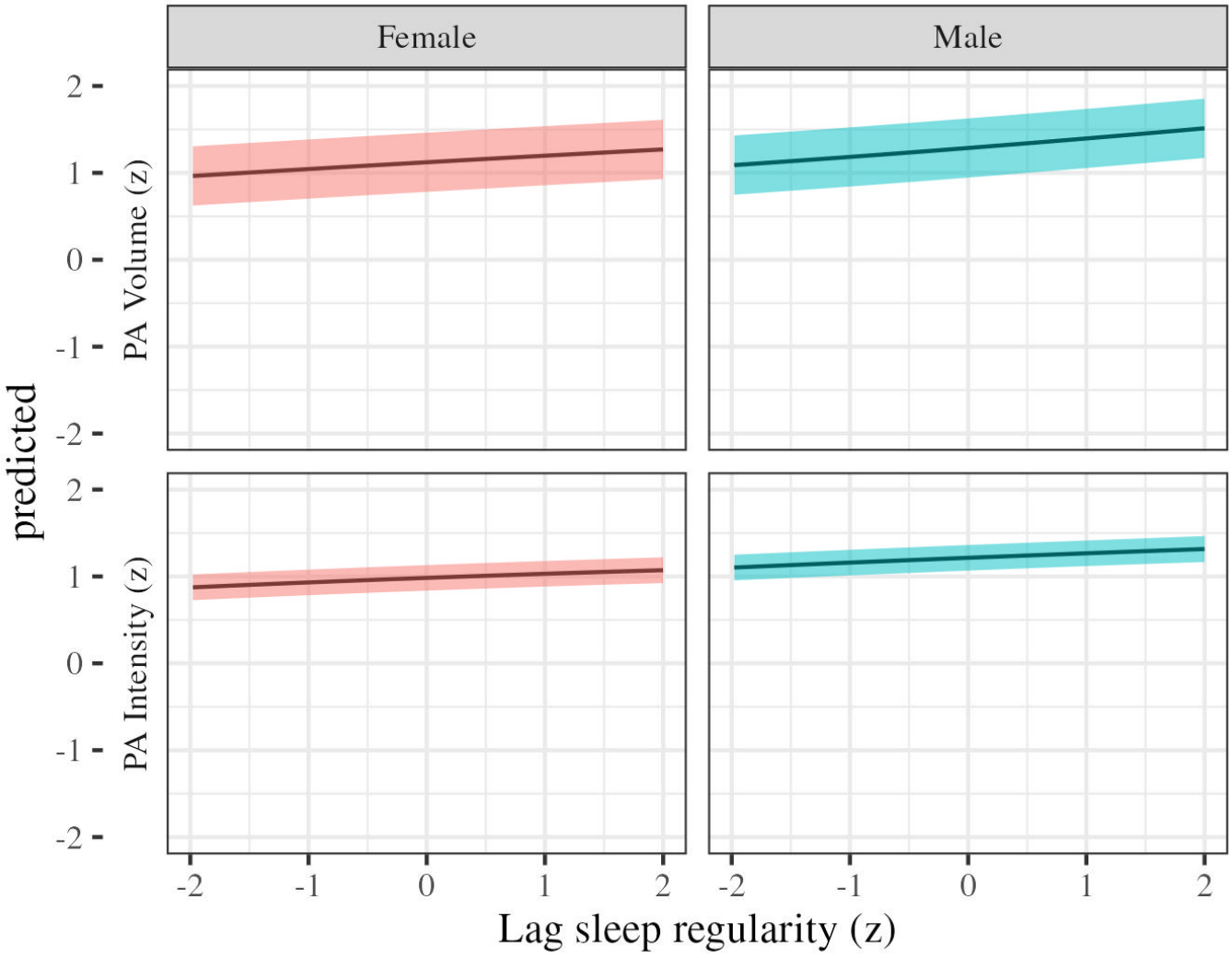

Figure 41. Physical activity by sleep regularity moderated by sex

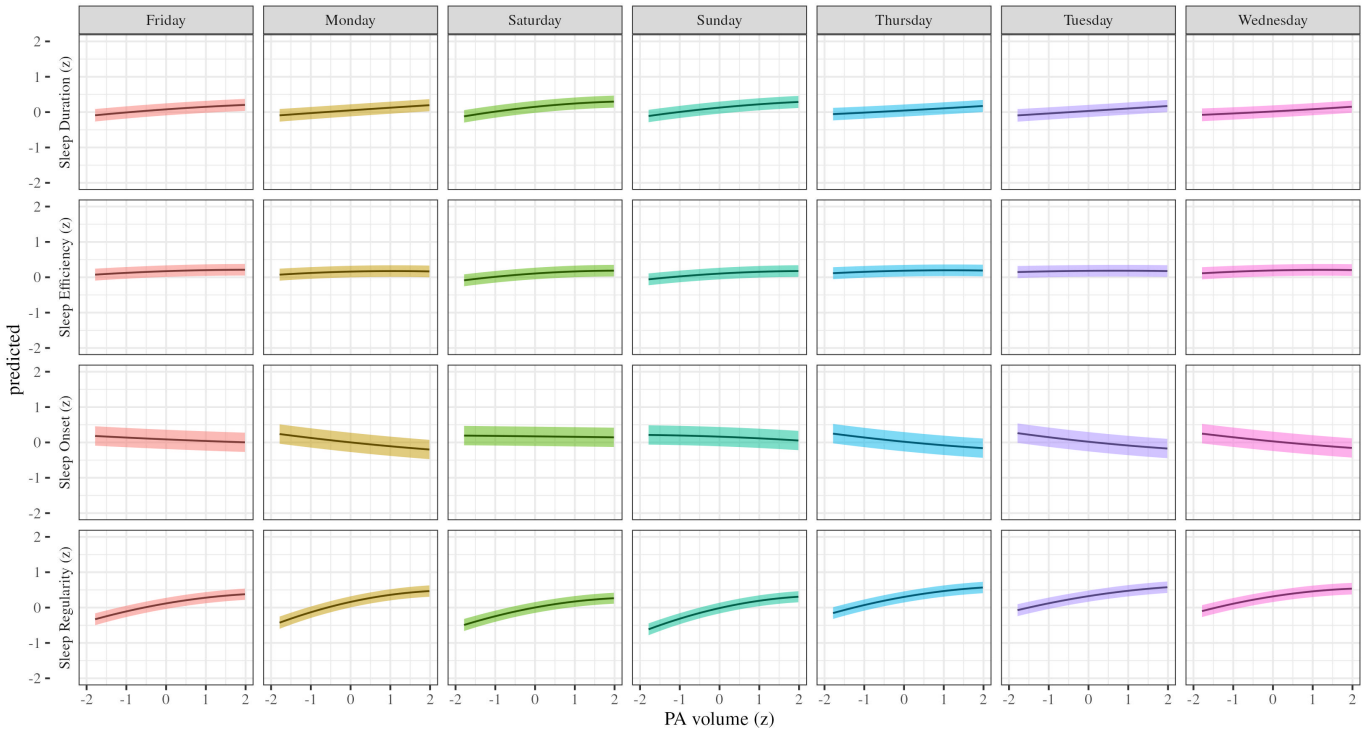

Figure 42. Sleep metrics on Physical activity volume by weekday

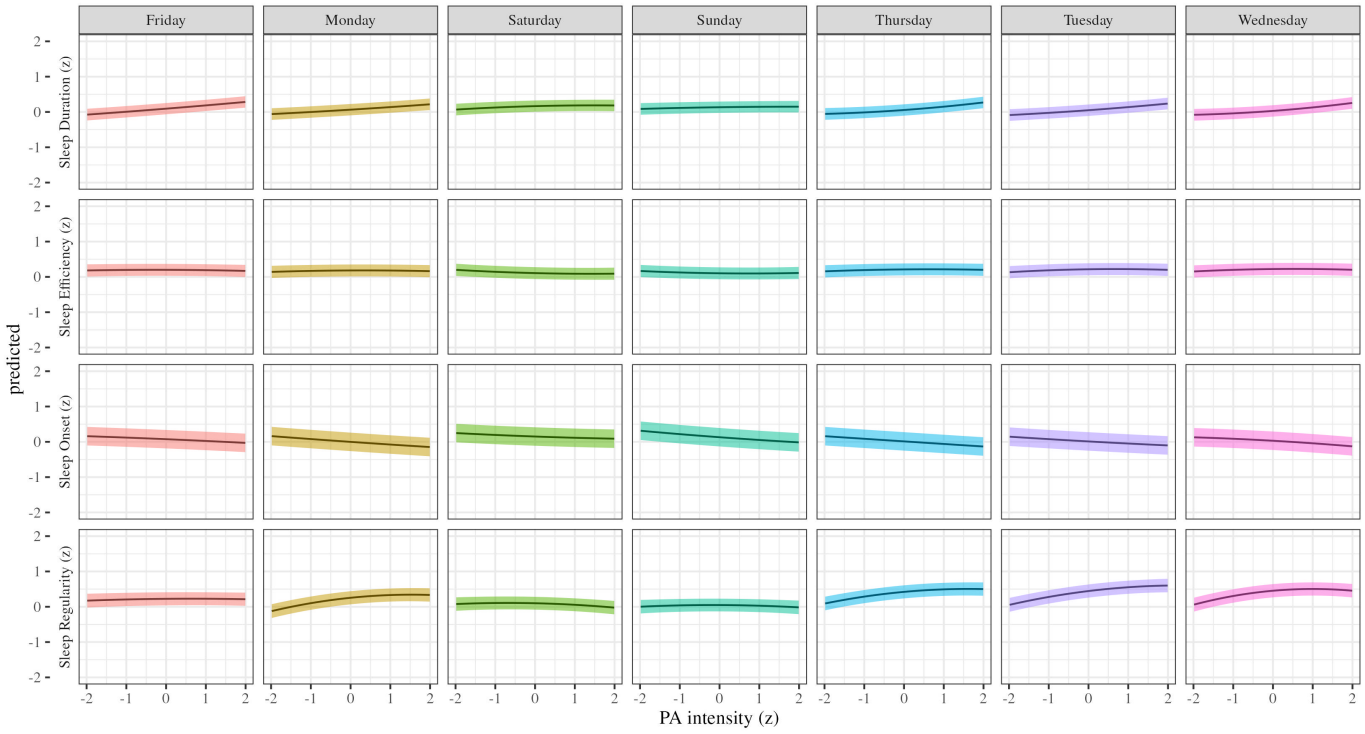

Figure 43. Sleep metrics on Physical activity intensity moderated by weekday

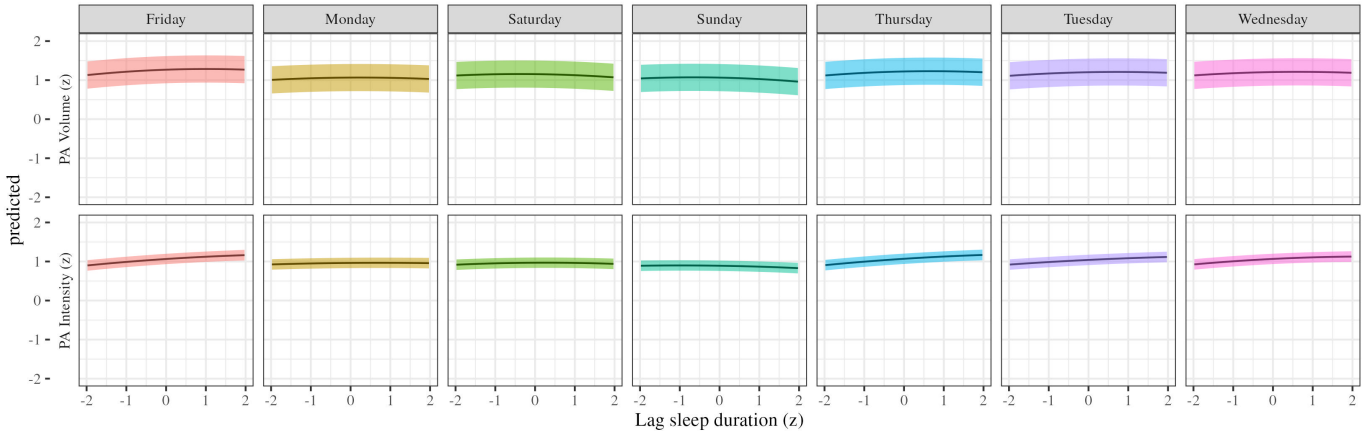

Figure 44. Physical activity by sleep duration moderated by weekday

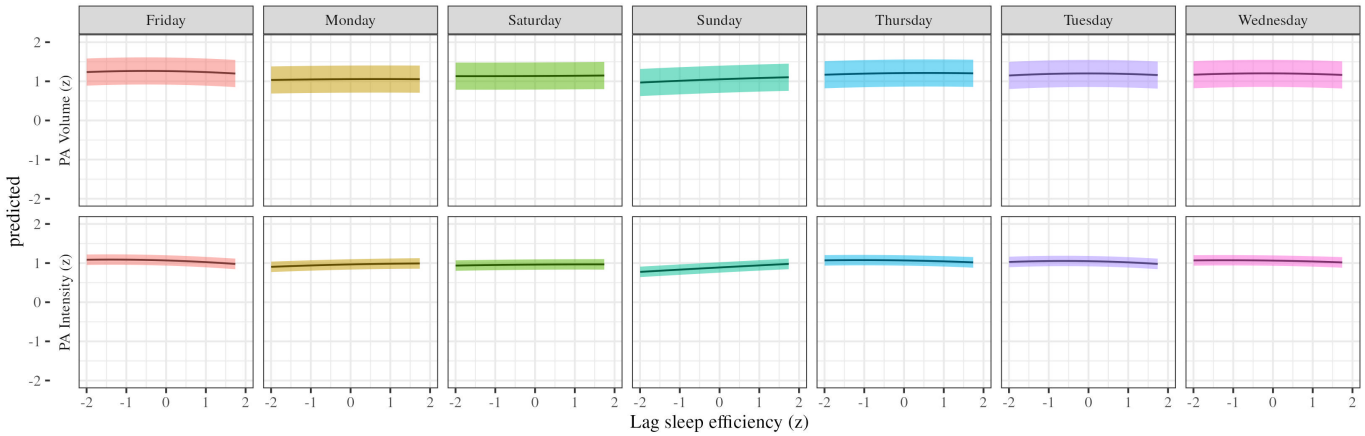

Figure 45. Physical activity by sleep efficiency moderated by weekday

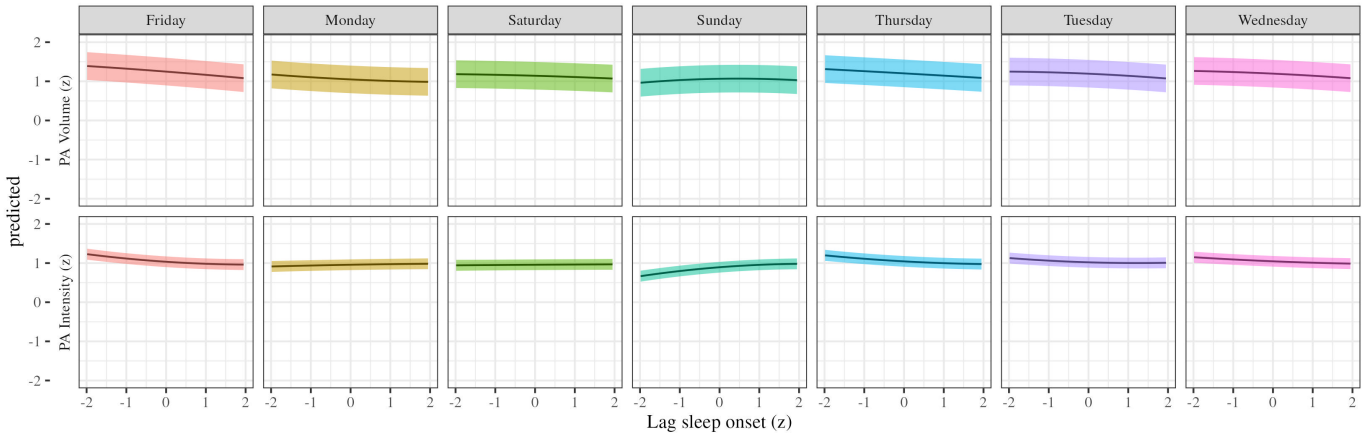

Figure 46. Physical activity by sleep onset moderated by weekday

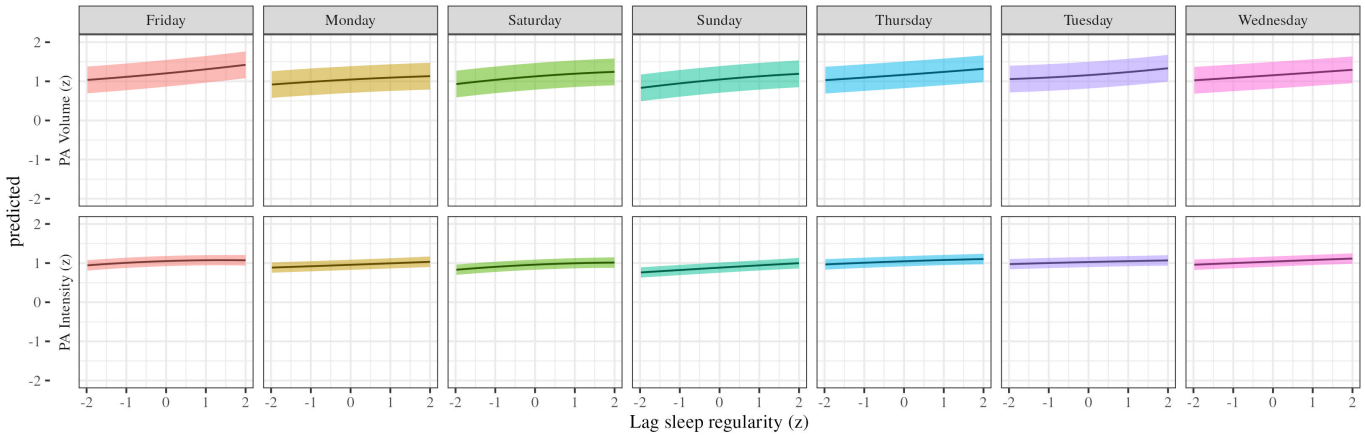

Figure 47. Physical activity by sleep regularity moderated by weekday

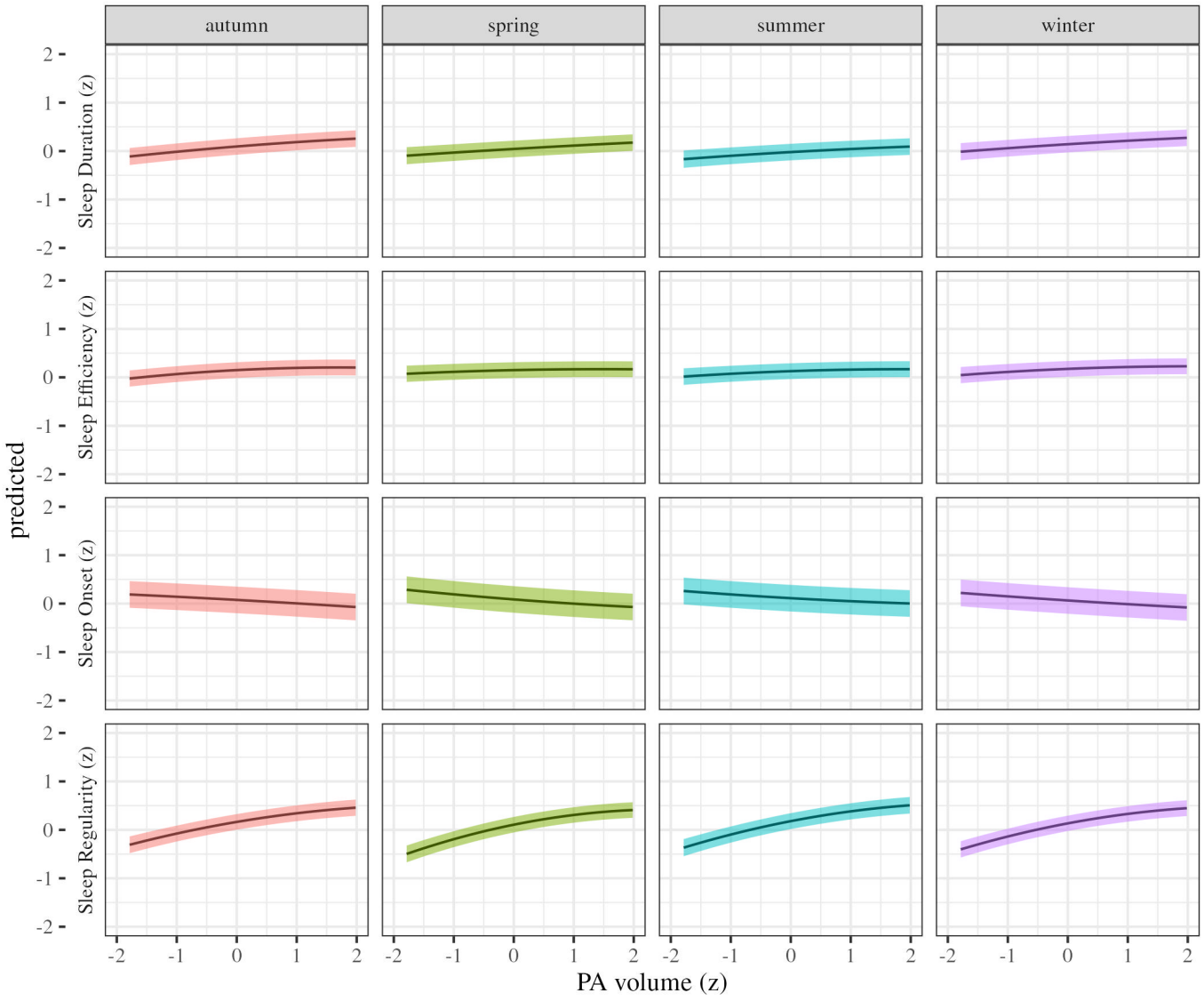

Figure 48. Sleep metrics on Physical activity volume by season

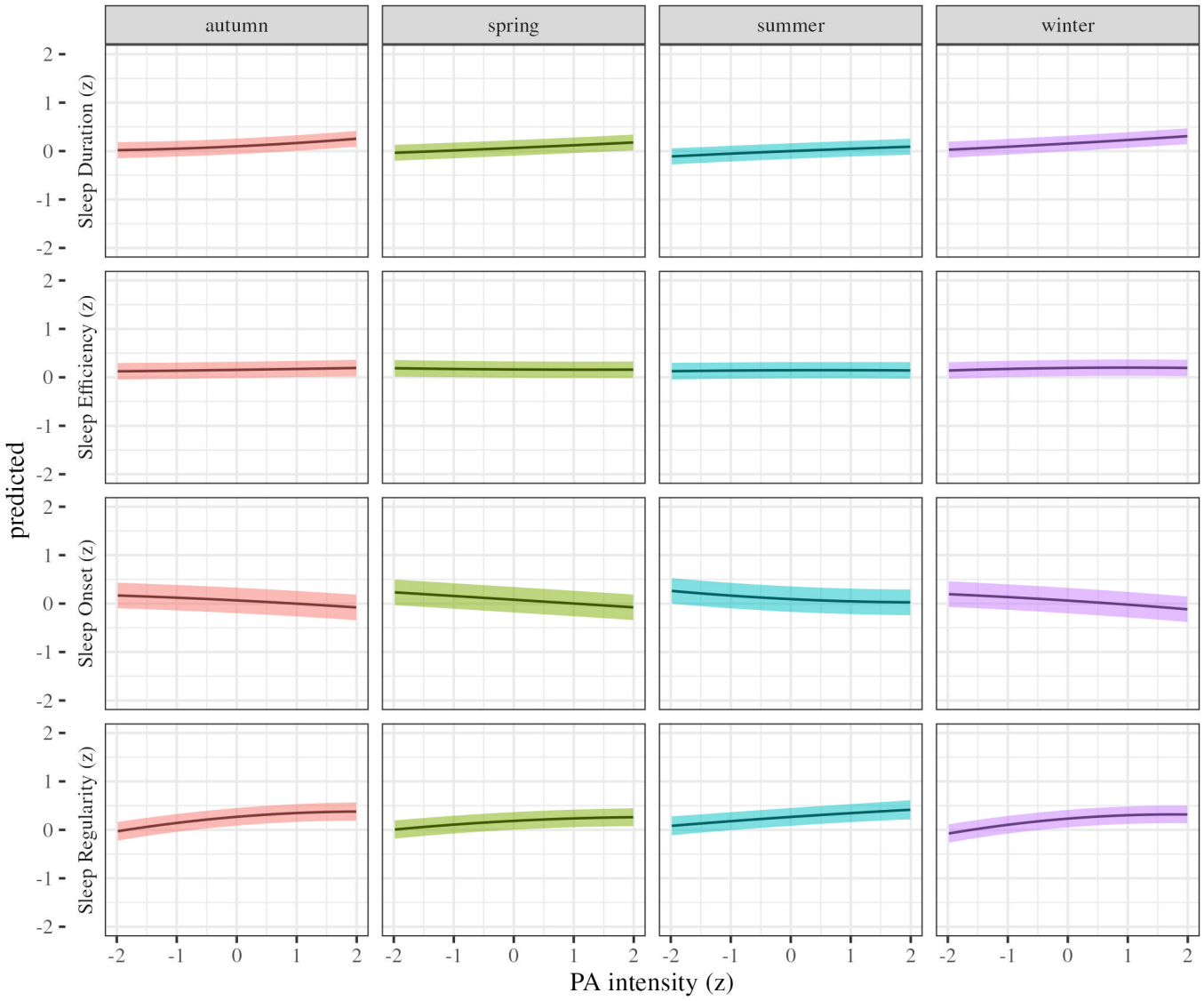

Figure 49. Sleep metrics on Physical activity intensity moderated by season

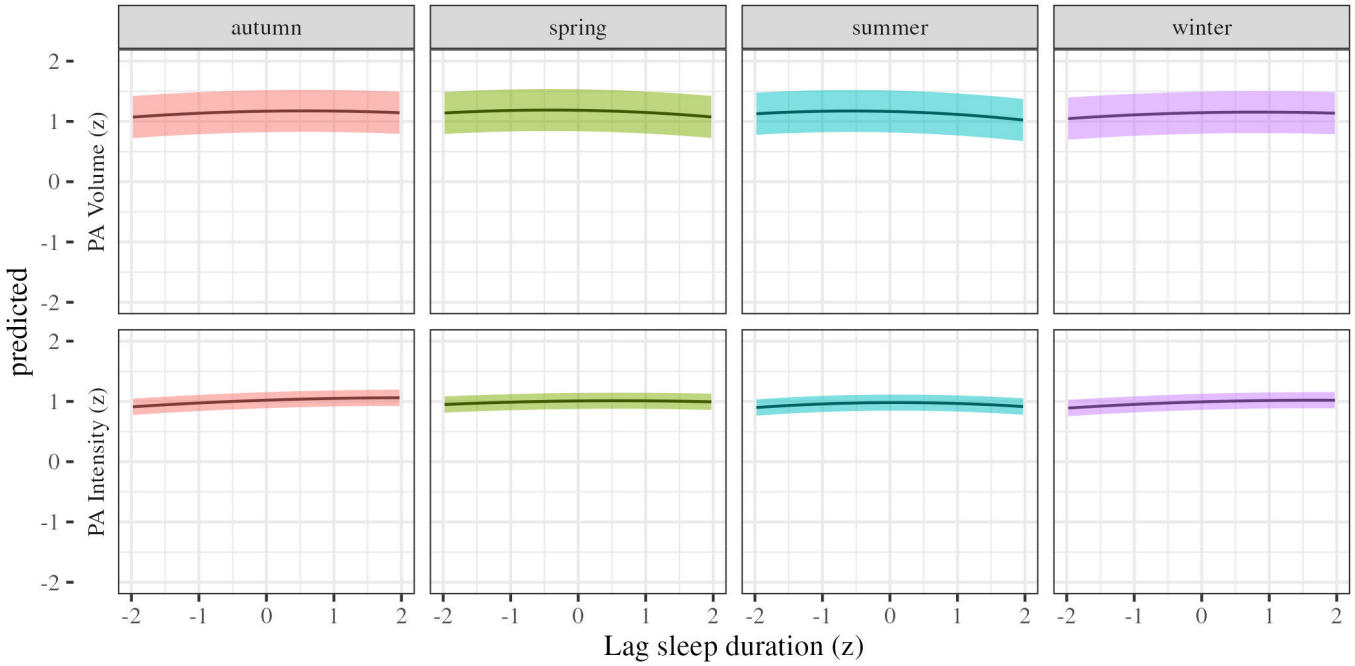

Figure 50. Physical activity by sleep duration moderated by season

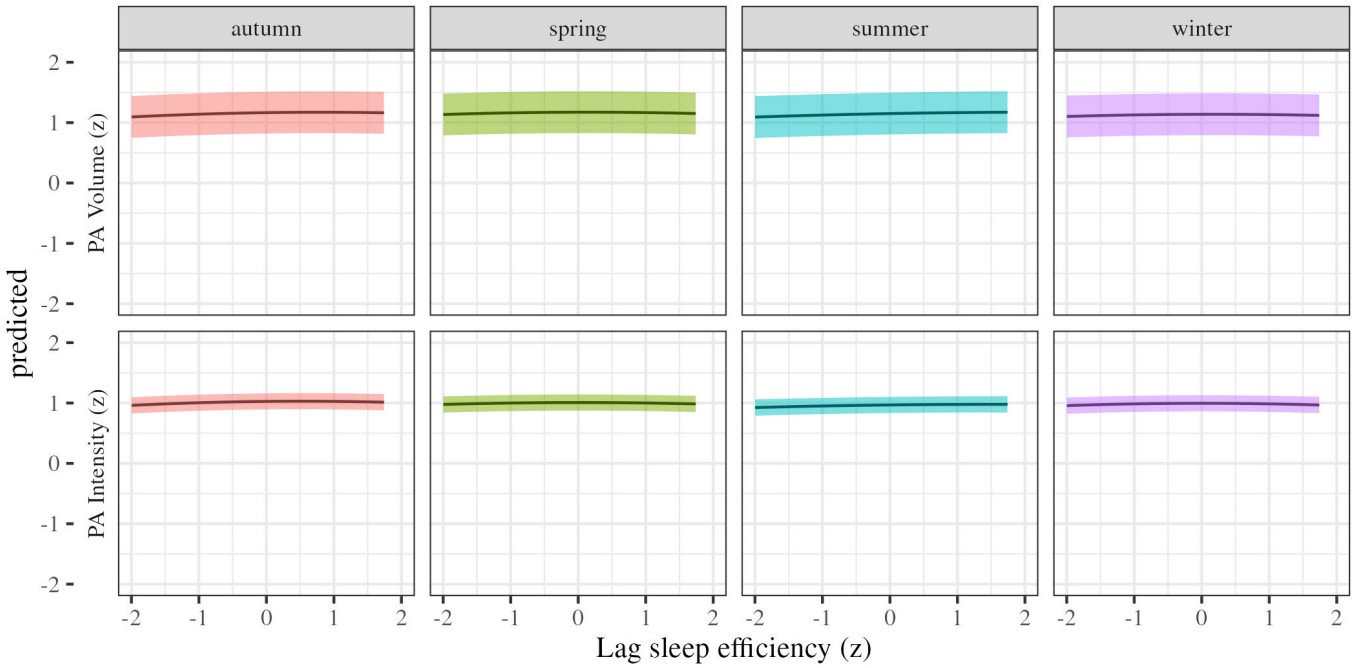

Figure 51. Physical activity by sleep efficiency moderated by season

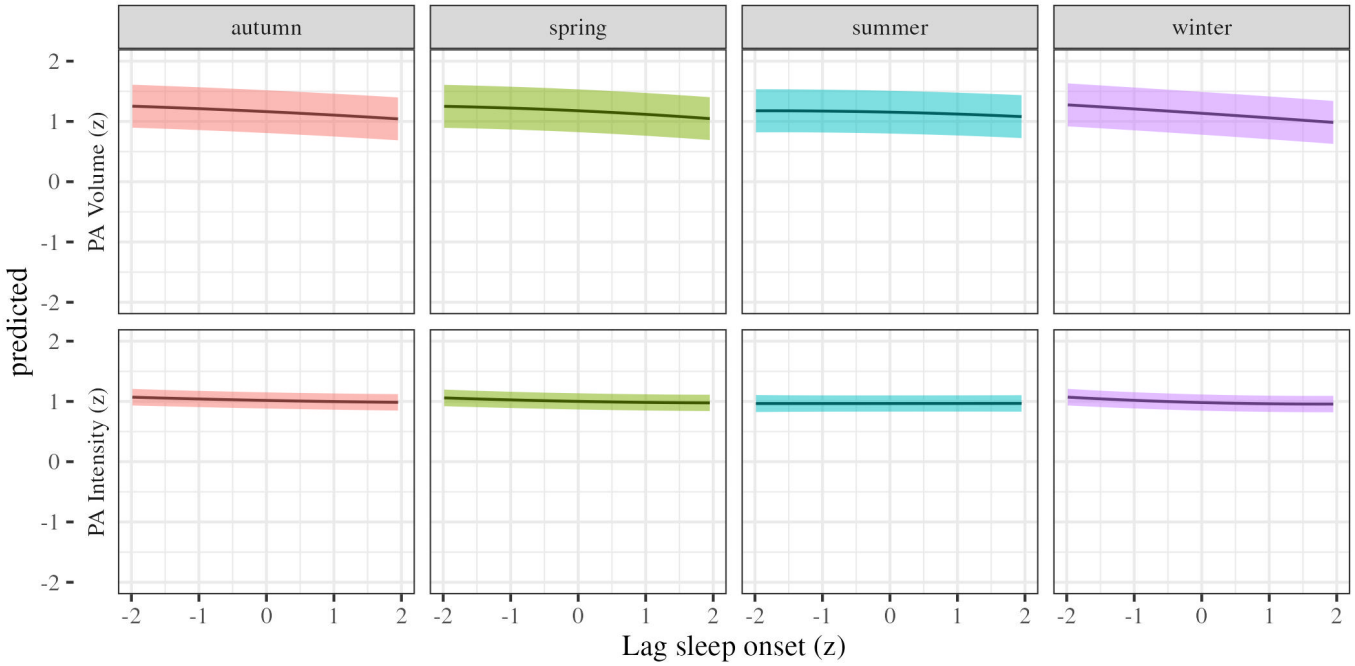

Figure 52. Physical activity by sleep onset moderated by season

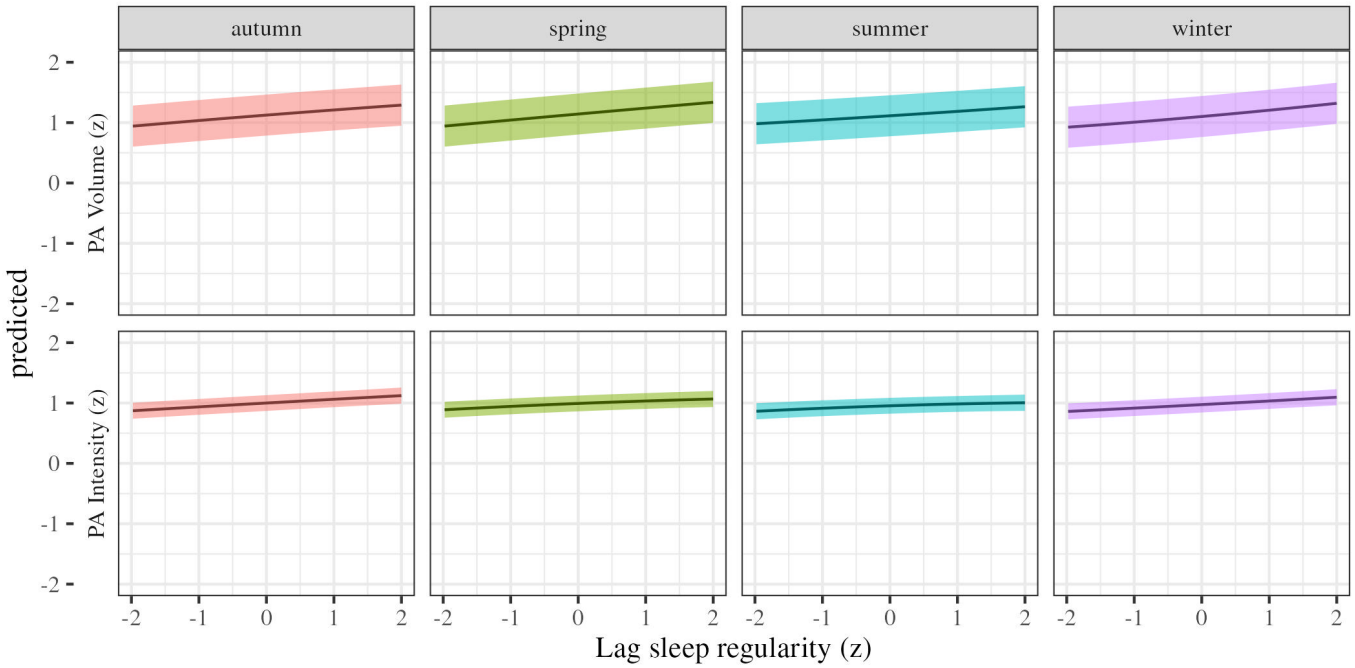

Figure 53. Physical activity by sleep regularity moderated by season

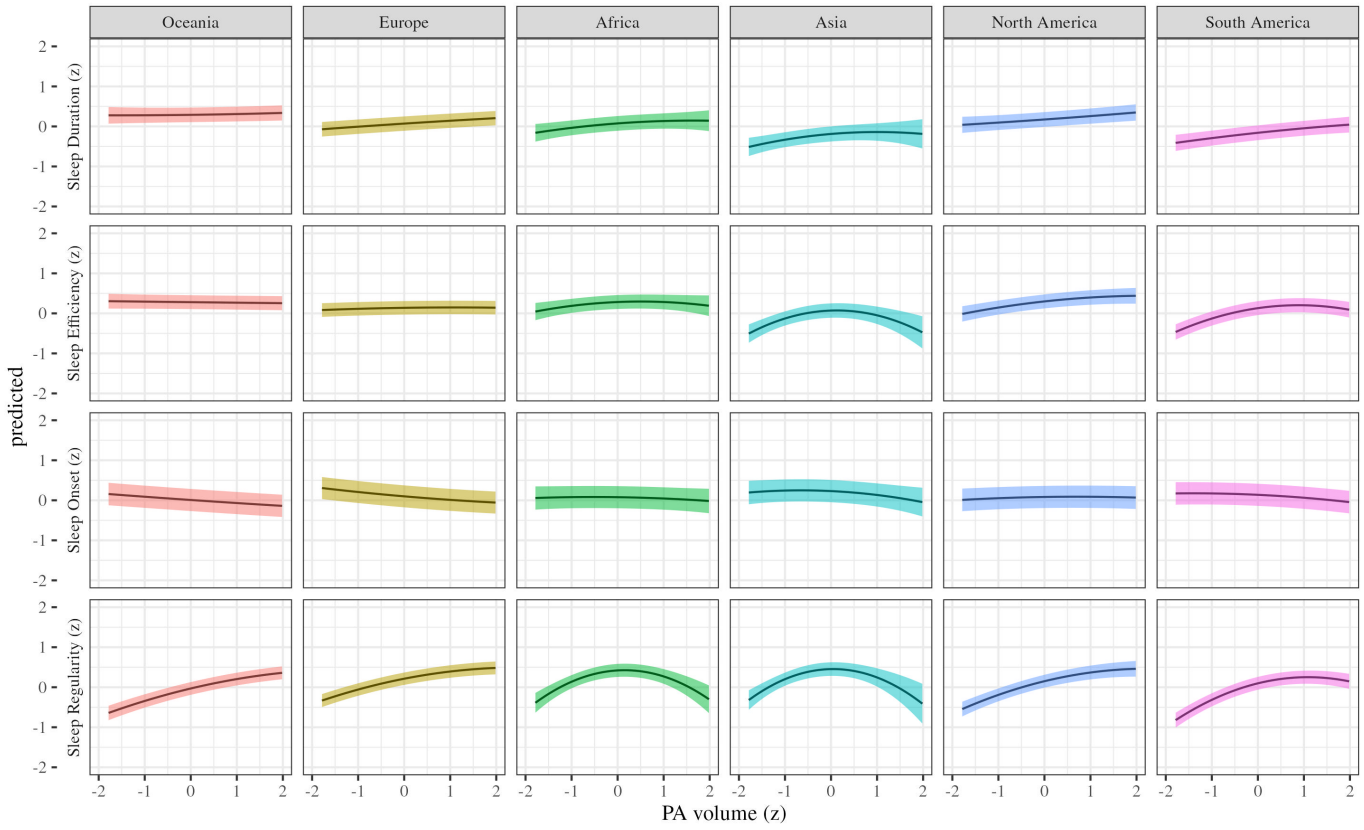

Figure 54. Sleep metrics on Physical activity volume by region

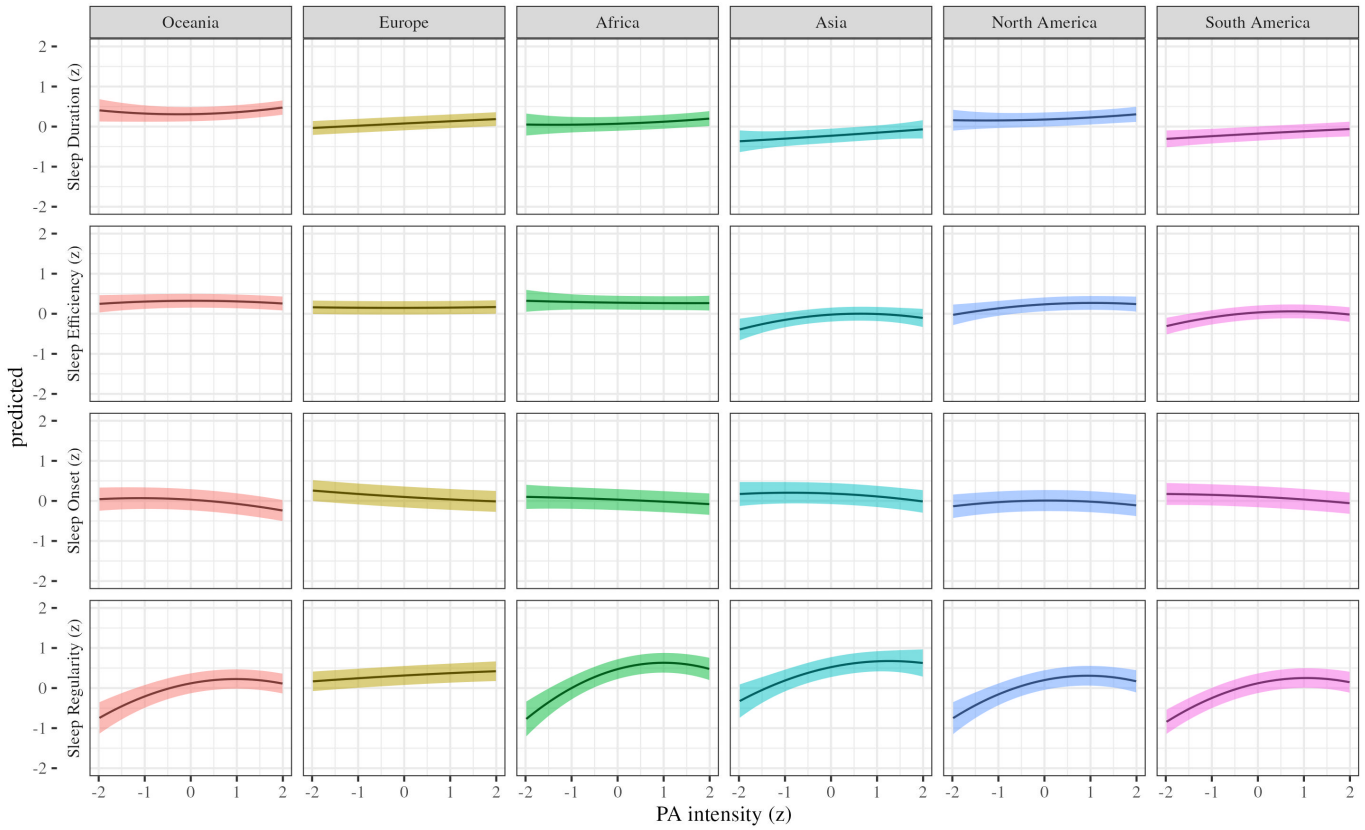

Figure 55. Sleep metrics on Physical activity intensity moderated by region

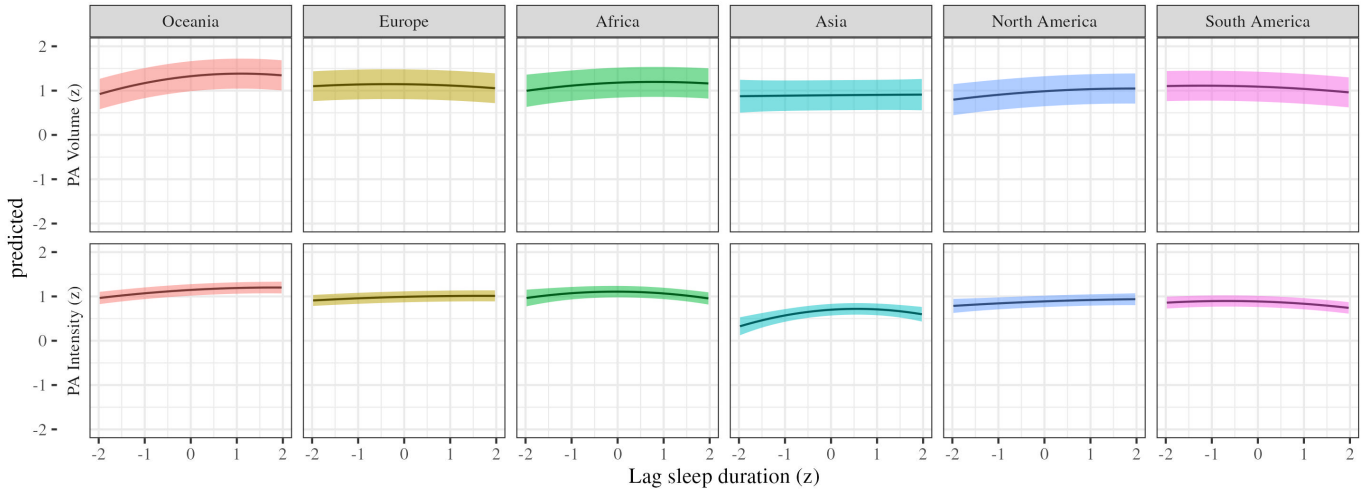

Figure 56. Physical activity by sleep duration moderated by region

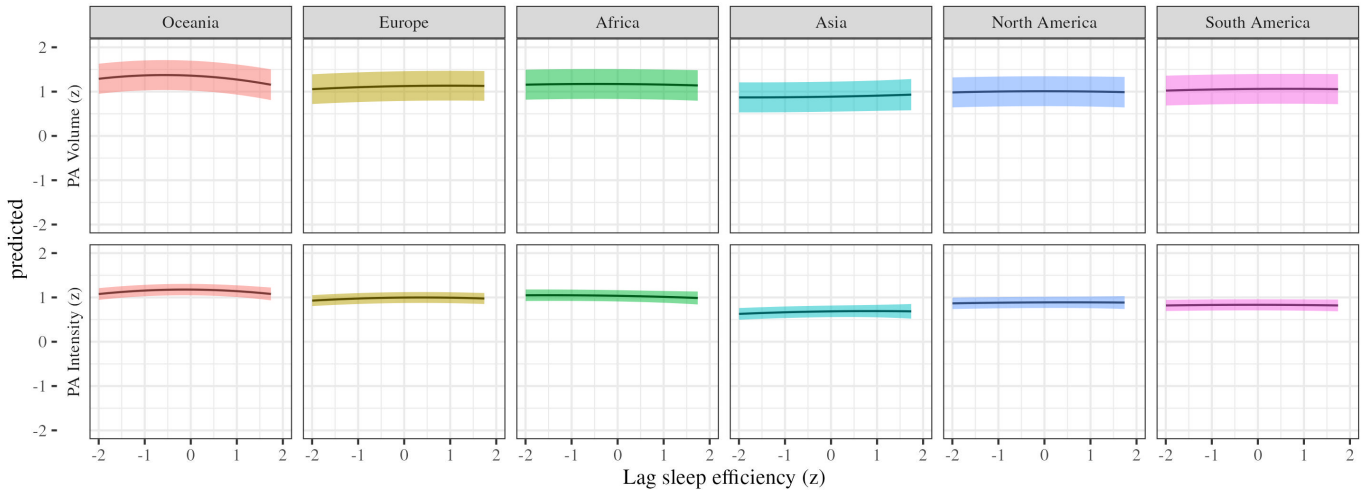

Figure 57. Physical activity by sleep efficiency moderated by region

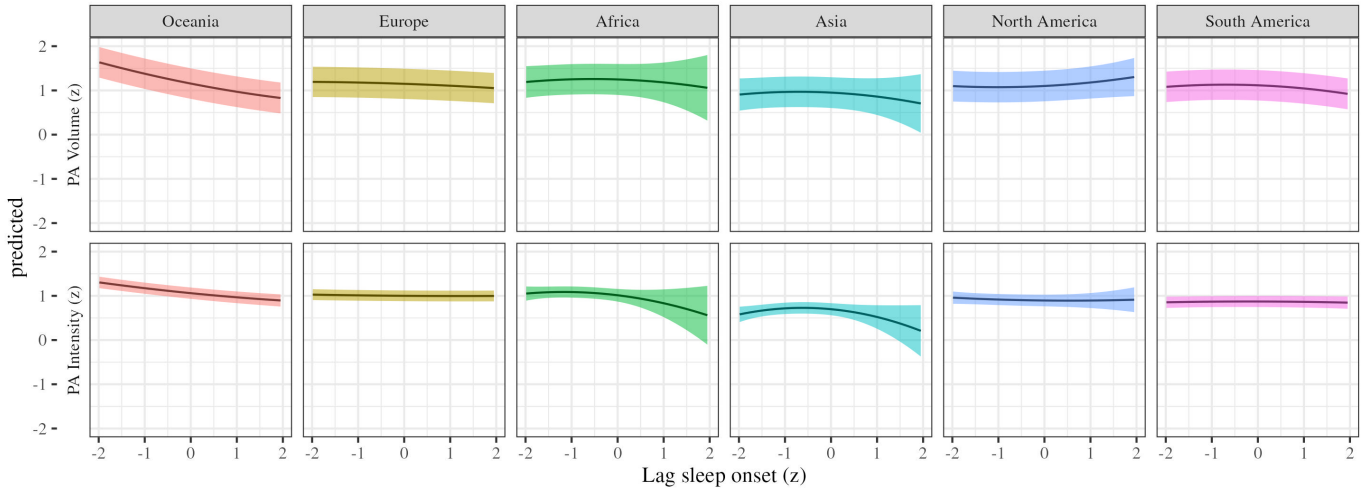

Figure 58. Physical activity by sleep onset moderated by region

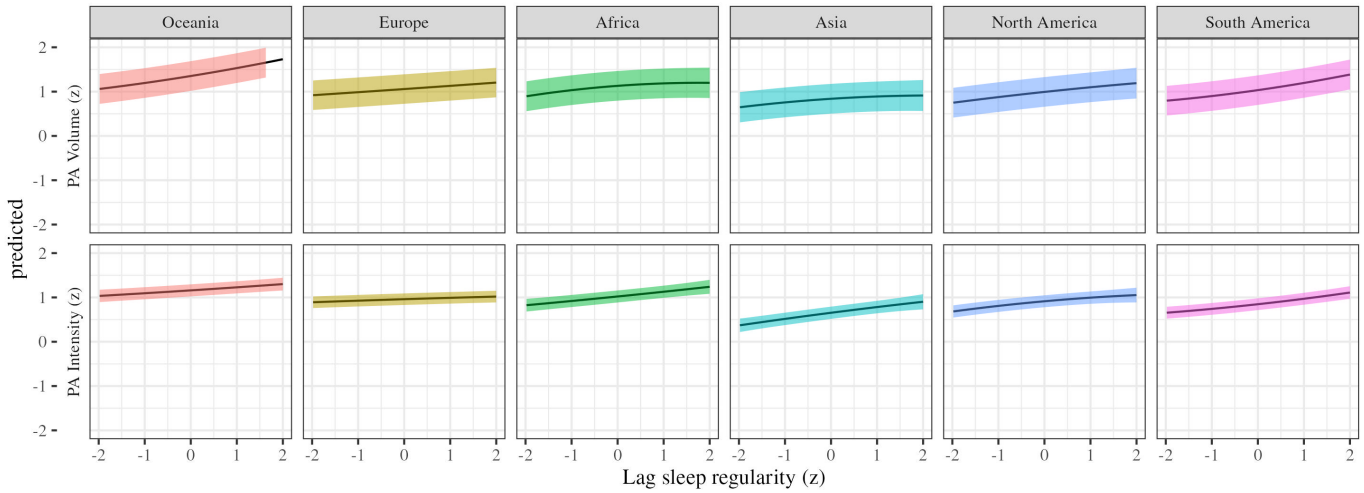

Figure 59. Physical activity by sleep regularity moderated by region

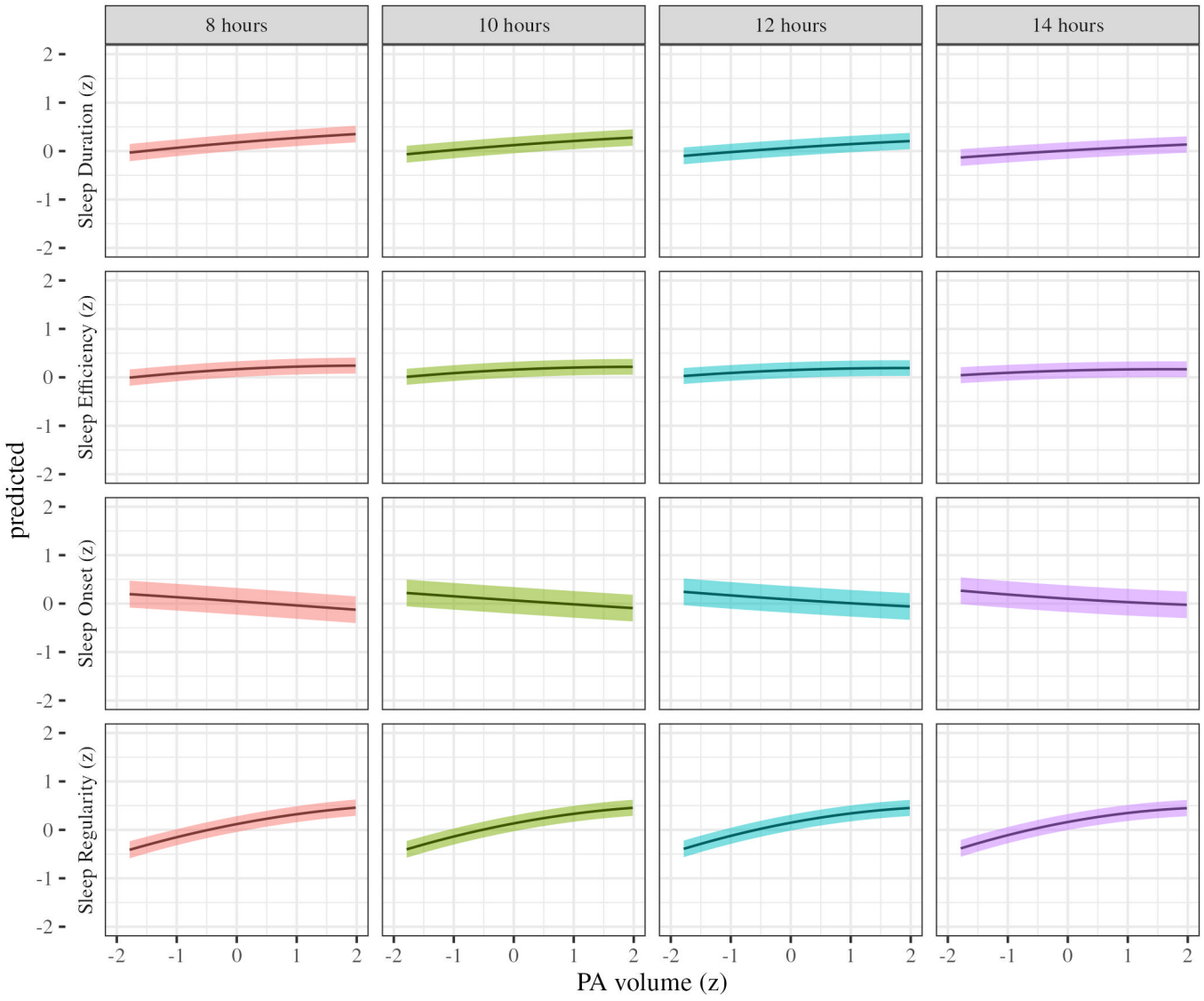

Figure 60. Sleep metrics on Physical activity volume by daylight hours

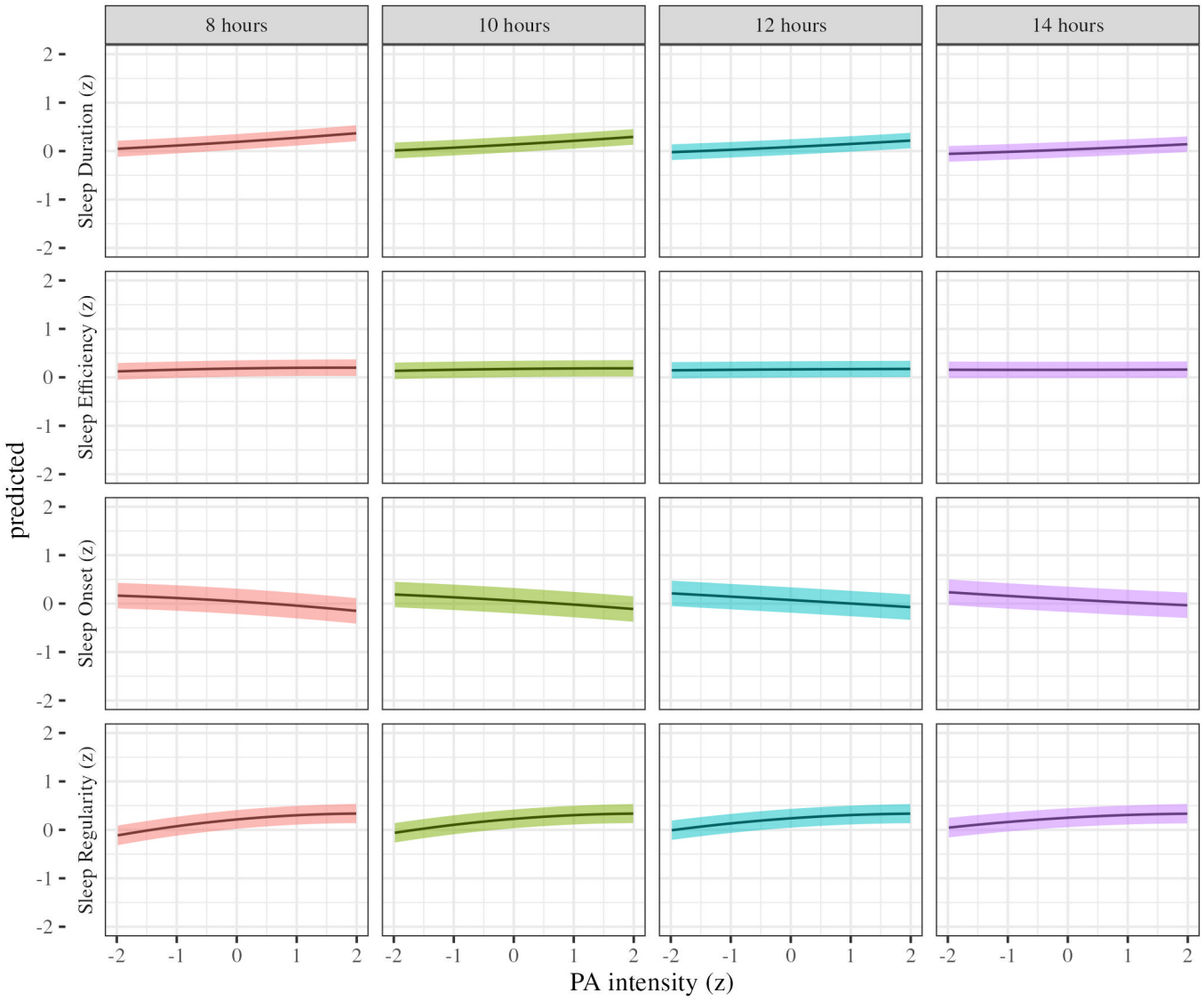

Figure 61. Sleep metrics on Physical activity intensity moderated by daylight hours

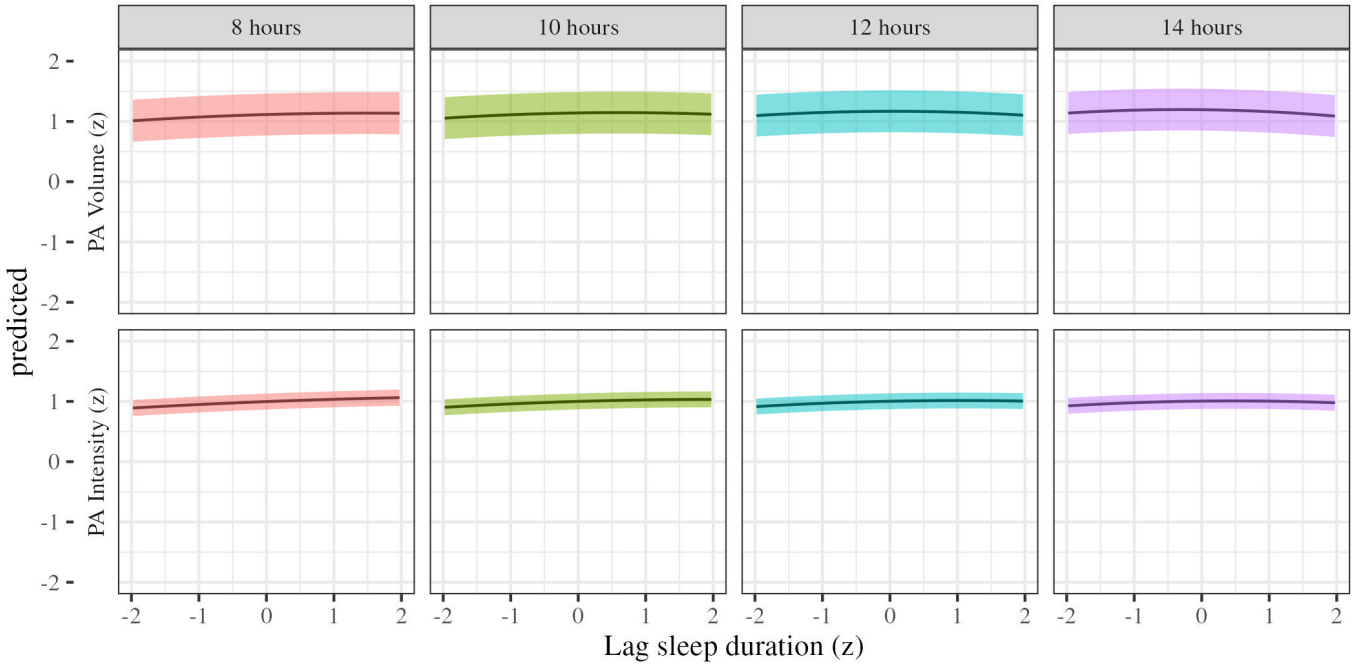

Figure 62. Physical activity by sleep duration moderated by daylight hours

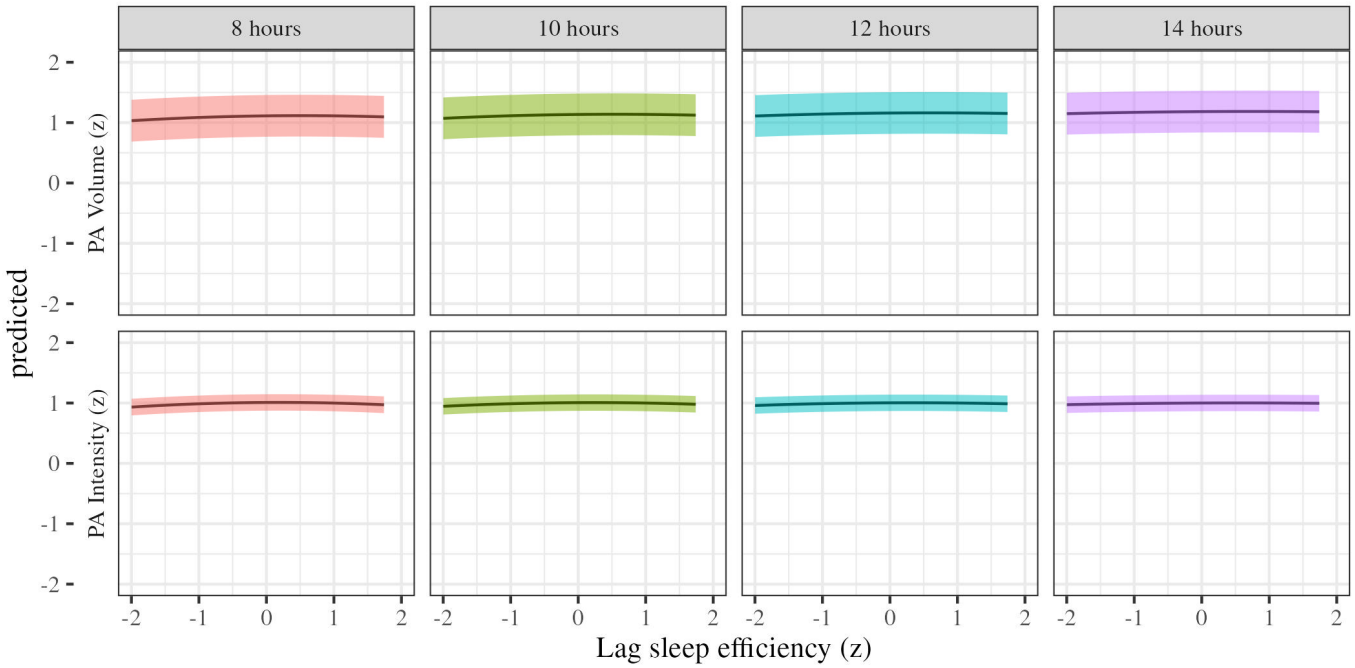

Figure 63. Physical activity by sleep efficiency moderated by daylight hours

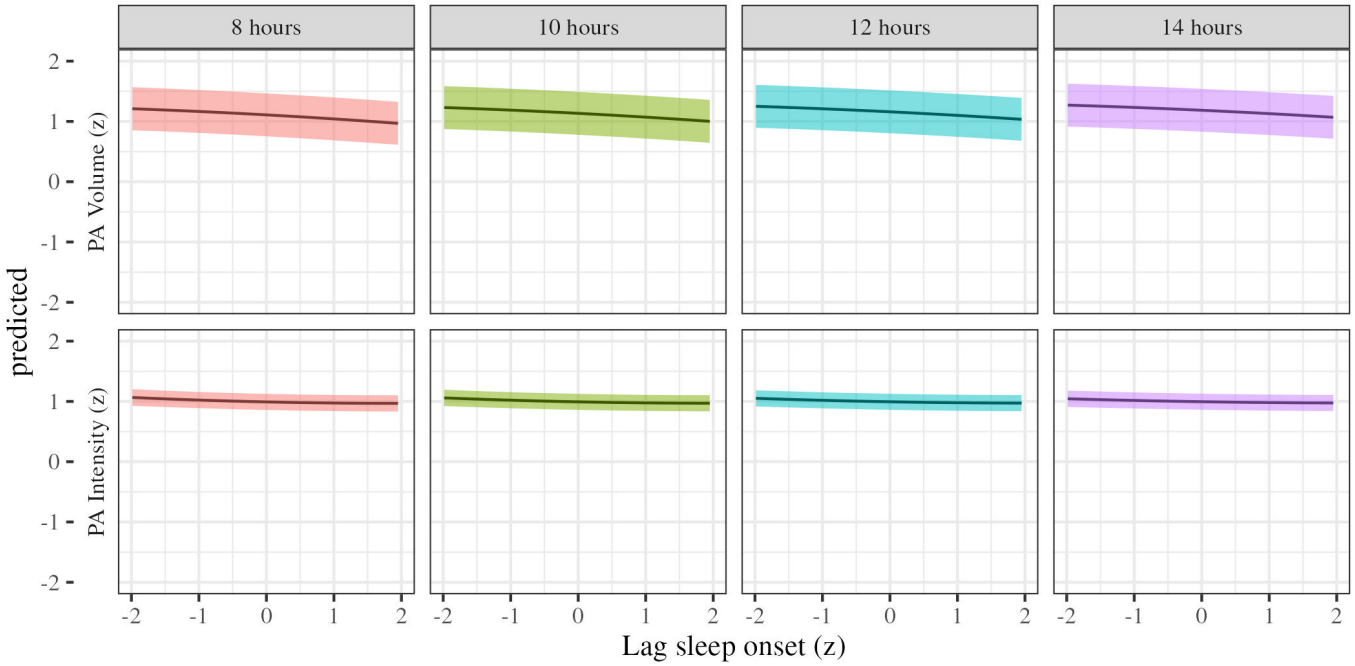

Figure 64. Physical activity by sleep onset moderated by daylight hours

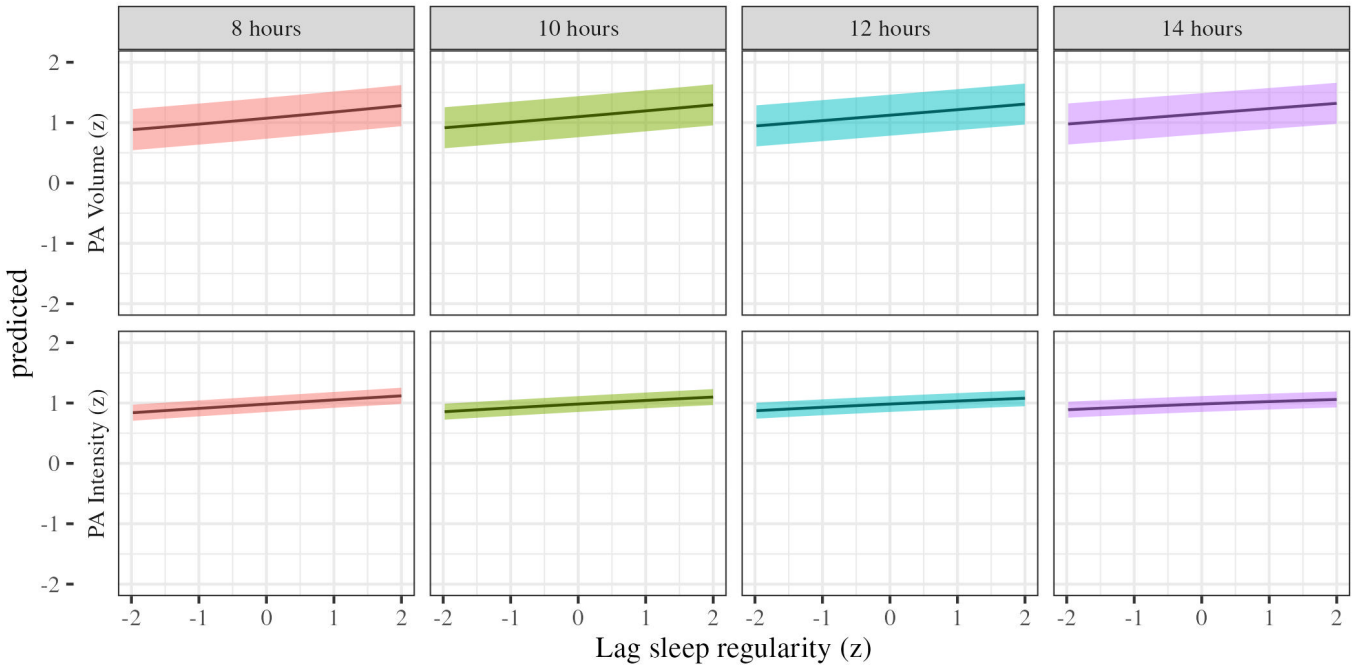

Figure 65. Physical activity by sleep regularity moderated by daylight hours

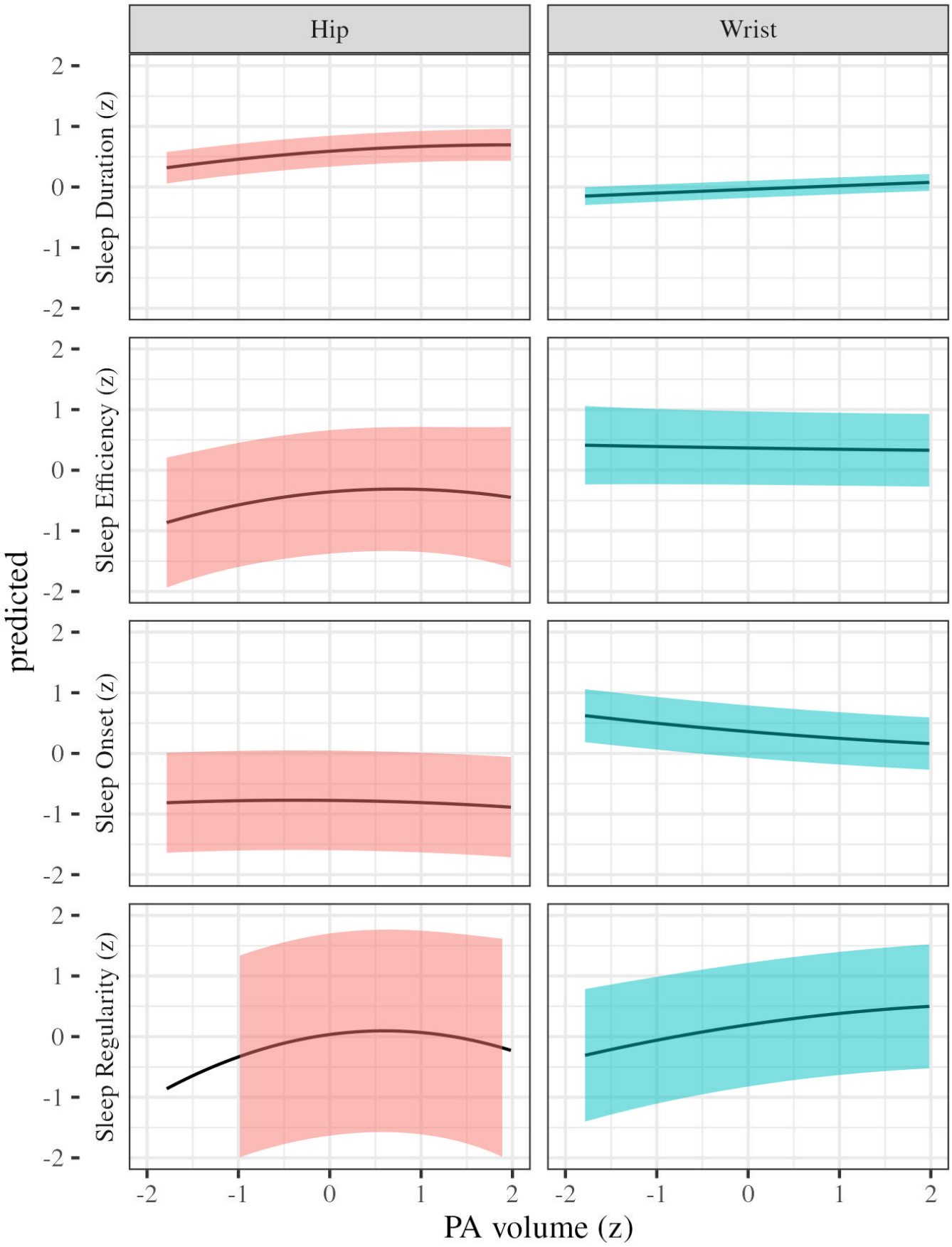

Figure 66. Sleep metrics on Physical activity volume by wear location

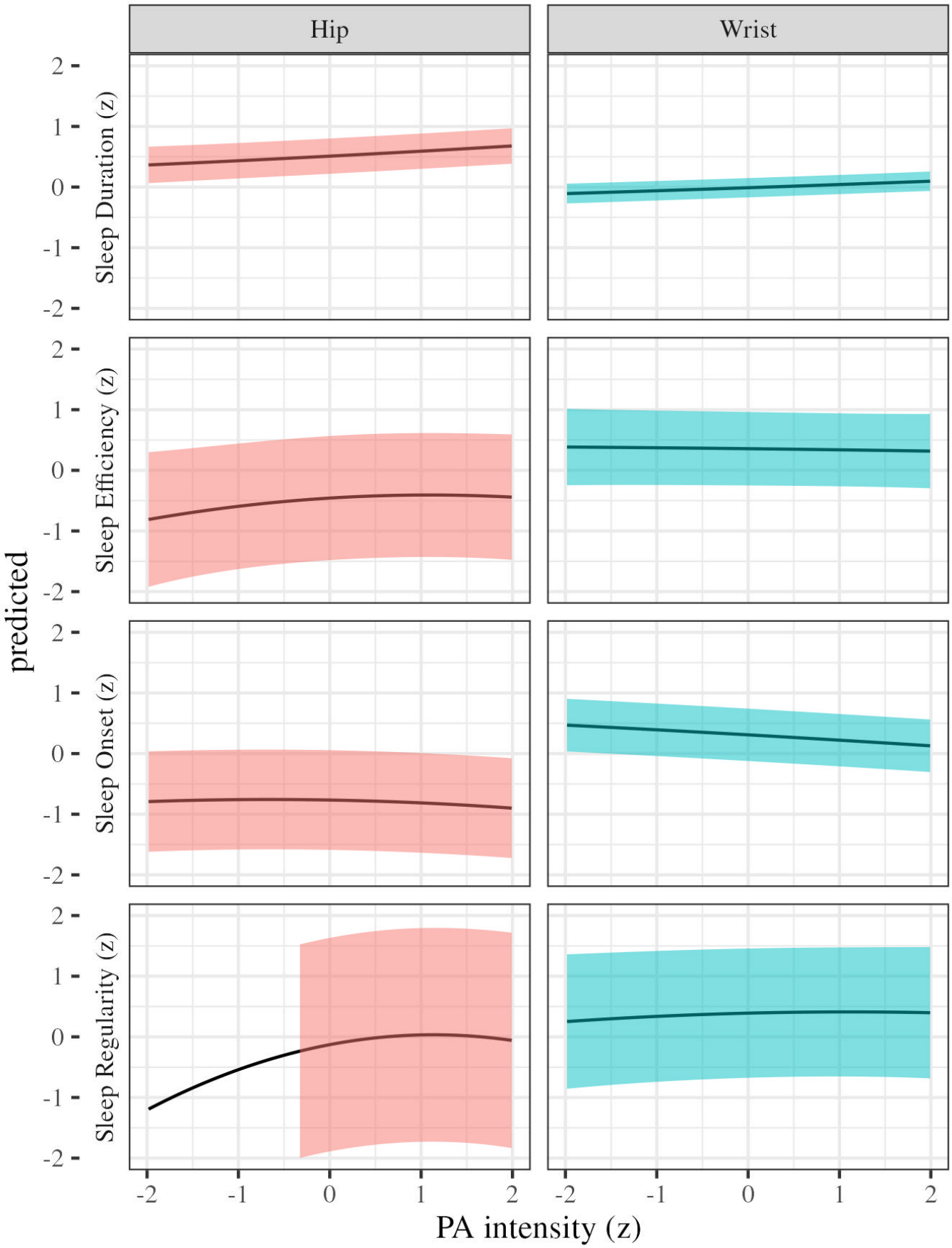

Figure 67. Sleep metrics on Physical activity intensity moderated by wear location

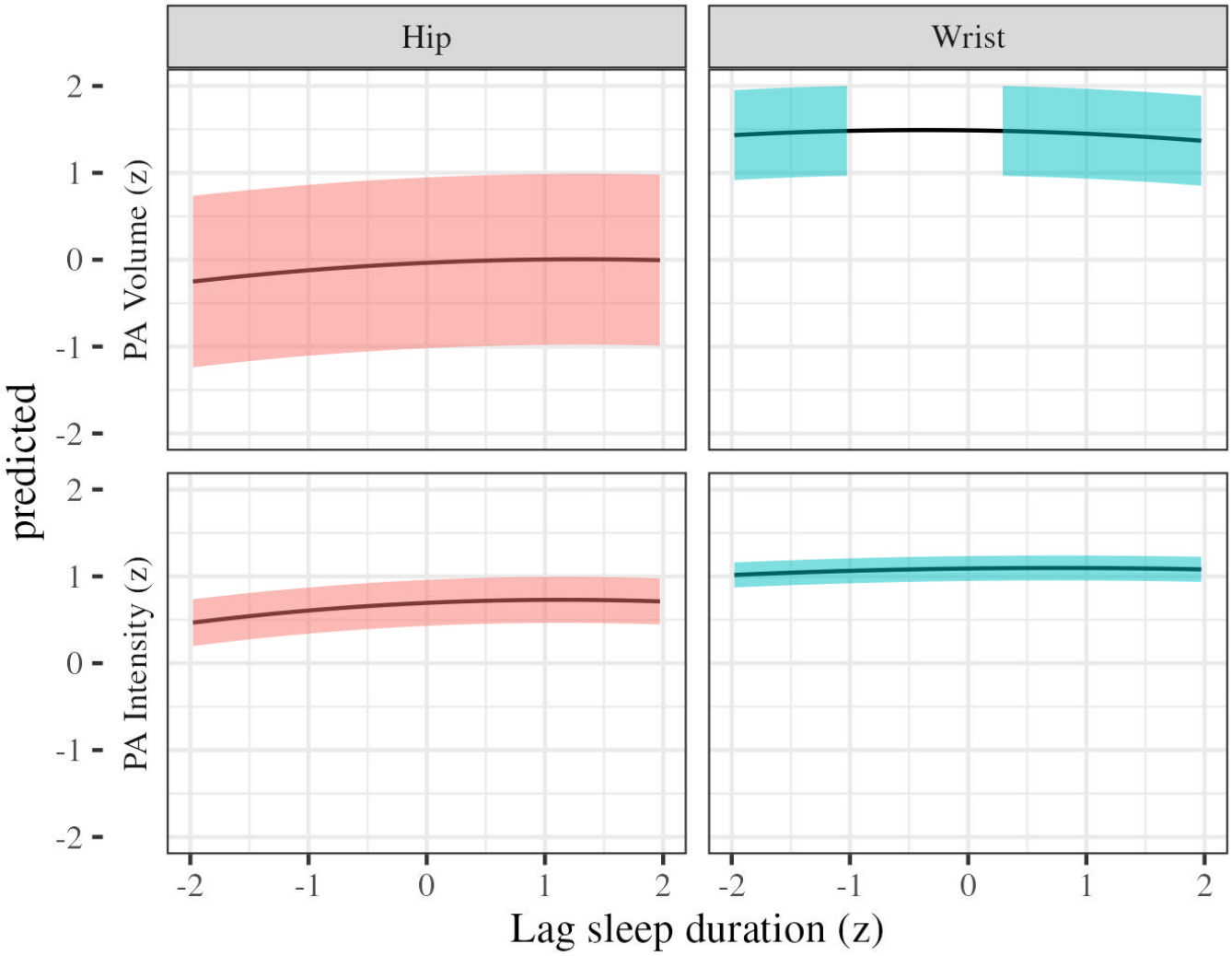

Figure 68. Physical activity by sleep duration moderated by wear location

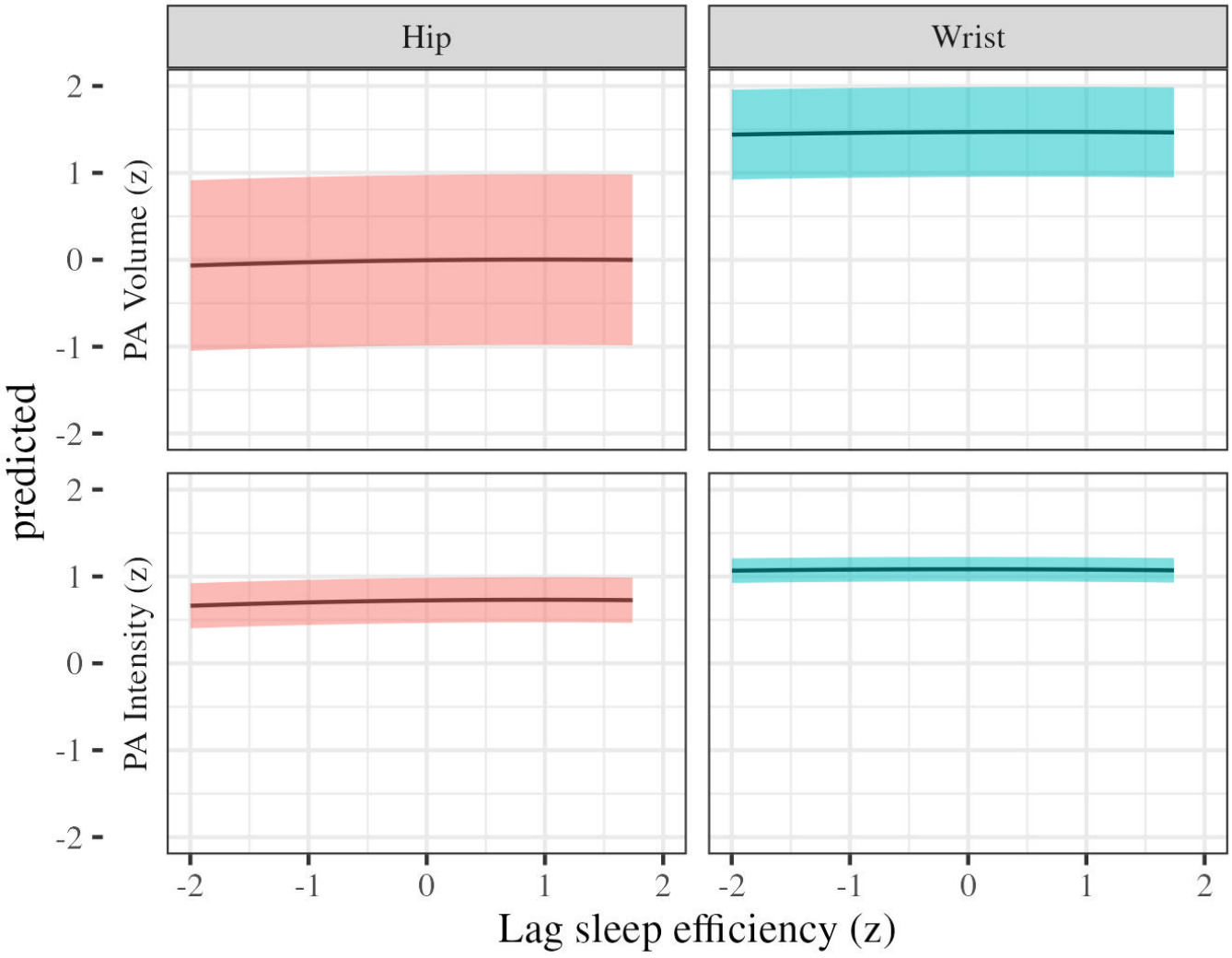

Figure 69. Physical activity by sleep efficiency moderated by wear location

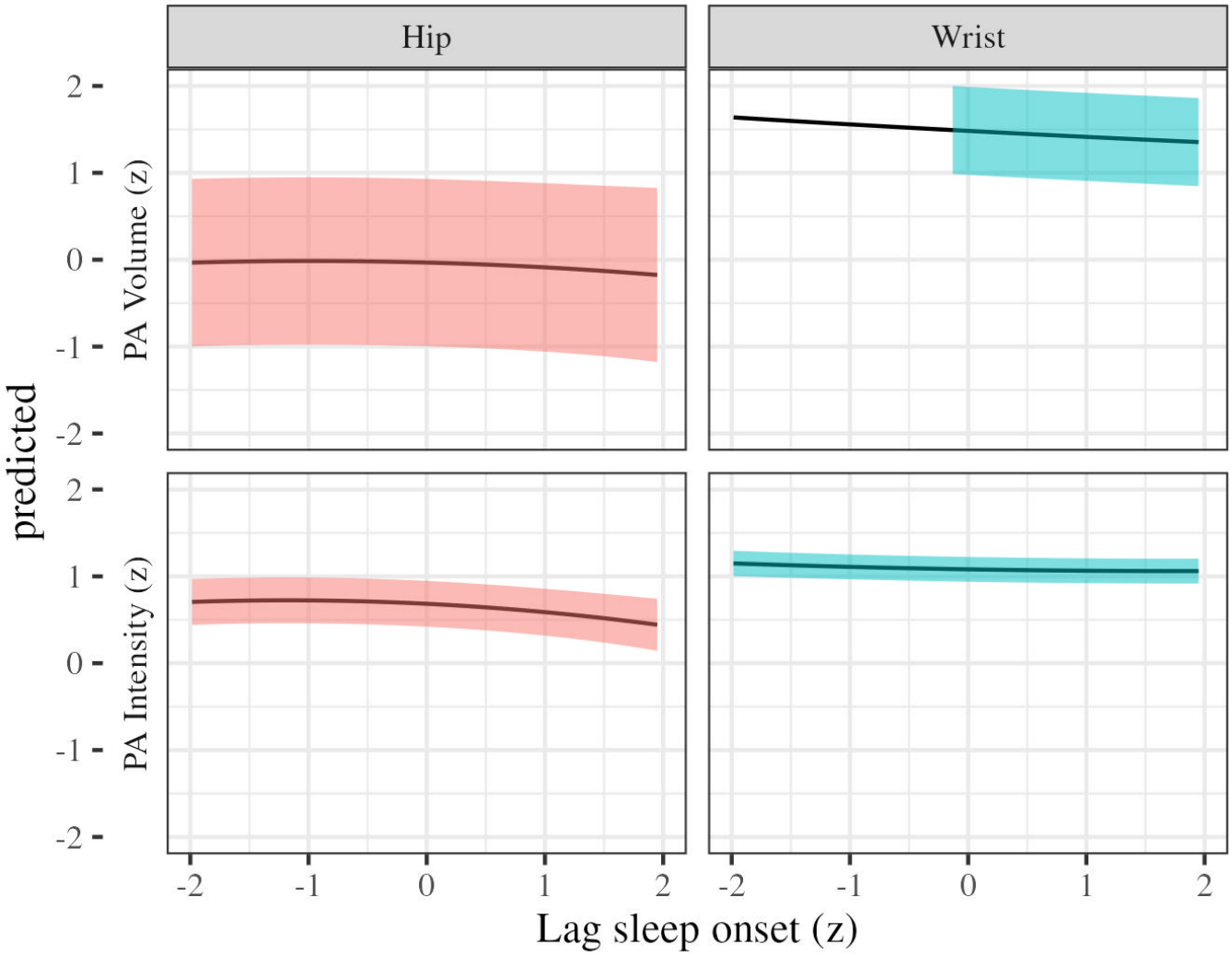

Figure 70. Physical activity by sleep onset moderated by wear location

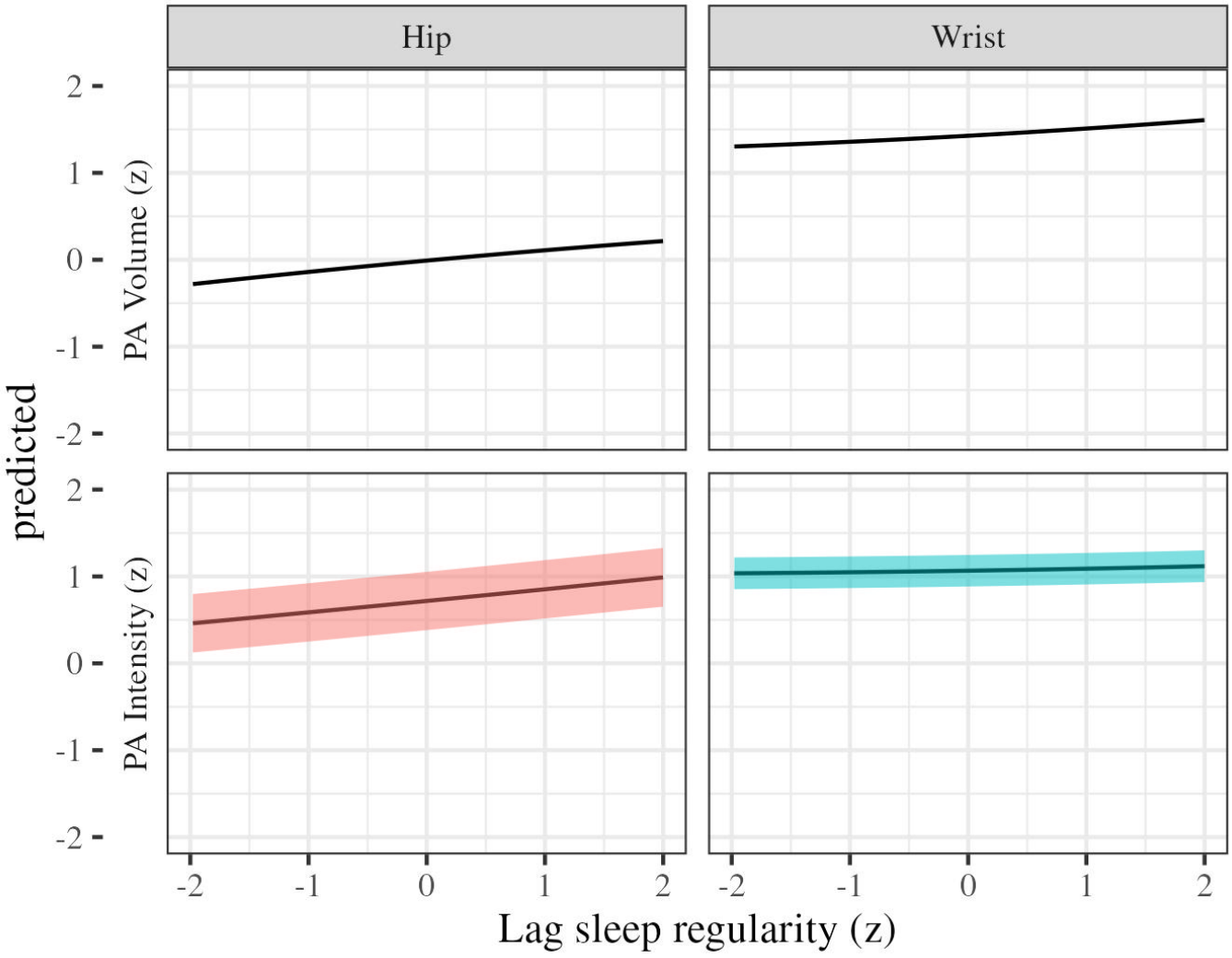

Figure 71. Physical activity by sleep regularity moderated by wear location

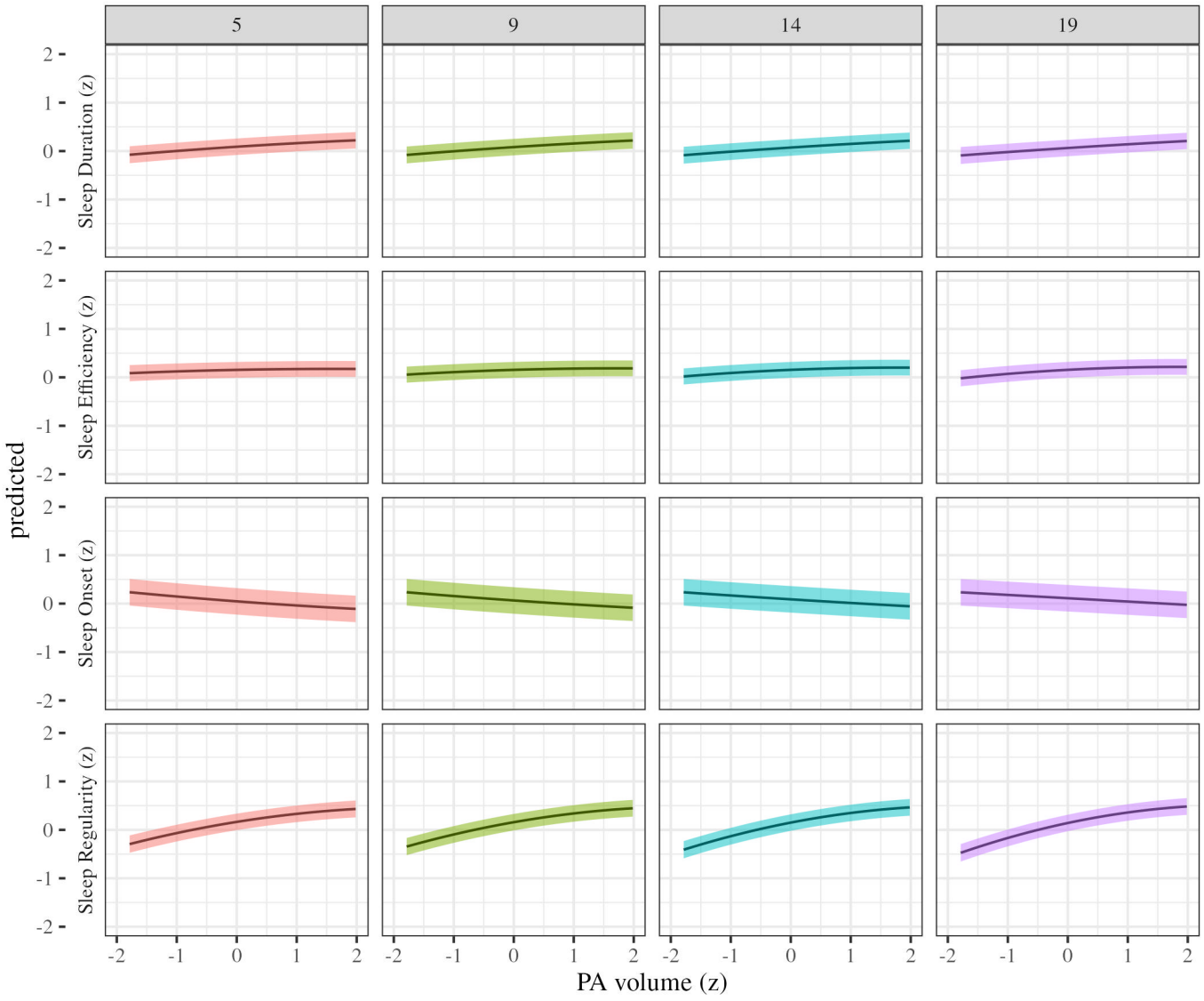

Figure 72. Sleep metrics on Physical activity volume by most active hour

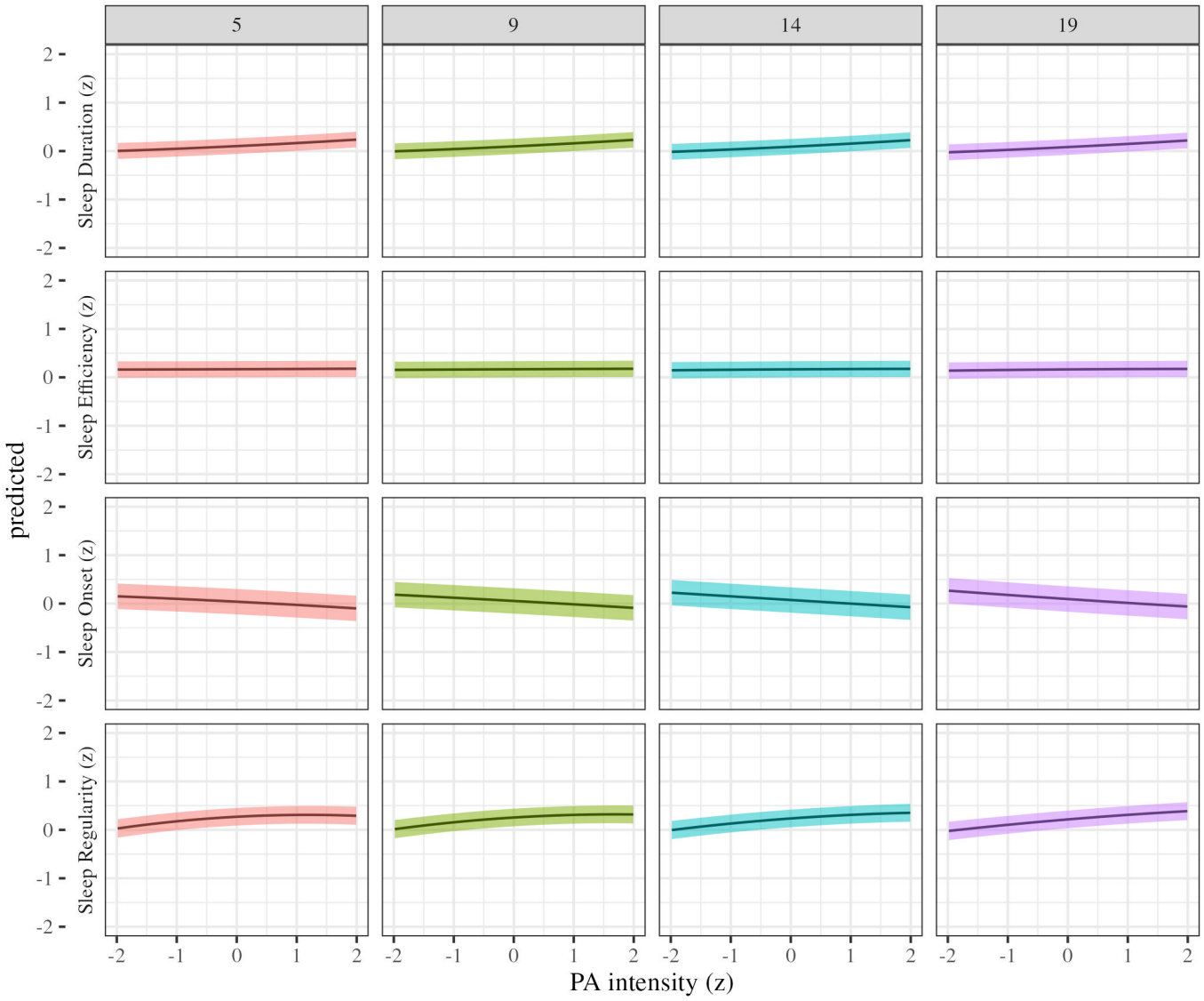

Figure 73. Sleep metrics on Physical activity intensity moderated by most active hour

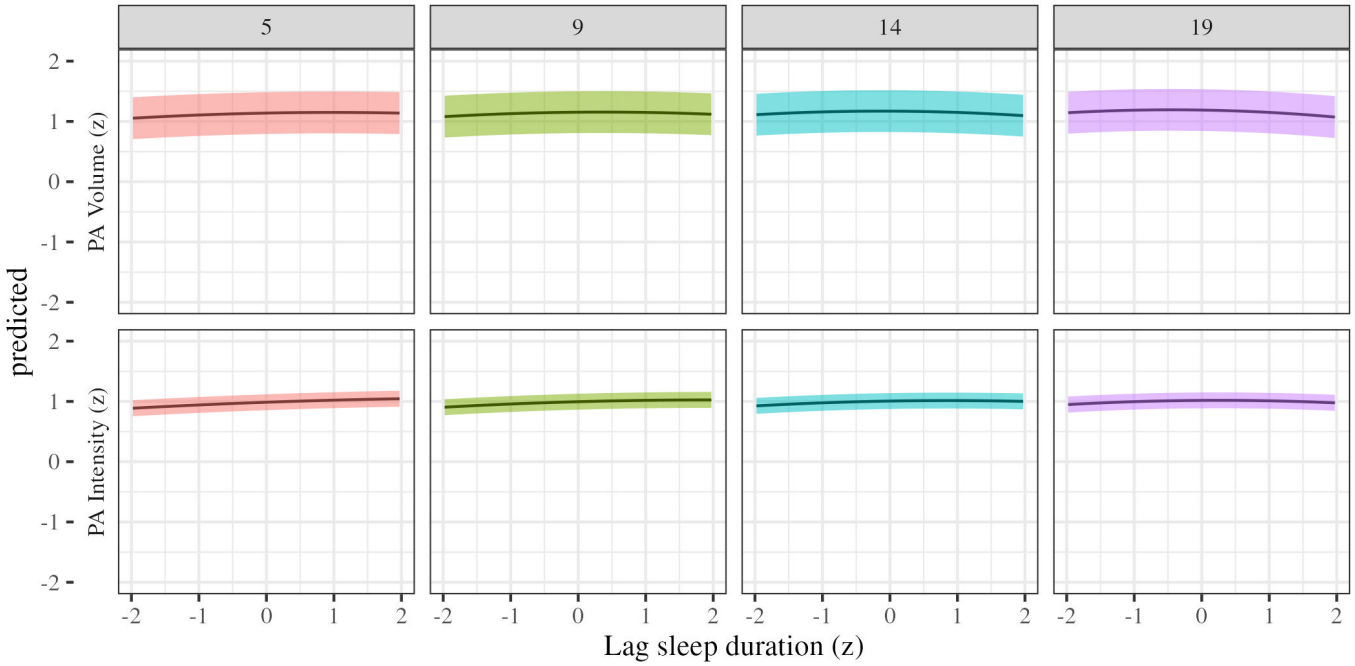

Figure 74. Physical activity by sleep duration moderated by most active hour

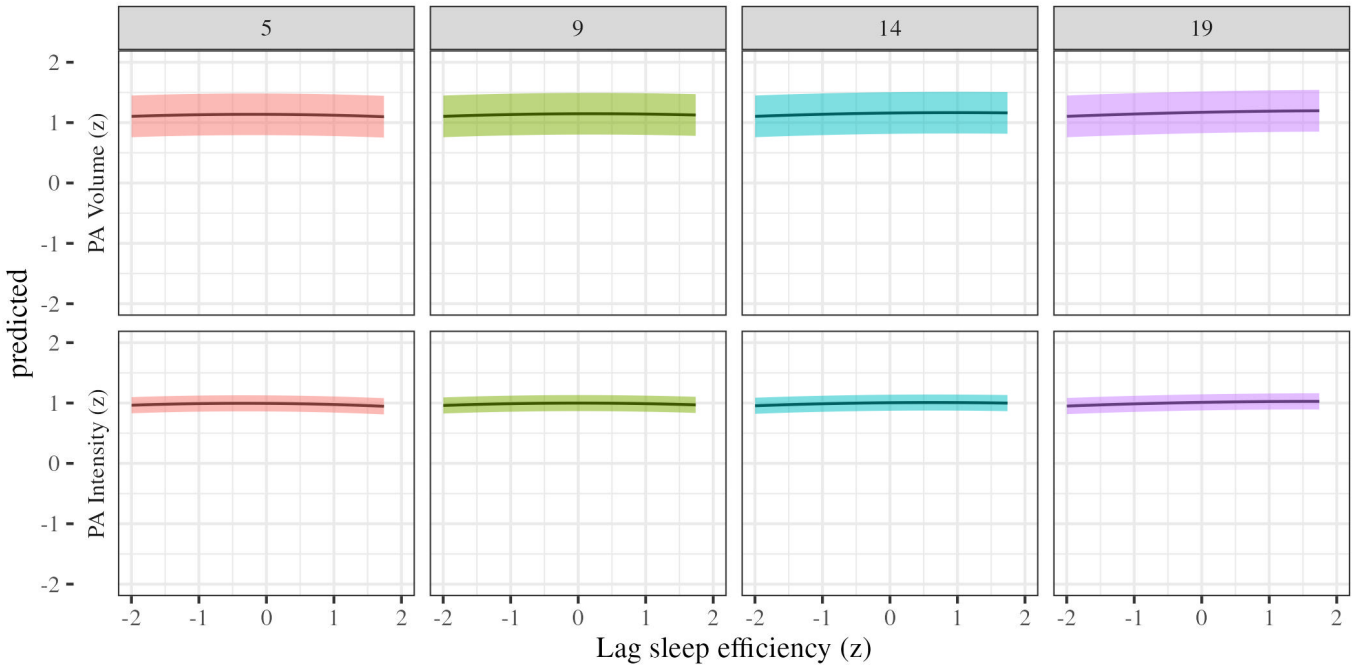

Figure 75. Physical activity by sleep efficiency moderated by most active hour

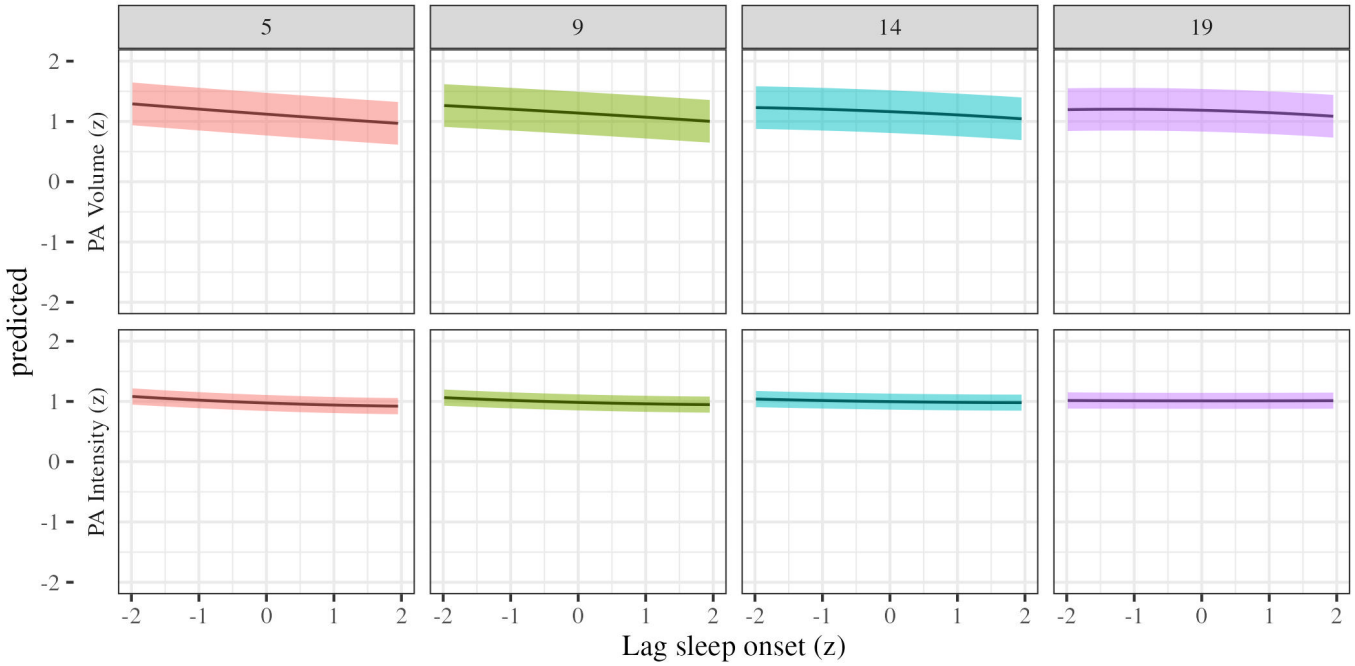

Figure 76. Physical activity by sleep onset moderated by most active hour

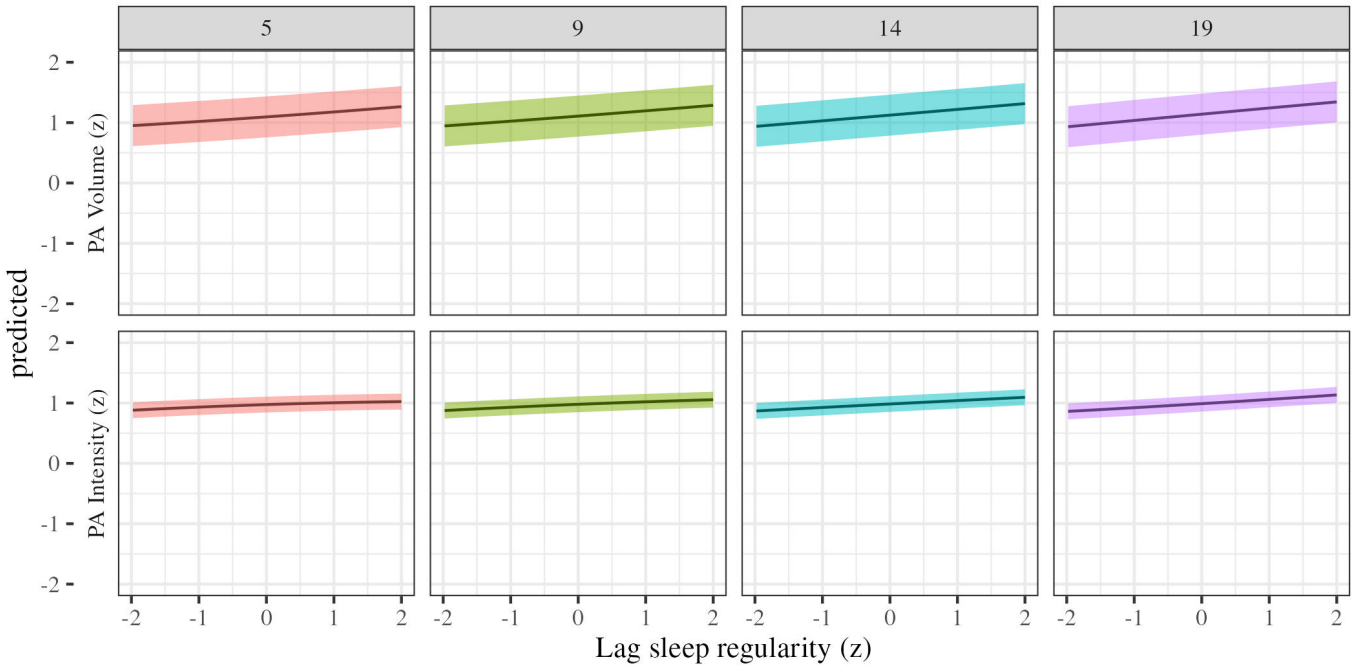

Figure 77. Physical activity by sleep regularity moderated by most active hour

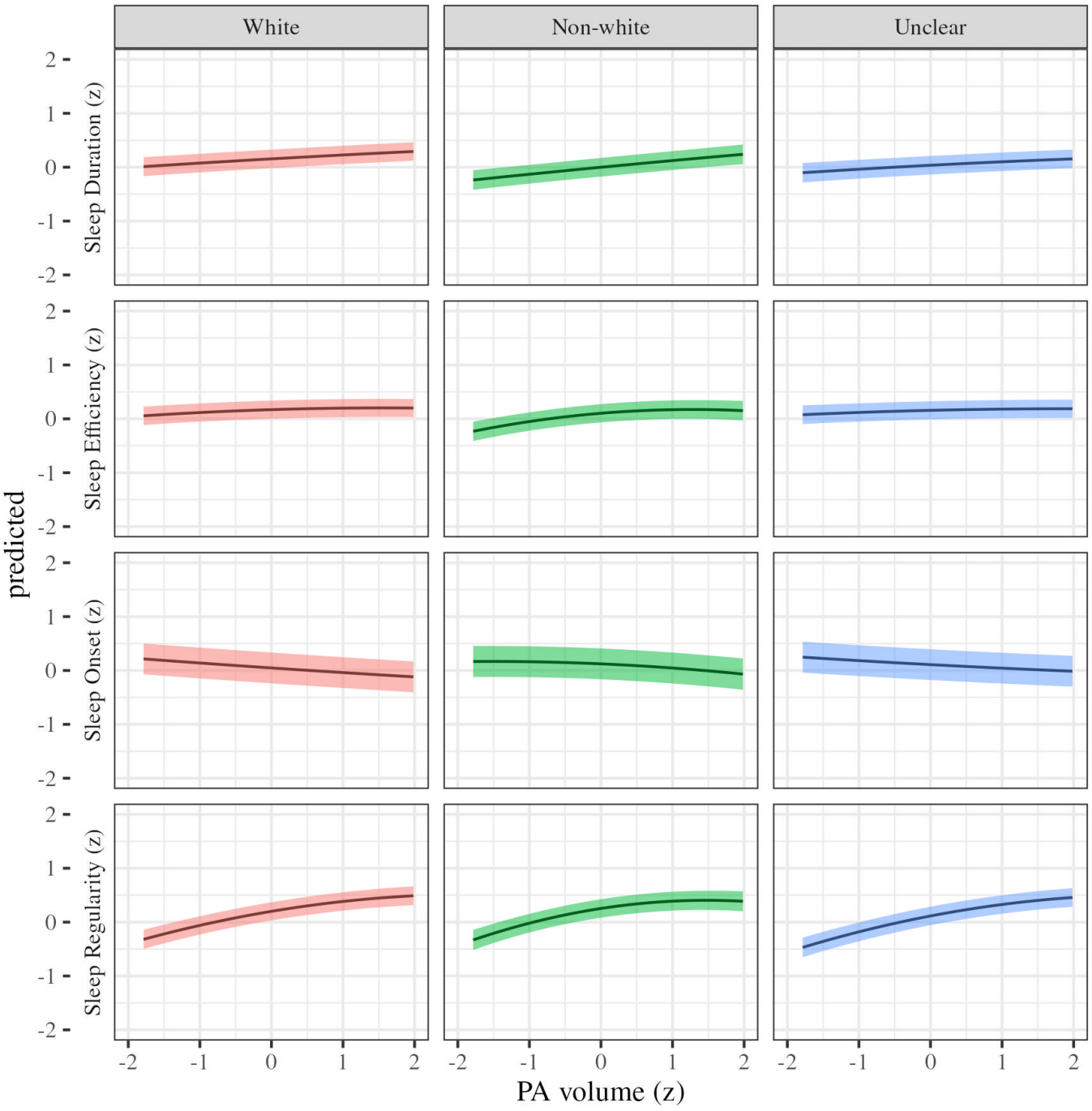

Figure 78. Sleep metrics on Physical activity volume by ethnicity

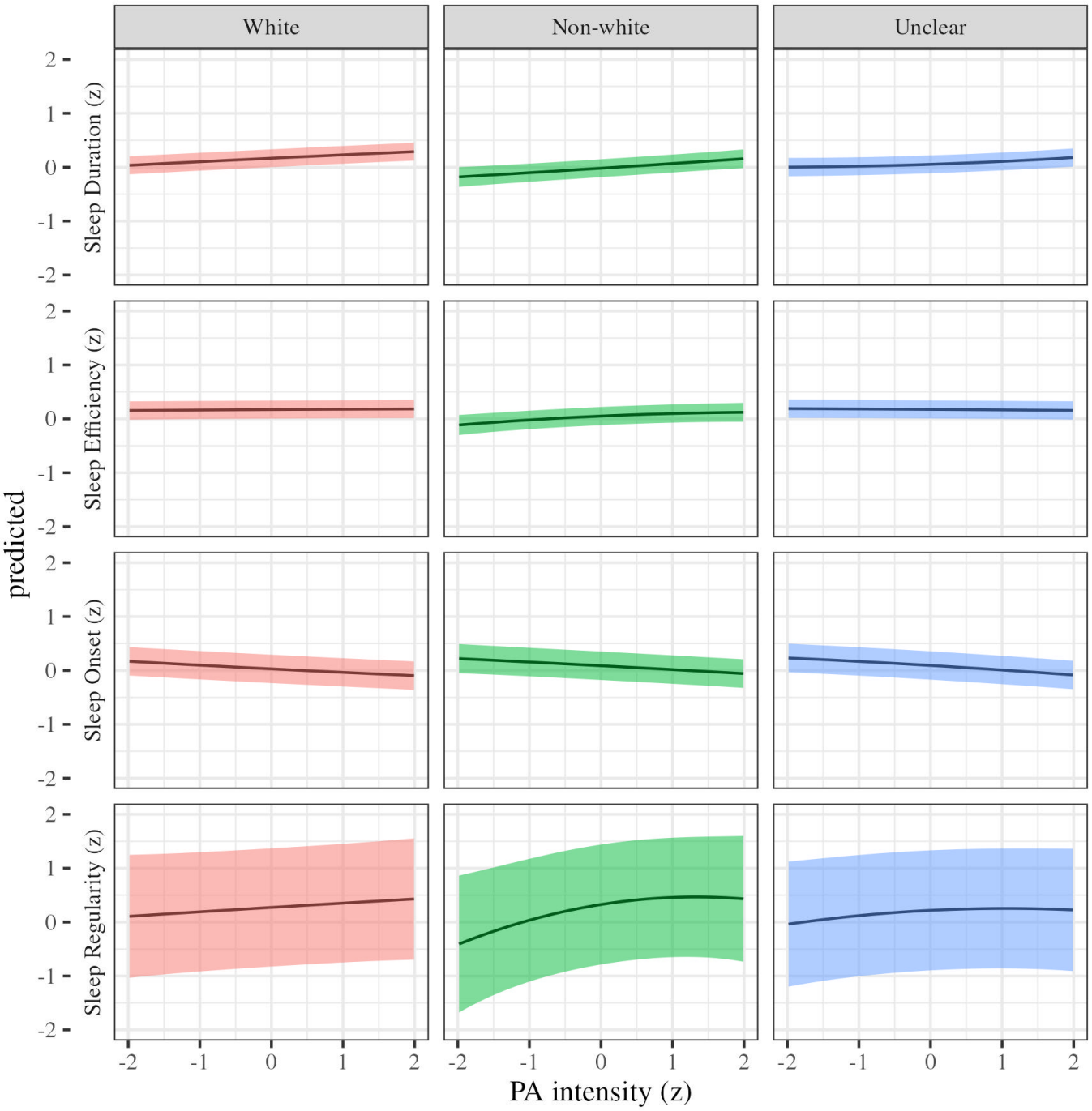

Figure 79. Sleep metrics on Physical activity intensity moderated by ethnicity

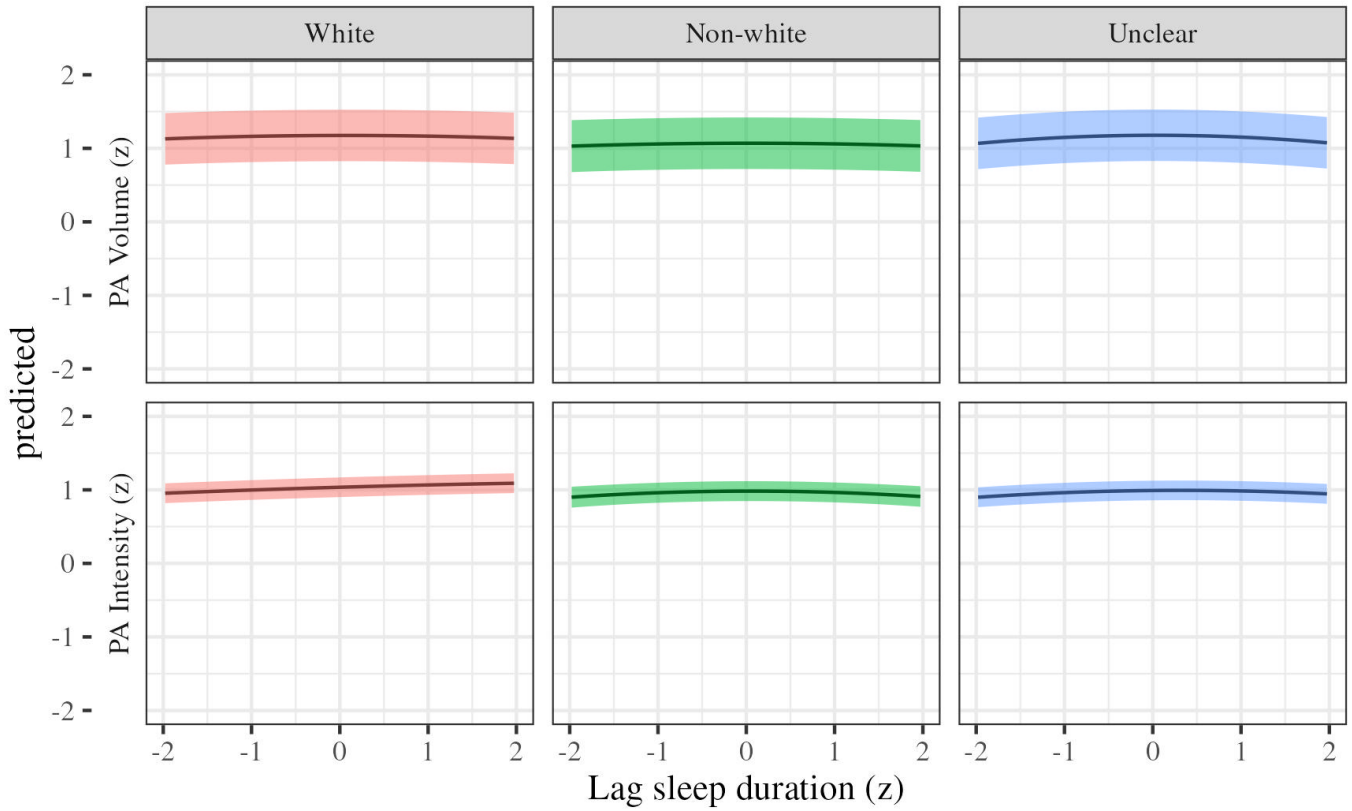

Figure 80. Physical activity by sleep duration moderated by ethnicity

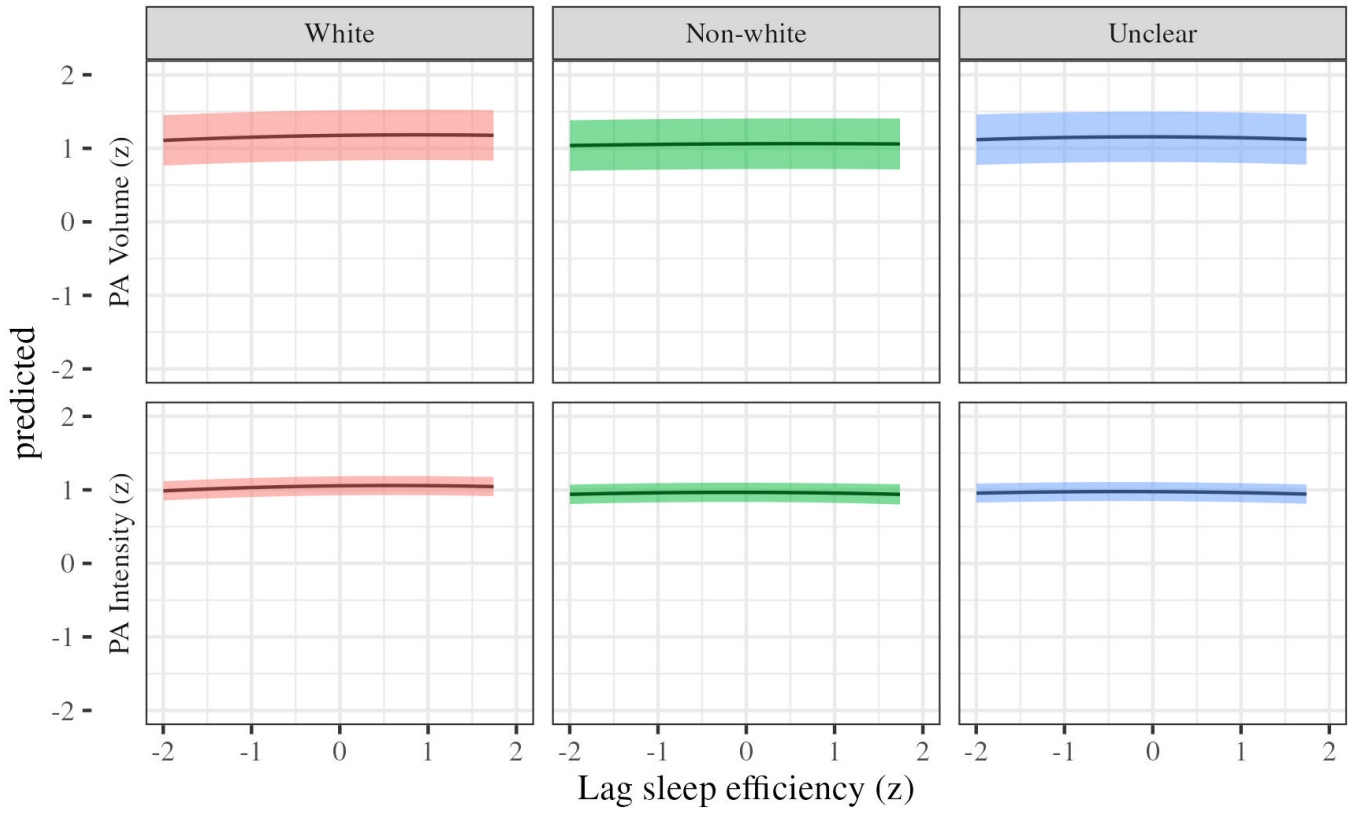

Figure 81. Physical activity by sleep efficiency moderated by ethnicity

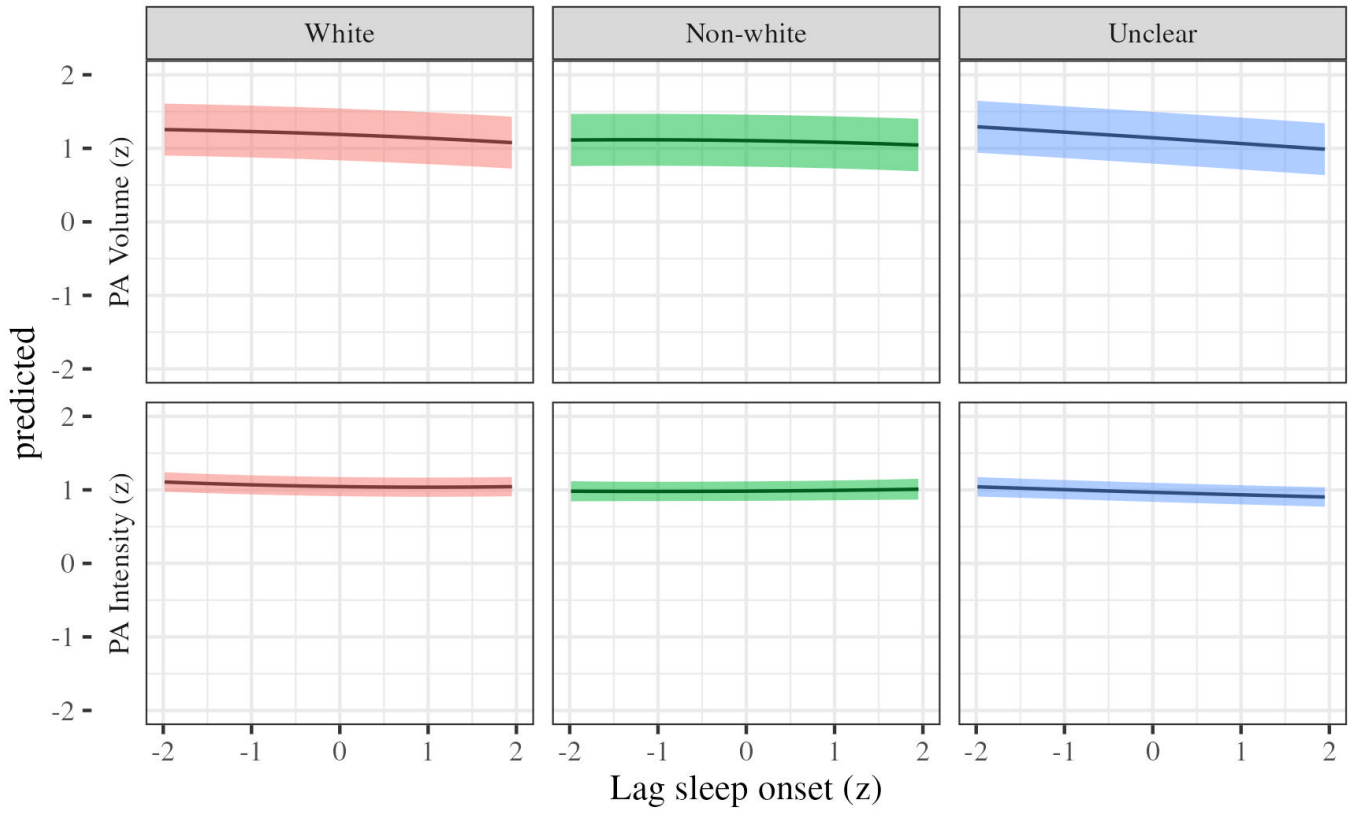

Figure 82. Physical activity by sleep onset moderated by ethnicity

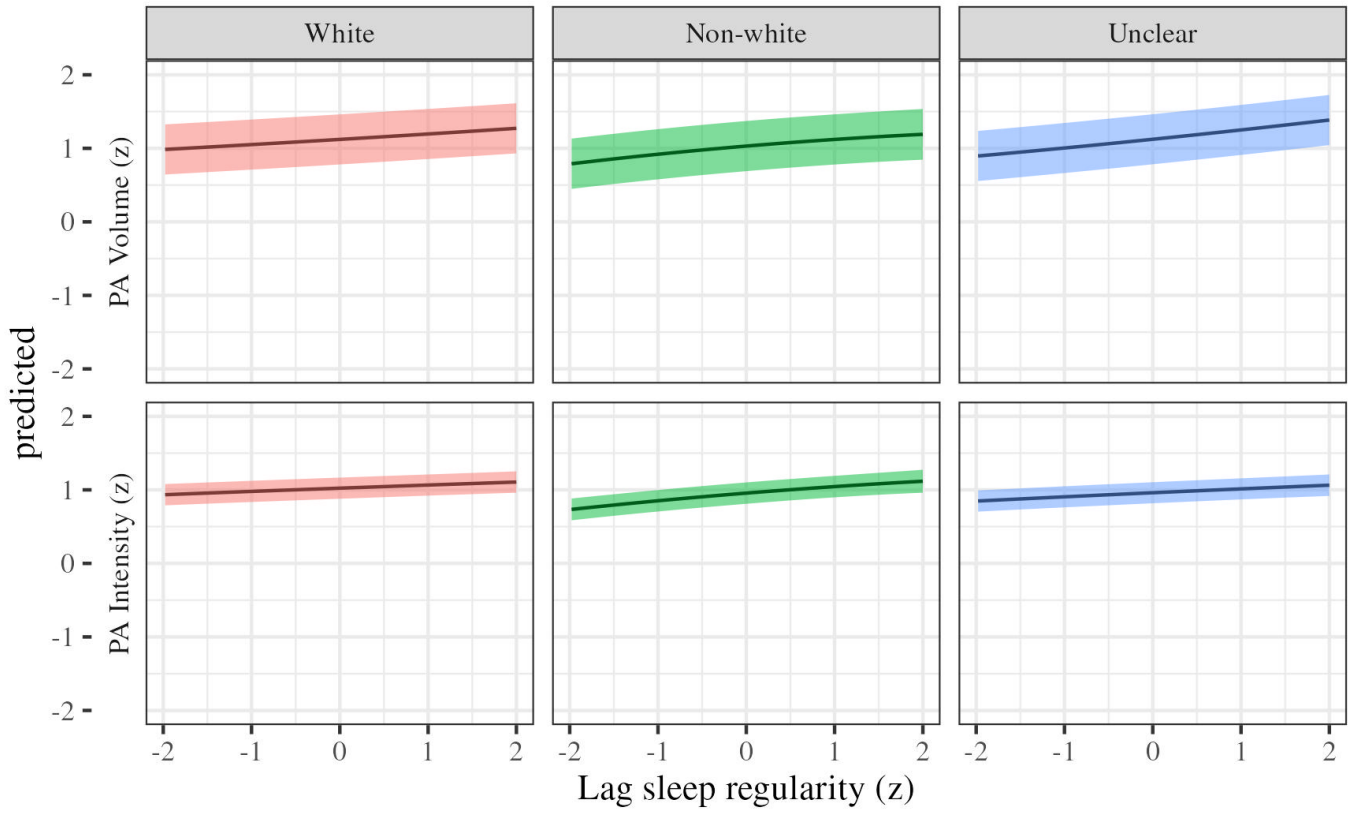

Figure 83. Physical activity by sleep regularity moderated by ethnicity